Joint Venture Formation in Procurement Auctions*

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Abstract

We study how joint venture formation affects competition in procurement auctions. Using Japanese public works data that identify joint venture members, we estimate a model of participation, partner choice, and bidding. Under equal sharing of profits and costs, the stable partition is unique for given beliefs about rival numbers. Equilibrium additionally requires these beliefs to match the distributions generated by stable partitions. This conditional uniqueness makes maximum likelihood estimation subject to equilibrium constraints and counterfactual computation tractable. Counterfactuals indicate that allowing joint ventures can lower winning prices and raise winning technical scores, particularly in larger projects. With many bids near the price floor, price effects alone miss an important benefit of joint bidding.

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1 Introduction

In many procurement auctions, firms may bid jointly. Public works agencies in Japan admit project-specific joint ventures on large construction projects, European procurers routinely receive bids from consortia, and joint bidding has a long history in auctions of offshore oil leases. The case for joint bidding is that some projects exceed what a single firm can undertake, and that firms with complementary capabilities can offer more together than apart. The case against it is that two firms that could each bid alone present one bid instead of two. Competition authorities accordingly assess joint bidding by asking whether the members could have bid individually: a consortium that lets firms take on a project none of them could undertake alone is considered harmless, whereas one among firms that could each compete for the contract may restrict competition (European Commission, 2023; Albano, Spagnolo and Zanza, 2009).

Whether joint bidding helps or hurts the procurer turns on a counterfactual that is never observed: what the members of a joint venture (JV) would do if they could not bid together. If they would not bid at all, JVs add bidders. If they would bid alone, JVs remove independent bidders—but a JV may still be a stronger bidder than either of its members, and other firms may respond to its presence. Whether a firm bids alone depends on the competition it expects, which depends on which other firms form JVs. Answering the question requires knowing who forms JVs with whom, and a model of formation at the level of firms and pairs of firms that can be combined with a model of bidding.

This paper builds such a model and estimates it with data in which the member firms of JVs are identified. In the formation stage, each potential entrant of an auction stays out, bids alone, or bids jointly with a specific partner. Firms weigh expected auction profits against the costs of participating alone or jointly. The outcome is a stable partition of the potential entrants into non-participants, solo bidders and JVs: no firm would gain by switching to solo bidding or staying out, and no available pair would both gain by forming a JV. Equal sharing makes each possible JV equally valuable to its two members; this structure yields a unique stable partition that can be computed by considering potential partnerships from highest to lowest value, making the likelihood of the observed partition tractable even in

auctions with more than a hundred thousand potential pairs. In the bidding stage, solo bidders and JVs compete in a scoring auction with a price floor, and firms differ persistently in bidding strength, both in their costs and in the technical scores of their proposals. Firms' beliefs about the number of rivals are required to be consistent with the model's equilibrium, conditional on unobserved auction-level states that make some auctions attract many bidders and others few.

Section 2 describes procurement by the Regional Development Bureaus of Japan's Ministry of Land, Infrastructure, Transport and Tourism. On projects that admit JVs, firms can bid alone or submit a joint proposal, and the winning bid is chosen using both price and technical score. The rules governing JV eligibility and a frequently binding price floor shape the participation and bidding choices in our model.

Section 3 builds a dataset that follows firms across solo and joint bids. We identify JV members in auction records from 2006 to 2022 and use the earlier records to construct each firm's bidding history and the set of potential participants in each auction. This lets us study which firms bid alone, which pair up, and which stay out in 396 large procurement auctions.

Section 4 documents the patterns that guide the model. JVs tend to undertake projects large relative to their members' past solo experience, often reunite previous partners, and win more frequently than solo bidders in the same auctions. These patterns motivate allowing firms to pool capacity, benefit from past partnerships, and differ in their bidding performance.

Section 5 formalizes the two-stage model outlined above. Joint bidding can change both the capabilities of the bidders and the number of competing bids, while firms choose whether and with whom to participate. The model lets us compare changes in the JV rules after accounting for the participation responses of members and other firms.

Section 6 estimates the bidding and formation stages in sequence. We first use prices and technical scores to recover persistent differences between firms and the profits they expect from each participation option, accounting for bids constrained by the price floor. We then estimate participation and partnership incentives from the observed combinations of solo bidders, JVs and non-participants, updating firms' expectations of competition along with the estimates.

Section 7 reports the estimated differences between JV and solo bidding and the incentives that explain partnership choices. The bidding estimates separate members' persistent capabilities from variation in their performance together, and show that measured differences in cost per unit of technical score depend on the solo comparison. Past joint bids lower the estimated cost of forming a JV, helping explain the persistence of partnerships. We also assess how well the model reproduces prices and participation.

Section 8 compares the current JV rules with a ban for projects below 10 billion yen, allowing former members and other firms to adjust their participation. The model predicts that permitting JVs benefits the procurer through both lower winning prices and higher winning technical scores in projects from 5 to below 10 billion yen. Across the broader sample, prices change little while winning scores fall under the ban. This comparison shows why price effects alone can miss consequences of joint bidding when a price floor binds. These are conditional model predictions; projects of 10 billion yen or more are excluded because participation fit is poor.

Related literature. The competitive effects of joint ventures have been studied in the merger literature (Shapiro and Willig, 1990; Estache and Iimi, 2009; Miller and Weinberg, 2017). Modeling the formation of mergers and joint ventures is difficult because the value of a coalition depends on the coalitions that others form; existing approaches use dynamic models (Gowrisankaran, 1999; Igami and Uetake, 2020) and matching models (Akkus, Cookson and Hortaçsu, 2016; Fox, 2018; Uetake and Watanabe, 2020). Our formation stage is a coalition formation game with pairs in which equal sharing makes preferences aligned (Farrell and Scotchmer, 1988; Banerjee, Konishi and Sönmez, 2001; Pycia, 2012) and pairs globally ranked (Abraham et al., 2008), which delivers a unique stable outcome conditional on firms' beliefs about competition. We exploit this property to estimate participation and partner choice at the level of firms and pairs by maximum likelihood, building on the empirical literature on matching and coalitions (Choo and Siow, 2006; Sørensen, 2007; Gordon and Knight, 2009; Agarwal, 2015; Menzel, 2015; Weese, 2015). Externalities across coalitions enter through beliefs about rival numbers that are consistent with equilibrium, as in entry models with incomplete information (Seim, 2006; Bajari et al., 2010).

Joint bidding has been studied in theory (Cho, Jewell and Vohra, 2002; Cantillon, 2008;

Chatterjee, Mitra and Mukherjee, 2017) and in procurement (Gugler, Weichselbaumer and Zulehner, 2021), subcontracting (Bouckaert and Van Moer, 2021) and patent auctions (Asker, Baccara and Lee, 2021); see Asker and Nocke (2021) for a review of synergies in mergers and joint ventures.[1] Empirical studies of joint bidding in procurement (Iimi, 2004; Estache and Iimi, 2009; Branzoli and Decarolis, 2015) generally find pro-competitive effects, and Gugler, Weichselbaumer and Zulehner (2021) endogenize the formation of bidding groups in reduced form. We complement this work by identifying the member firms of JVs and estimating which firms participate jointly and with whom. This lets us compare a joint bid with the solo bids that could replace it, accounting for changes in both the number and the capabilities of the bidders.

We also contribute to the empirical literature on procurement auctions with entry (Li and Zheng, 2009; Krasnokutskaya and Seim, 2011; Athey, Levin and Seira, 2011); here, entry by other firms partly offsets the loss of bidders when JVs are banned. The bidding stage builds on the first-price inversion of Guerre, Perrigne and Vuong (2000) adapted to scoring auctions (Che, 1993; Asker and Cantillon, 2008; Hanazono et al., 2020; Perrigne and Vuong, 2019; Lewis and Bajari, 2011; Kong, Perrigne and Vuong, 2022) and accounts for a binding price floor (Decarolis, 2014; Chassang and Ortner, 2019). We use this connection between entry and bidding to assess joint bidding through both winning prices and technical scores. When the price floor binds, changes in participation can alter the winning proposal even if its price changes little. The auction-level states relate to unobserved heterogeneity in auctions (Krasnokutskaya, 2011) and in games (Aguirregabiria and Mira, 2007), and the role of past joint bids echoes the persistence of relationships between firms (Kellogg, 2011).

2 Institutional Background

This section describes the procurement auctions we study and the rules that shape joint bidding in them. We first describe the auction format, a first-price sealed-bid scoring auction with a secret reserve price. We then describe project-specific joint ventures—what they

[1] In common-value auctions of natural resources, joint bidding can act as a bidding ring (Mead, 1967; Levin, 2004), but empirical studies find pro-competitive effects through information pooling, relaxed budget constraints and risk sharing (Millsaps and Ott, 1985; Moody and Kruvant, 1988; Hendricks and Porter, 1992).

are, when the procurer admits them, and how they bid. Finally, we describe the low-bid investigation threshold, which acts as a price floor.

2.1 Scoring auctions at the Regional Development Bureaus

We study procurement auctions conducted by the Regional Development Bureaus of the Ministry of Land, Infrastructure, Transport and Tourism (MLIT), which build and maintain national highways, rivers, dams, ports and other infrastructure under the direct control of the national government. Contracts are classified into some twenty work categories, including civil engineering, steel-bridge construction and prestressed-concrete construction.[2] To bid, a firm must be registered as a qualified bidder of the bureau in the work category; categories with many firms also rank firms into grades by size, certified engineers and technical capacity, and the tender notice states which grades may bid, the technical requirements of the project, and whether joint ventures may bid. The reserve price is the engineer's estimate of the cost of the project; it is not disclosed before bidding, and bids above it are invalid. Contracts above roughly 680 million yen are covered by the WTO Agreement on Government Procurement and are let by general competitive tendering.[3] Our formation sample also includes auctions below this threshold; Section 3.4 describes its construction.

Large projects are awarded by scoring auction. Each bidder submits a technical proposal and a price. The proposal is scored by a rule announced in the tender notice: every bidder that meets the minimum requirements receives 100 points, and up to 100 additional points are awarded for the firm's track record and engineers and for the content of the proposal, such as construction methods and quality control. The procurer awards the contract to the bidder with the highest *effective bid*, the ratio of the score to the price. Scores are lower in the early years of our data: the median score in the bid panel rises from 110 in 2006 to about 150 in 2011 and is stable afterwards. A qualification stage fixes the set of eligible bidders, and a joint venture is examined as a single bidder against the same requirements as a firm. Eligible bidders submit their technical proposals, which are scored over several weeks, and then sealed prices. The bureau discards bids above the reserve, awards the contract, and

[2]Table A2 in Appendix A lists the categories of the bid records.
[3]The threshold is revised every two years (Table A1 in Appendix A).

Table 1. Size thresholds for project-specific joint ventures

Facility	Project-specific JV allowed (approximate project cost)	Projects requiring special technology
Bridges	2 bn yen	above 1 bn yen
Buildings	3 bn yen	above 1.5 bn yen
Weirs, sluices, pumping stations, tunnels, utility tunnels, etc.	5 bn yen	above 2.5 bn yen
Dams	10 bn yen	above 5 bn yen

Notes: MLIT circular on joint ventures in direct construction (Kensetsushō Kōhatsu No. 176, 1988, last revised 2006), Part 1, sections 1(1)–(2). The amounts are guidelines; the bureaus decide whether a project admits JVs and state it in the tender notice. The bridge threshold also applies to electrical and air-conditioning works separated from buildings.

publishes the prices and scores of all bidders.

2.2 Project-specific joint ventures

A project-specific joint venture (*tokutei kensetsu kōji kyōdō kigyōtai*) is a consortium formed to carry out one particular project and dissolved when the project is completed. Other types of joint ventures, registered for a period of years, bid on smaller recurring contracts and rarely in the auctions we study. Throughout the paper, "JV" means a project-specific JV.

When joint ventures may bid. A circular of the former Ministry of Construction, issued in 1988 and last revised in 2006, allows project-specific JVs only on large and technically demanding projects, with a size threshold that depends on the type of facility (Table 1): about 2 billion yen for bridges, 3 billion yen for buildings, 5 billion yen for weirs, sluices, tunnels and similar structures, and 10 billion yen for dams, and half of these amounts for projects that require special technology. The thresholds are guidelines, and the bureaus apply them differently. In Kanto bridge construction the 2 billion yen threshold is applied sharply, while other bureaus admit JVs at lower sizes or rarely at all (Section 4).

Who may form a joint venture and how it bids. A JV has two or three members, each qualified for the work category; each member's equity share must be at least 60 percent of an equal share—at least 30 percent in a two-firm JV—and the representative member

must be the one with the largest capacity and share. Since 2003, single firms may also bid in competitions that admit JVs, so a firm in a JV-eligible auction can bid alone, bid jointly with a partner, or not bid. The partners agree on shares and the division of work and register the JV by the deadline for applications. The JV then bids as one bidder, with one technical proposal and one price; its score reflects the members' combined track records and engineers. In our estimation sample all JVs have two members.

2.3 The low-bid investigation threshold

For each auction, the bureau computes a low-bid investigation threshold from the components of the engineer's estimate.[4] A bid below the threshold triggers an investigation of whether the bidder can perform the contract at that price. The bidder must cooperate with inquiries and may be asked to provide a price breakdown; an adverse finding can prevent it from receiving the contract.[5] In practice, the threshold works as a price floor: in our estimation sample, only 1.8 percent of valid bids fall below it, while 56 percent of valid bids and 75 percent of winning bids are at or below 101 percent of the threshold. Its median ratio to the engineer's estimate among the winners in our sample rose from 0.80 in 2006 to 0.92 in 2022. A floor that binds this often limits what a low-cost bidder gains by bidding aggressively and matters for the profits that firms expect from bidding; we include it in the bidding stage of the model.

3 Data

This section describes the data. We start from the bid records published by seven Regional Development Bureaus and construct three objects that the analysis needs and that the records do not directly contain: the member firms of each JV, the histories of firms and pairs of firms, and the set of potential entrants of each auction. We then describe the estimation sample.

[4]See MLIT, "Revision of the Formula for the Low-Bid Investigation Threshold" (February 24, 2022), p. 2 (in Japanese).

[5]See Ministry of Construction, "Administrative Procedures for Applying the Criteria under Article 85 of the Budget, Settlement of Accounts and Accounting Ordinance," Notice No. 248 (March 19, 1976), sections 2(4), 4, and 6–8 (in Japanese).

3.1 Bid records and joint-venture members

For each auction, the bureaus publish the project name, office, date and work category, the engineer's estimate, the low-bid investigation threshold, and, for every bidder, its name, price, score and whether it won; withdrawn and invalid bids are also recorded. We collect these records for the Chugoku, Shikoku, Chubu, Kanto, Kyushu, Hokuriku and Kinki bureaus. Between 2006 and 2022 the panel contains 86,485 auctions and 574,553 bidder records; about 84,400 auctions are competitive and the rest negotiated contracts. The full panel supplies the histories used to construct the formation sample and the bids used to estimate the bidding stage. Section 3.4 describes the formation sample.

The records list a JV under its own name, which usually strings together abbreviated names of its members.[6] We parse these names into member tokens and link each token to a firm in the roster of firms that bid alone, using exact name matches within the same bureau, work category and period and a curated dictionary of abbreviations with recorded evidence; ambiguous cases are left unresolved (Appendix A.2). Across the whole panel, all named members are linked for 4,591 JV bids of all types, which we use to measure the joint-bidding histories of pairs of firms.

3.2 Firm histories

We use past participation to measure firms' capacities and the histories of pairs of firms. Both use only auctions held before the auction in question.

Capacity. We measure the capacity of firm i by K_i, the largest engineer's estimate among the auctions in which it previously submitted a valid solo bid in the same bureau and work category. The *capacity shortfall* of firm i in auction t is $g_{it} = \max\{0, \log(R_t/K_i)\}$, where R_t is the engineer's estimate; the shortfall of a pair is $g_{ijt} = \max\{0, \log(R_t/(K_i + K_j))\}$, so that a JV pools its members' capacities.

Joint-bidding history and contact. For each pair, n_{ijt} counts the times the two firms previously bid together in a two-member JV in the same bureau and work category, and

[6]The published records do not otherwise list the members, and JV names occasionally omit a member.

$h_{ijt} = \log(1 + n_{ijt})$. A JV is *repeat* if $n_{ijt} > 0$ and *new* otherwise. Two firms have *contact* if they previously bid in the same auction of the bureau and work category, alone or in JVs.

3.3 Potential entrants

To study participation, we need to identify firms that could bid even when they do not appear in the current auction's records. We define the *potential entrants* of auction t as firms listed, alone or as identified JV members, in at least one earlier auction of the same bureau and work category during the preceding three years, on a project whose engineer's estimate was at least $R_t/10$. These requirements use recent activity and project-scale experience to identify firms plausibly able to participate in the auction. The set is constructed before observing participation in auction t; firms are not added to it because they participate in the current auction.

The three-year window allows firms that bid infrequently to remain potential entrants, while the size screen excludes firms whose recent experience is confined to much smaller projects. Comparisons of these rules show that shorter windows or tighter size screens exclude more auctions with observed participants, whereas looser screens substantially expand the sets with little gain in coverage (Appendix A.3). We retain firms with relatively little capacity because they may participate through a JV; the model lets a capacity shortfall lower their incentive to bid rather than ruling them out by an additional capacity cutoff.

3.4 Estimation sample

We estimate the formation model on large projects in bureau–work categories in which project-specific JVs are observed, retaining auctions both with and without JVs. We begin with 547 competitive auctions held between 2006 and 2022, using the smallest project in an initial sample of identified JV bids to set the minimum size for each bureau and work category.[7] We retain the 477 auctions in which all listed firms, including the members of every JV, are identified, then the 462 in which all JVs have two members, and finally the

[7]The initial sample contains 450 JV bids in 196 auctions across 13 bureau–work categories. The size thresholds describe the support of that sample; the model separately restricts the availability of joint bidding (Section 4). Appendix A.3 gives the construction details.

396 in which every listed participant belongs to the set of potential entrants.[8] The final restriction allows us to represent every observed solo bidder and JV member within a set determined from prior records. It limits the analysis to auctions whose participants are covered by the historical rule.

A firm *participates* if it is listed in the bid record, alone or as a JV member, including firms that later withdrew or whose bids were invalid. Of the 6,471 listed bids, solo or joint, 4,697 have a positive price. The sample contains 232 general civil engineering auctions, 161 steel-bridge construction auctions, one prestressed-concrete construction auction, and two air-conditioning auctions. These are the work categories used in the administrative records.

Table 2 describes participation in these auctions. The 396 auctions have 6,159 solo bids and 312 JVs, of which 145 are new and 167 repeat; 127 auctions have at least one JV. The median engineer's estimate is 2.45 billion yen, and six auctions exceed 10 billion yen. Across the sample, the sets of potential entrants contain 2,106 distinct firms, of which 171 participate at least once: 83 only alone, nine only as JV members, and 79 in both roles. The remaining 1,935 firms qualify through their earlier participation but are never listed as participants in these 396 auctions.

The median auction has 58 potential entrants and the largest 521. In total, the sample contains 50,136 firm–auction observations and 6,680,124 pairs of potential entrants, of which 19.6 percent have had contact.[9] These sets include both participants and non-participants, allowing the model to explain the decision to bid as well as the choice of partner.

Participation and joint bidding differ across bureaus and work categories, so we allow their participation costs and JV formation costs to vary across six groups. The four bureau–work categories with the most observed JVs—Kanto steel bridges and civil engineering in Kanto, Kinki and Kyushu—each have their own cost effects. The remaining categories have at most 16 observed JVs each; we pool their cost effects into an other-bridge group and an other-civil-engineering group to limit the number of parameters estimated from sparse JV observations.[10] JVs bid in 69 percent of Kanto bridge auctions, mostly between repeat

[8] A separate check of member identification covers WTO-covered civil engineering and bridge auctions in 2012–2022: all named members are linked for 499 of 533 project-specific JV bids (93.6 percent). This check uses a different sample from the formation sample; see Appendix A.2.

[9] For 1,449 of the 50,136 firm–auction observations, no earlier valid solo bid is available to measure capacity. We impute K_i by the median among the other potential entrants of the auction whose capacity is observed.

[10] The other-bridge group comprises steel-bridge auctions in Kyushu, Kinki, Shikoku and Chugoku and the

Table 2. Estimation sample

	Auctions	Median estimate (bn yen)	Potential entrants (median)	Solo bidders	JVs	Auctions with a JV (%)	JVs: new / repeat
Panel A. By bureau–work-category group							
Kanto, civil engineering	69	3.53	117	14.1	0.96	43	48 / 18
Kanto, bridges	58	2.24	49	13.4	2.31	69	34 / 100
Kinki, civil engineering	99	3.06	130	18.3	0.24	12	18 / 6
Kyushu, civil engineering	50	2.89	63	17.8	0.74	40	22 / 15
Other bridges	104	0.66	38	14.7	0.29	18	16 / 14
Other civil engineering	16	4.77	41	11.6	1.31	38	7 / 14
Panel B. By size of the engineer's estimate							
Q1 (0.38–1.53)	99	0.62	39	14.8	0.21	14	9 / 12
Q2 (1.58–2.44)	99	2.00	313	16.6	0.68	29	31 / 36
Q3 (2.45–3.63)	99	2.92	129	16.9	0.89	38	30 / 58
Q4 (3.63–30.56)	99	4.95	59	13.9	1.37	46	75 / 61
≥5 bn yen	46	5.75	56	9.7	1.72	63	54 / 25
≥10 bn yen	6	20.08	46	0.7	3.00	100	16 / 2
All	396	2.45	58	15.6	0.79	32	145 / 167

Notes: Formation sample of 396 competitive auctions, 2006–2022. "Solo bidders" and "JVs" are means per auction of listed participants. A JV is *repeat* if the two members bid together in the same bureau and work category before the auction, and *new* otherwise. "Other bridges" are steel-bridge auctions in Kyushu, Kinki, Shikoku and Chugoku and one prestressed-concrete auction in Kyushu; "other civil engineering" covers Hokuriku and Shikoku civil engineering and two Kanto air-conditioning auctions. Size quartiles in billion yen.

partners, but in only 12 percent of Kinki civil engineering auctions, which attract many solo bidders.

Collusion is a known concern in Japanese procurement. Appendix A.5 applies the screen of Kawai and Nakabayashi (2022) and a battery of standard screens to the bureaus' auctions; neither shows the patterns associated with collusion in the markets of our sample, except for a weak marker in Shikoku bridge construction, which contributes 10 of the 396 auctions. We treat bidding as competitive.

prestressed-concrete auction in Kyushu. The other-civil-engineering group comprises civil engineering in Hokuriku and Shikoku and the two Kanto air-conditioning auctions. The groups share cost coefficients; potential entrants are defined separately for each auction within its own bureau and work category.

4 Stylized Facts and a Case Study

This section documents six facts that motivate the model in Section 5: JVs become more common above project-size thresholds (Fact 1), pool the capacities of their members (Fact 2), win more often than solo bidders (Fact 3), are formed repeatedly by the same pairs (Fact 4), and cluster within auctions (Fact 5); most winning bids sit at the price floor (Fact 6). A separate case study of the Kanto bridge threshold then examines whether allowing JVs attracts additional firms or changes how existing participants bid.

4.1 Stylized facts

Fact 1: Joint ventures become more common above size thresholds. Figure 1 pools projects across bureaus and facility types in the formation sample. To compare projects facing different size guidelines, we express each engineer's estimate relative to the reference threshold for its facility in Table 1. The share of auctions with a JV bidder rises around these thresholds. JVs also appear below them: the guidelines allow exceptions for technically demanding projects, and practices differ across bureaus (Appendix A.4). The figure describes this pooled pattern; the case study in Section 4.2 examines the sharper Kanto bridge threshold. In the model, joint bidding is unavailable below the WTO threshold and in Kanto bridge auctions below 2 billion yen without an observed JV bidder.

Fact 2: Joint ventures pool capacity and may create synergies. Firms that form JVs bid on larger projects relative to their past solo bidding experience than firms that bid alone. The members' combined capacity—the sum of the largest projects on which they previously bid alone—covers the current project in 91 percent of JVs. This pattern is consistent with joint bidding helping firms participate in larger projects by pooling capacity.

JVs also form when at least one member already has solo bidding experience at the current project's scale. In 70 percent of JVs, at least one member has previously bid alone on a project at least as large as the current one. Their choice to bid jointly suggests that JVs may offer synergies between the members' capabilities, in addition to the benefit of expanding capacity.

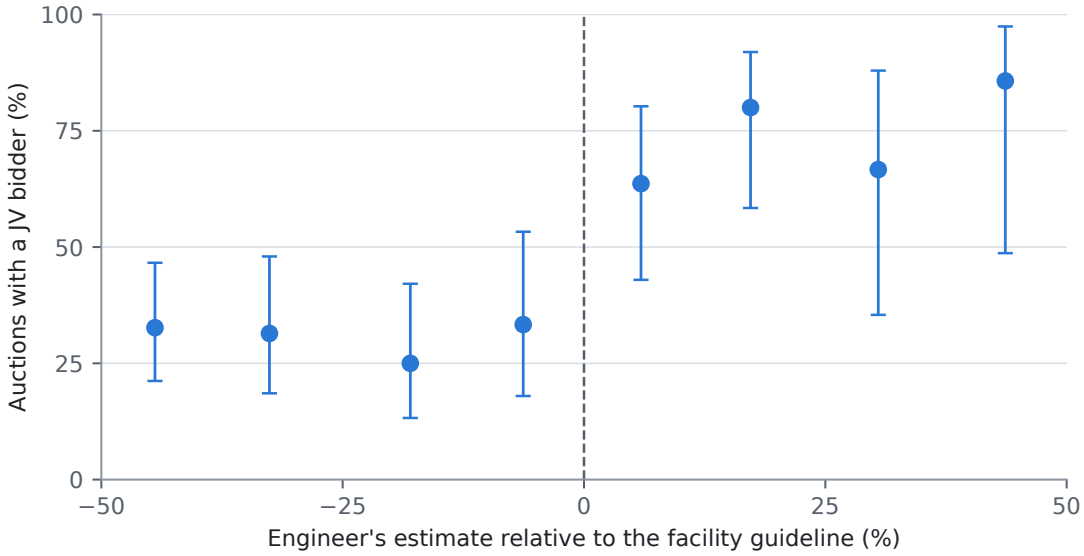


Figure 1. JV participation around the facility-specific size guidelines. Formation sample, 2006–2022. Of the 396 auctions, 322 can be assigned a reference threshold from their work category and project name; the figure shows the 198 within 50 percent of that threshold. The horizontal axis is $100(R_t/c_t - 1)$, where c_t is the reference threshold; zero marks the guideline. Points are the shares of auctions with a JV in equal-width bins, placed at the mean horizontal-axis value; bars are 95 percent Wilson intervals. Appendix A.4 describes the classification.

Fact 3: Joint ventures win more often than solo bidders. We compare the winning probabilities of JVs and solo bidders competing in the same auction. A regression of the contract-award indicator on a JV indicator and auction fixed effects gives a difference of 4.1 percentage points.[11] This comparison describes the performance of the JVs that form, combining the selection of their members with any gains from joint bidding. It motivates allowing JVs and solo bidders to differ in their bidding behavior.

Fact 4: Joint ventures are formed by repeat partners. Of the 312 JVs, 167 are between firms that bid together before in the same bureau and work category, and the 312 JVs involve only 144 distinct pairs. JV partners also tend to have prior contact: 296 of the

[11] The regression uses all 6,471 bidder listings in the 396 formation-sample auctions, counting each JV as one bidder. The standard error is 2.1 percentage points, clustered by auction. Restricting the comparison to the 4,697 positive-price bids that were not withdrawn or invalid gives a difference of 4.9 percentage points, with a standard error of 2.8 points. The two-sided p-values are 0.048 and 0.079, respectively.

312 JVs are between firms that previously participated in the same auction, either alone or as JV members. Among all pairs of potential entrants, counted separately for each auction, only 19.6 percent have such prior contact. In the model, past joint bids lower the cost of forming a JV, and pairs without contact are less likely to be able to form one.

Fact 5: Joint ventures cluster within auctions. The 312 JVs bid in only 127 auctions, an average of 2.46 per auction with a JV. Solo participation can also differ substantially between projects in the same bureau, work category, and size quartile. In the 34 Kinki civil engineering auctions in the second size quartile, for example, the number of solo bidders ranges from 1 to 38. These patterns motivate allowing unobserved project characteristics to affect the participation payoffs of multiple firms in the same auction. The model therefore includes one auction-level shock that shifts the value of bidding jointly for all potential pairs, and another that shifts the value of bidding alone for all potential entrants.

Fact 6: The price floor binds. Seventy-five percent of the winning bids are at or below 101 percent of the low-bid investigation threshold (Section 2.3). Firms with low costs cannot translate them into lower bids beyond the floor, which compresses the gains from winning, and the cost of a bidder at the floor cannot be read off its bid. The bidding stage of the model includes the floor, and its estimation treats the costs of bidders at the floor as censored.

4.2 Case study: JV eligibility in Kanto bridge auctions

The Kanto bridge threshold provides a case study of whether allowing JVs attracts additional firms or changes how existing participants bid. JVs are admitted above an engineer's estimate of 2 billion yen and rarely appear below it. Comparing auctions around this threshold therefore sheds light on what JV members might do without the option of joint bidding.

Figure 2(a) shows a rise in JV bidding alongside a fall in solo bidding. Panel (b) shows the same pattern among firms that join JVs above the threshold, consistent with these firms switching from solo to joint participation. Panels A and B of Table 3 report the corresponding regression estimates. They show no clear increase in the total number of participating firms; the estimated change in independent bidders, counting each JV as one

Table 3. Kanto bridges at the 2 billion yen eligibility threshold

	Below	Above	Jump at threshold	(s.e.)
Panel A. Per auction (112 auctions, 1.25–3.5 bn yen)				
Solo bidders	14.59	10.69	−3.73	(1.62)
JVs	0.23	2.80	+1.40	(0.60)
Independent bidders (solo + JVs)	14.82	13.49	−2.33	(1.40)
Participating firms (solo + JV members)	15.05	16.29	−0.93	(1.43)
Panel B. Firm participation (firm-auction pairs)				
Firms that join JVs above the threshold (47 firms)				
Bids alone (%)	31.5	23.5	−11.1	(4.2)
Bids in a JV (%)	1.2	14.6	+7.8	(3.0)
Participates (%)	32.8	38.2	−3.3	(4.0)
Other firms (106 firms)				
Bids alone (%)	5.1	3.6	+0.3	(1.9)
Bids in a JV (%)	0.0	0.0	−0.1	(0.1)
Participates (%)	5.2	3.6	+0.1	(1.9)

	Auctions	$\hat{d}$	95% confidence set for d
Panel C. Displacement ratio $d = -\Delta solo/\Delta JV$ of JV-prone firms, by window			
1.25–3.5 bn yen (main)	112	1.42	$[0.48, \infty)$
1.0–4.0 bn yen	158	1.28	[0.44, 2.71]
1.58–5.13 bn yen	96	0.64	[−0.18, 2.02]
1.25–5.13 bn yen	128	0.76	[−0.31, 3.50]
Formation sample, 1.58–5.13 bn yen	57	0.61	[−1.00, 1.59]

Notes: Kanto bridge auctions in the full bid panel, 2006–2022. "Below" and "above" are raw means on each side of 2 billion yen. Jumps are estimates of the above-threshold coefficients in equations (16) and (17). Both regressions include year fixed effects and a linear term in the log engineer's estimate, with the same slope below and above the threshold. The firm-level regression also includes firm fixed effects, allowing each firm its own baseline participation rate. The potential entrants of an auction are the firms that bid on a Kanto bridge within two years; JV-prone firms join a JV in some Kanto bridge auction above the threshold. Standard errors are clustered by auction. The confidence set for d inverts the test of $\Delta\text{solo} + d\,\Delta\text{JV} = 0$ (Fieller). Source: Section 4.2 and Appendix B.

bidder, is negative but imprecise. The comparison is local and rests on few auctions near the threshold; Appendix B.1 specifies the regressions and Appendix B.2 reports robustness checks.

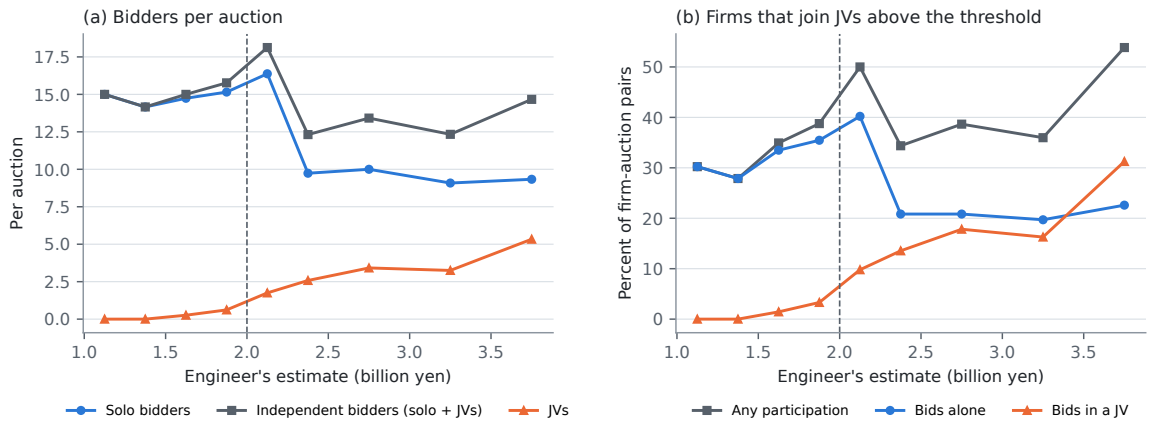


Figure 2. Kanto bridge auctions around the 2 billion yen threshold for JVs, full bid panel, 2006–2022, with engineer's estimates from 1.0 to 4.0 billion yen. Panel (a): solo bidders, JVs and independent bidders per auction. Panel (b): participation of the 47 firms that join a JV in some Kanto bridge auction above the threshold, in the auctions in which they are potential entrants (firms that bid on a Kanto bridge within two years). Bin means; the dashed line marks 2 billion yen.

5 Model

We model each auction as a separate game. An auction t has an engineer's estimate R_t, a price floor $\tau_t R_t$, a market $g(t)$, and a set of potential entrants $\mathcal{N}_t$. We treat the engineer's estimate as known to firms.[12] We write $g = g(t)$ for the market in which the auction takes place.

The potential entrants play a two-stage game. First, in the *formation stage*, each chooses to stay out, bid alone, or bid with one available partner. Second, in the *bidding stage*, each bidder learns its score and cost and submits a price; the highest score-to-price ratio wins. We describe bidding first because it determines the expected profits used in the formation decisions. Figure 3 shows the sequence of nature's draws and firms' decisions, together with the information available at each stage.

[12]The reserve is not formally disclosed before bidding (Section 2.1). MLIT's July 23, 2004 report on contracts awarded at the reserve price, section 3, reports that bidders use published cost-estimation standards and commercially available software capable of producing estimates close to the reserve. We approximate this ability to reconstruct the engineer's estimate by treating R_t as known, abstracting from residual prediction error.

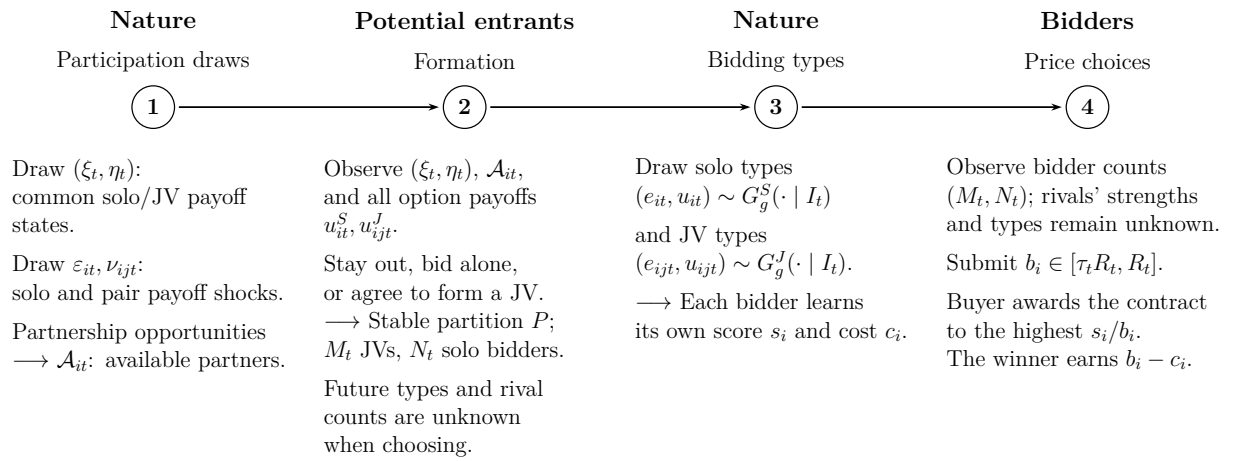


Figure 3. Game flow in the two-stage model. Events proceed from left to right.

5.1 Bidding stage

Bidder i submits a technical proposal with score s_i and a price b_i, and wins if its effective bid s_i/b_i is the highest; its profit is then $b_i - c_i$, where c_i is its cost of carrying out the project as proposed. We express effective bids using prices relative to the engineer's estimate. We denote the resulting log effective bid of bidder i by y_i:

$$y_i \equiv \log[s_i/(b_i/R_t)] = \log(s_i/b_i) + \log R_t.$$

A *valid price* lies between the floor and the reserve: $\tau_t R_t \leq b_i \leq R_t$, or equivalently, $\log s_i \leq y_i \leq \log(s_i/\tau_t)$.

Types and bidding strength. Following the reduction of scoring auctions to a one-dimensional problem (Che, 1993; Asker and Cantillon, 2008; Hanazono et al., 2020), we treat a bidder's score and cost as drawn after formation and observed by the bidder before it chooses its price.

Each auction has a common score level μ_t and a common cost condition δ_t, which firms know before formation. We collect the common auction conditions in $I_t = (X_t, \mu_t, \delta_t)$, where

X_t contains the observed auction characteristics. The econometrician observes X_t but not μ_t or δ_t.

Consider a solo bidder. We model the bidder's score and cost using two components. The first captures persistent differences across firms in the cost of carrying out a proposal of a given score and in the scores of their proposals. We represent these differences by firm-specific cost strength α_i and score strength σ_i, which together form the firm's *bidding strength* $a_i = (\alpha_i, \sigma_i)$. The firm knows its own strength before formation. The second captures variation specific to the bidder and auction: firm i draws a score residual e_{it} and a cost residual u_{it}. We call (e_{it}, u_{it}) the bidder's *type*. Types are drawn independently and identically across solo bidders from a joint distribution $G_g^S(\cdot \mid I_t)$, independently of their strengths; S denotes solo bidding.

Let $\tilde{c}_{it} \equiv c_{it}/R_t$ denote the *normalized cost*, measured relative to the engineer's estimate. The function q_g captures the contribution of the proposal's log score to log normalized cost.

Assumption 1 (Scores and costs)**.** *For solo bidders in market g,*

$$\log s_{it} = \mu_t + \sigma_i + e_{it},$$
$$\log \tilde{c}_{it} = \delta_t + q_g(\log s_{it}) - \alpha_i + u_{it},$$

and

(a) the residual u_{it} is independent of the score s_{it};

(b) the joint distribution of (e_{it}, u_{it}) does not depend on the auction beyond its market.

The firm's cost of carrying out its proposal is

$$c_{it} = R_t \tilde{c}_{it}.$$

Thus a higher α_i lowers the cost of a given proposal, while a higher σ_i raises its score and changes its cost through q_g. The type records the remaining variation in scores and costs after the firm strengths (α_i, σ_i) and auction score and cost levels (μ_t, δ_t) are taken into account. We leave its distribution unspecified at this stage.

How can we model the bidding strength of a JV? A JV of firms i and j is assumed to use, in each respect, the strength of its stronger member; that is,

$$a_{ij} = (\max\{\alpha_i, \alpha_j\}, \max\{\sigma_i, \sigma_j\}).$$

This is analogous to the optimized-value case of Asker, Baccara and Lee (2021) with one patent application, in which a coalition's value equals its highest member valuation. Here the maximum applies to each component of the members' bidding strength: the JV can draw on the better technical team and on the cheaper production of its members.

A JV draws its type (e_{ijt}, u_{ijt}) from $G_g^J(\cdot \mid I_t)$. These draws are independent across JVs and independent of their strengths; J denotes joint bidding. Its score and normalized cost are

$$\log s_{ijt} = \mu_t + \max\{\sigma_i, \sigma_j\} + e_{ijt},$$
$$\log \tilde{c}_{ijt} = \delta_t + q_g^J(\log s_{ijt}) - \max\{\alpha_i, \alpha_j\} + u_{ijt}.$$

The JV's score-cost relation q_g^J and type distribution can differ from those of solo bidders. Given the outcome of formation, the type distributions G_g^S and G_g^J, together with these score-cost relations, determine the bidders' scores and costs.

Best responses and rival bids. At the bidding stage, a bidder knows its own type and strength, I_t, and the numbers of rival JVs and solo bidders, but not their strengths or types. It treats the strengths of its rivals of each mode as independent draws from the distribution of strength among bidders of that mode in auctions with information I_t and the same numbers of bidders. It treats their types as draws from the corresponding type distributions, G_g^S and G_g^J. Observing rival numbers at bidding is a simplifying assumption: in practice, firms submit prices without observing the realized number of bidders.[13]

Under this competitive environment, what can the distribution of rival bids depend on? A rival's log effective bid is a function of its type, its strength and the information common to all bidders. The bidder knows neither the rival's type nor its strength and regards both as independent draws, so it regards the rival's bid as an independent draw as well. We restrict

[13] Auctions in which the number of bidders remains unknown at bidding are studied by McAfee and McMillan (1987) and Harstad, Kagel and Levin (1990). In such a formulation, a bidder would choose its bid to maximize expected profit averaged over its beliefs about rival numbers.

attention to equilibria in which bidders of the same mode follow the same bidding rule, which maps a bidder's type, its strength and the common information into its log effective bid. The log effective bids of a bidder's rivals of mode d are then independent draws from a distribution $F^d(\cdot \mid I_t, M_t, N_t)$, where M_t and N_t are the numbers of JVs and solo bidders in the auction. We write F_t^S and F_t^J for the two distributions in auction t.

A solo bidder that faces m JVs and n other solo bidders wins with log effective bid y with probability $F_t^J(y)^m F_t^S(y)^n$. Among valid prices, it chooses the one whose log effective bid solves

$$\max_{\log s \le y \le \log(s/\tau_t)} R_t \left(s\, e^{-y} - \tilde{c} \right) F_t^J(y)^m F_t^S(y)^n, \tag{1}$$

where $\tilde{c} = c/R_t$ is the project cost normalized by the engineer's estimate. An interior solution satisfies

$$\frac{1}{1 - \tilde{c}\, e^y / s} \;=\; n\, \frac{f_t^S(y)}{F_t^S(y)} + m\, \frac{f_t^J(y)}{F_t^J(y)}, \tag{2}$$

which depends on the type and strength only through the *effective cost* $\tilde{c}/s$.

When the floor binds, the bidder bids the floor, and a range of costs leads to the same bid. A lone bidder bids the reserve. A JV solves the same problem against $m-1$ JVs and n solo bidders. In a Bayesian Nash equilibrium, F_t^S and F_t^J are the distributions of the log effective bids that these best responses generate.[14]

Synergies. We use *synergies* to describe how a JV's effective cost differs from that of a specified solo benchmark in the same auction. A reduction in effective cost represents positive synergy, and an increase represents negative synergy. The comparison includes the members' persistent cost and score strengths as well as the JV's score-cost relation and type distribution. It therefore asks both whether joint bidding improves cost per unit of score and where any improvement comes from. Section 7.1 decomposes mean log effective-cost differences into contributions from α, σ, and the conditional distribution of scores and costs generated by q_g^d and G_g^d. Because types are drawn after formation, favorable persistent strengths can coexist with an unfavorable realization of joint performance.

[14] Lebrun (1999) establishes existence under regularity conditions and gives sufficient conditions for uniqueness in asymmetric independent-private-value first-price auctions. Our scoring formulation has score-dependent bid bounds, so these results do not directly establish existence or uniqueness here. We interpret the estimated bid distributions as describing the equilibrium observed in the data.

Expected profits. Let $\pi^S_{it}(m,n)$ be firm i's expected profit from bidding alone against m JVs and n other solo bidders—the maximized value of (1) averaged over types, with the distributions of rival bids evaluated at these numbers of bidders, $F^d(\cdot \mid I_t, m, n+1)$—and $\pi^J_{ijt}(m,n)$ that of the JV of i and j against m other JVs and n solo bidders, with $F^d(\cdot \mid I_t, m+1, n)$. Before formation is resolved, firms know neither their future bidding types nor the numbers of rivals they will face. Let ξ_t and η_t denote the common states affecting solo and JV participation payoffs, respectively. Firms observe both states before formation, but the econometrician observes neither. Given beliefs $H^S_t(m,n \mid \xi,\eta)$ and $H^J_t(m,n \mid \xi,\eta)$—the probabilities assigned to facing m rival JVs and n rival solo bidders, conditional on the common states and the bidder's mode—the expected profits are

$$\begin{aligned} V^S_{it}(\xi,\eta;H) &= \sum_{m,n} H^S_t(m,n \mid \xi,\eta)\, \pi^S_{it}(m,n), \\ V^J_{ijt}(\xi,\eta;H) &= \sum_{m,n} H^J_t(m,n \mid \xi,\eta)\, \pi^J_{ijt}(m,n). \end{aligned} \tag{3}$$

5.2 Formation game

5.2.1 Formation game description

Each potential entrant $i \in \mathcal{N}_t$ can stay out, bid alone, or bid jointly with one available partner. Let $\mathcal{A}_{it}$ be the set of firms with which i can form a JV in auction t. The auction's eligibility rules and the realized partnership opportunities determine this set; if the auction does not admit JVs, it is empty for every firm. Availability is symmetric: $j \in \mathcal{A}_{it}$ if and only if $i \in \mathcal{A}_{jt}$. Potential entrants observe these partnership opportunities before making their choices. Forming a JV requires both members to agree.

The outcome of these choices is a *partition*. A partition P is a complete assignment of the potential entrants to these options. Each firm has exactly one assignment, and a JV must be mutual and available: if i is assigned to bid with j, then j is assigned to bid with i and $j \in \mathcal{A}_{it}$.

Given a partition, each firm's payoff is the payoff of its assigned option:

$$U_i(P) = \begin{cases} 0, & \text{if } i \text{ stays out under } P, \\ u_{it}^S, & \text{if } i \text{ bids alone under } P, \\ u_{ijt}^J, & \text{if } i \text{ bids with } j \text{ under } P. \end{cases}$$

Here u_{it}^S is firm i's net payoff from bidding alone, and u_{ijt}^J is the net payoff that each member receives from the JV of i and j; staying out yields zero. These payoffs combine expected bidding profits with participation costs and other benefits or costs of each option. Potential entrants observe one another's realized option payoffs before choosing whether and with whom to participate. Their bidding types have not yet been drawn, so the bidding-profit component is an expectation over types and rival numbers. We assume that the two members receive the same net payoff from their JV, including both their shares of bidding profits and all other payoff components.[15] The particular functions used to construct these payoffs are introduced in Section 5.2.3. Thus P determines which of the option payoffs a firm receives. When referring to a fixed partition, we also write U_i for $U_i(P)$.

5.2.2 Model solution

The model's solution combines two requirements. Let H collect the beliefs about rival numbers, H_t^S and H_t^J, introduced in (3). For given beliefs H and realized payoffs, the assignment must be stable against individual and pair deviations. Across realizations, the assignments must generate the distributions of rival numbers that firms used to evaluate their options. We first consider stability and then impose consistency of beliefs to obtain an equilibrium of the model.

Stability. For given beliefs, we consider a *stable partition*. Potential entrants evaluate deviations using these realized option payoffs and holding H fixed. A firm can leave its

[15] The MLIT circular on JVs in direct construction, Part 1, sections 2(4)–(5), requires each member of a two-firm JV to hold at least 30 percent and the representative to hold the largest share. MLIT's standard Type A JV agreement, Articles 8, 13 and 14, allocates profits and losses according to the agreed contribution shares. Equal sharing here is a symmetric benchmark that, together with the common payoff terms, yields the common ranking used in Section 5.2.2.

assigned option to bid alone or stay out without another firm's agreement. Forming a new JV, however, requires both prospective members to agree; either can leave an existing partner to do so. A stable partition requires that no firm can gain on its own and no available pair can make both members better off by joining. This is the core stability criterion in coalition formation, restricted to single firms and available pairs (Ray and Vohra, 2015, Section 5.3.2), with each firm's outside option given by the better of bidding alone and staying out.

Definition 1 (Stable partition and blocking pair)**.** *A partition P is stable if*

(i) *no potential entrant prefers an option it can take alone: $U_i(P) \geq \max\{0, u_{it}^S\}$ for every i; and*

(ii) *no two potential entrants prefer to bid together: there is no available pair (i, j), not bidding together under P, with $u_{ijt}^J > U_i(P)$ and $u_{ijt}^J > U_j(P)$.*

A pair satisfying these two strict inequalities is a blocking pair*.*

Example 1 (Four-firm example)**.** Return to firms A, B, C, D, and suppose that all six pairs are available. Suppress the auction index and consider the illustrative solo payoffs

$$(u_A^S, u_B^S, u_C^S, u_D^S) = (2, 1, 3, -1).$$

Staying out yields zero for every firm. The JV payoffs, received by *each* member of the pair, are

$$u_{AB}^J = 5, \quad u_{AC}^J = 4, \quad u_{BC}^J = 2.5,$$
$$u_{AD}^J = -2, \quad u_{BD}^J = -3, \quad u_{CD}^J = -4.$$

For instance, when A and B bid jointly, each receives five. Consider the following three partitions, with payoffs listed in the order A, B, C, D:

Partition	JV	Solo bidders	Non-participants	Payoffs
P^*	AB	C	D	$(5, 5, 3, 0)$
P^1	AB	None	C, D	$(5, 5, 0, 0)$
P^2	AC	B	D	$(4, 1, 4, 0)$

The first partition, P^*, is stable. Neither A nor B would leave their JV to bid alone or stay out; C prefers bidding alone to staying out; and D prefers staying out to bidding alone. There is also no blocking pair. Although C would gain by joining A, receiving four instead of three, A would lose, receiving four instead of five. The agreement of both firms is required, so AC does not block. Pair BC would give each member only 2.5, below both firms' current payoffs, and every pair involving D has a negative payoff.

The other two partitions illustrate the two distinct ways stability can fail. In P^1, firm C can raise its payoff from zero to three simply by bidding alone, violating condition (i). In P^2, every firm receives at least as much as its best individual option, so condition (i) holds. But A and B can form a JV and each receive five: A gains relative to four and B relative to one. Thus AB blocks P^2, violating condition (ii). Firm C would lose its partner in this deviation, but its consent is not required for A to leave.

Moreover, P^* is the *only* stable partition. Both A and B obtain their highest possible payoff by joining each other. In any partition without AB, firm A receives at most four and firm B at most 2.5, so AB would block. Hence every stable partition must contain this JV. Neither remaining firm can form a blocking pair with A or B, and CD gives both members a negative payoff. Condition (i) then requires C to bid alone and D to stay out. The example therefore has a stable partition, and that partition is unique.

Existence and uniqueness in this example are not a coincidence. The key is that both members receive the same payoff from any given JV. When pair values are distinct, the highest-valued pair that both members prefer to their individual options is therefore unavoidable in a stable partition, just as AB was unavoidable in the example. Removing its members leaves a smaller problem with the same property, so the argument can be repeated. Under this common-payoff assumption, preferences are *aligned* (Pycia, 2012), and pairs are *globally ranked* (Abraham et al., 2008). The following proposition extends the example to any number of potential entrants and any set of available pairs, holding beliefs about rival numbers fixed.

Proposition 1. *Suppose that the payoffs u_{it}^S and u_{ijt}^J are continuously distributed. Then, with probability one, there is exactly one stable partition. It is obtained by the following greedy procedure. Let $\bar{u}_i = \max\{0, u_{it}^S\}$. Among the available pairs with $u_{ijt}^J > \max\{\bar{u}_i, \bar{u}_j\}$,*

form the pair with the highest value u^J_{ijt} and remove both members; repeat until no such pair remains. Every remaining potential entrant bids alone if $u^S_{it} > 0$ and stays out otherwise.

The proof is in Appendix C.1. It formalizes the repeated selection of an unavoidable pair and verifies that the resulting partition has no profitable individual deviation or blocking pair. For given beliefs about rival numbers, the proposition gives a unique formation outcome with probability one, so the probability of the observed partition is well defined without a rule for selecting among stable partitions. The greedy procedure also provides a fast way to simulate the formation stage in auctions with hundreds of potential entrants.

Belief consistency. A stable partition alone does not close the model, because the expected profits used to determine it depend on H. Different beliefs can produce different participation outcomes, even when each set of beliefs yields a unique stable partition. Stability places no requirement on whether firms' beliefs correctly predict rival numbers. The definition of *equilibrium* below supplies this additional requirement: the beliefs must agree with the distribution of rival numbers generated by the stable partitions themselves.

For given payoff functions, distributions of shocks and partnership availability, and policy governing which auctions admit JVs, let $\Gamma(H)$ be the distributions of rival numbers generated when firms use the expected profits $V(H)$. Hold both common states (ξ, η) fixed, solve for the stable partition at each realization of the idiosyncratic shocks and availability, and record the resulting bidder counts. Specifically, $\Gamma^S_t(m, n \mid \xi, \eta)$ is the probability that the auction has m JVs and $n+1$ solo bidders, conditional on (ξ, η) and on having at least one solo bidder; $\Gamma^J_t(m, n \mid \xi, \eta)$ is the probability of $m+1$ JVs and n solo bidders, conditional on (ξ, η) and on having at least one JV. These conditional distributions are required only when the corresponding mode occurs with positive probability. The beliefs about rival numbers condition on the common states and participation mode: realized option payoffs and availability determine the assignment, but do not further refine the distributions used to evaluate bidding profits.

Definition 2 (Equilibrium)**.** *An equilibrium consists of beliefs H and their induced stable partitions such that $H = \Gamma(H)$ for almost every pair of common states (ξ, η): every potential entrant evaluates its options with the expected profits $V(\xi, \eta; H)$, the outcome of each auction is the stable partition given those payoffs, and the beliefs are the state-conditional distributions of rival numbers that the stable partitions generate.*

Thus a stable partition describes the outcome for a particular realization given beliefs, while equilibrium determines beliefs that are consistent with these outcomes across realizations. This consistency accounts for the feedback from beliefs about rival numbers to participation: a state that attracts more solo bidders also leads firms to expect more rivals and lower bidding profits, which dampens entry. Proposition 1 establishes uniqueness only of the conditional stable partition, not of the beliefs satisfying $H = \Gamma(H)$. We do not prove that the equilibrium is unique; we compute beliefs by damped iteration and compare outcomes from different starting beliefs (Appendix G.2).[16]

5.2.3 Empirical specification

For estimation, we specify partnership availability using the auction's JV eligibility and firms' prior contact. If the auction does not admit JVs, no pair can bid jointly. Otherwise, pairs whose members have previously participated in the same auction of the bureau and work category are always available. Each pair without such contact is available independently with probability $e^{-\tau}$, capturing the difficulty of finding a partner outside one's network. These draws determine the sets $\mathcal{A}_{it}$ introduced above.

The common states (ξ_t, η_t) directly shift the value of bidding alone and the value of every JV, respectively. Both states also affect expected bidding profits through the beliefs in (3). Firms observe both states before formation, but the econometrician observes neither. We assume that the states are independent standard normal variables.

We measure payoffs in units of $\sigma > 0$, the standard deviation of the solo payoff shock in yen. To express the two components of bidding strength on comparable scales, let σ_α and σ_σ be their cross-firm standard deviations, fixed at the first-stage estimates. Let ε_{it} denote a firm-specific shock to the payoff from bidding alone, capturing participation costs or benefits

[16] Appendix C.2 compares our model with Uetake and Watanabe (2020).

unobserved by the econometrician. We specify the net payoff from bidding alone, u_{it}^S, as

$$\begin{aligned} u_{it}^S &= \frac{V_{it}^S(\xi_t, \eta_t; H)}{\sigma} - f_t - \lambda_S g_{it} - \beta_S \ell_t + \kappa_{S,\alpha}\frac{\alpha_i}{\sigma_\alpha} + \kappa_{S,\sigma}\frac{\sigma_i}{\sigma_\sigma} + c_S\xi_t + \varepsilon_{it}, \\ \varepsilon_{it} &\sim N(0,1). \end{aligned} \tag{4}$$

The participation cost $f_t = \bar{f}\exp(\psi_{g(t)} + \psi_{q(t)})$ varies by market and size quartile $q(t)$; it captures the cost of preparing the proposal and the estimate and of committing engineers, net of any non-monetary benefits of bidding.

The capacity shortfall g_{it} captures the cost of taking on a project larger than the firm has done before, with coefficient λ_S. Project size can also affect participation even when a firm has sufficient capacity, through the risk, staffing and guarantees associated with a large project. The separate term $-\beta_S\ell_t$, where $\ell_t = \max\{0, \log(R_t/\bar{R})\}$, captures this effect for projects above a reference size $\bar{R}$; a positive β_S lowers the payoff from bidding alone as project size increases.

Cost and score strength enter separately, in addition to their effects through expected profits. The coefficients $\kappa_{S,\alpha}$ and $\kappa_{S,\sigma}$ measure the direct payoff effects of a one-standard-deviation increase in the cost and score strengths, respectively, holding expected profits fixed. These effects capture reasons for participating that are not proportional to bidding profits, such as the cost of preparing a proposal or the value of a performance record for future scores. We estimate the two coefficients freely, so their relative magnitude and signs are not fixed by the bidding-stage relation between costs and scores. The term $c_S\xi_t$ is the contribution of the auction-level solo state.

Let ν_{ijt} denote the pair-specific payoff shock shared by the two members of a JV. We specify the net payoff that each member receives, u_{ijt}^J, as

$$\begin{aligned} u_{ijt}^J = \frac{1}{2}\left[\frac{V_{ijt}^J(\xi_t, \eta_t; H)}{\sigma} - K_t e^{-\gamma h_{ijt}} - \lambda_J g_{ijt}\right] - f_t - \beta_J \ell_t \\ + \kappa_{J,\alpha}\frac{\max\{\alpha_i, \alpha_j\}}{\sigma_\alpha} + \kappa_{J,\sigma}\frac{\max\{\sigma_i, \sigma_j\}}{\sigma_\sigma} + c_J\eta_t + q\nu_{ijt}. \end{aligned} \tag{5}$$

The terms in brackets are the components of the JV's value that the members split equally: its expected profit, the additional cost of forming and running a JV, and the cost of the

pair's combined capacity shortfall. The additional cost $K_t = K_{q(t)} \exp(\delta_{g(t)})$ varies by size quartile and market; it captures the costs of agreeing on the division of work, of coordinating two organizations and of registering the JV, and any other reason why firms prefer to bid alone. It declines with the pair's history of joint bids h_{ijt} at rate γ. Each member pays its own participation cost f_t and has the size term $-\beta_J \ell_t$.

The JV's direct strength effects use the stronger member in each dimension, as do its bidding profits. The coefficients $\kappa_{J,\alpha}$ and $\kappa_{J,\sigma}$ enter each member's payoff and are estimated separately from the solo coefficients. The size coefficients β_S and β_J are also estimated separately, with no sign restriction or link to the capacity coefficients. The cost levels $\bar{f}$ and $K_{q(t)}$, the capacity coefficients λ_S and λ_J, and the repeat-partner discount γ are nonnegative. We take the common-state scales c_S and c_J to be nonnegative, while the relative pair-shock scale q is positive.

The pair-specific term $q\,\nu_{ijt}$ captures unobserved compatibility between the two firms on the project, beyond their observed characteristics and history of joint bids. Both members receive the same realization, which they observe before deciding whether to form the JV. We construct this term from an independent standard normal draw for each pair:

$$\nu_{ijt} = r_t \zeta_{ijt} - l_t, \qquad \zeta_{ijt} \sim N(0,1), \qquad r_t > 0.$$

The auction-specific constants r_t and l_t adjust the scale and location of these draws for the number of potential partners. They are fixed before estimation; the parameter q governs the remaining scale of the pair shocks relative to the solo shock. In the main specification, we fix $q = 1$, equating the two shock scales at the reference number of partners. This is a restriction on their relative dispersion in addition to measuring payoffs in units of σ; the pair shock's standard deviation in auction t remains r_t.

All shocks—ε_{it}, ζ_{ijt}, the availability draws, and the states—are independent of each other and of the observed characteristics.

Remark 1 (The number of potential partners and the pair shocks)**.** With independent pair shocks, a firm with more potential partners has more chances to find an exceptionally favorable match, even holding observed payoffs fixed. This issue motivates adjustments to outside options and idiosyncratic payoff scales as matching markets grow (Menzel, 2015).

We use a normal extreme-value approximation to choose r_t and l_t so that the maximum pair shock has approximately the same location and spread across auctions, using the average expected number of available partners. JV formation still varies with payoffs, availability, and mutual consent. Appendix C.3 gives the derivation and construction.

6 Estimation

This section describes how we estimate the model. Estimation has two steps. The bidding stage is estimated first from bids alone (Section 6.1): bidding strength, the distributions of rival bids and the distributions of types, which together give the profit π of every option against any number of rivals. The formation stage uses the likelihood of the observed partitions together with consistency of beliefs about rival numbers (Section 6.2).

6.1 Bidding stage

The bidding stage supplies two inputs to the formation stage: each firm's bidding strength and the expected profit of each participation option at every number of rivals. We use the scoring auctions in the bid panel and proceed in five steps: estimate the distributions of rival bids, recover costs from optimal bidding, estimate persistent firm strength (α_i, σ_i), recover auction score and cost levels (μ_t, δ_t) and types (e, u), and calculate expected profits.

For clarity, we explain this procedure for interior bids, whose prices lie strictly between the floor and the reserve. In the actual estimation, we also use bids constrained by these boundaries. Appendix D.2 derives the cost bounds for these observations; Appendix D gives the full implementation.

Step 1: distributions of rival bids. We first estimate the rival bid distributions that determine a bidder's winning probability. Section 5.1 conditions these distributions on the numbers of JVs and solo bidders and the common information $I_t = (X_t, \mu_t, \delta_t)$.

Let ω_t denote a shift common to solo and JV log effective bids in auction t. It captures unobserved auction heterogeneity in rival bids, as in Krasnokutskaya (2011). In our approximation, this shift summarizes the effects of the score level μ_t and cost condition δ_t

on the location of rival bids. Firms observe (μ_t, δ_t), but the econometrician does not, so we estimate ω_t from bids.

Let $\mu^d(X_t, m, n)$ be the mean log effective bid before adding this shift, and let σ_t^d be its within-auction standard deviation, for $d \in \{S, J\}$. The mean is a known function up to parameters, and the scale varies with the mode, market and year. The function μ^d describes the location of bids; it is distinct from the score level μ_t in the model. Write $\sigma^2_{\omega,t}$ for the variance of the common bid shift, which may vary with the year.

Assumption 2 (Normal rival bids)**.** *For $d \in \{S, J\}$,*

$$F^d(y \mid X_t, \mu_t, \delta_t, m, n) = \Phi\left(\frac{y - \mu^d(X_t, m, n) - \omega_t}{\sigma_t^d}\right).$$

Conditional on the year, $\omega_t \sim N(0, \sigma^2_{\omega,t})$, independently of the remaining characteristics in X_t and of M_t and N_t.

The normal specification approximates an equilibrium bid distribution for estimation; it does not impose normality on the model's primitive type distributions.

We estimate the parameters jointly by maximum likelihood, pooling solo and JV bids and integrating over the common bid shift ω_t. The means depend on the market, year, floor ratio, engineer's estimate and bidder numbers; coefficients and scales differ between solo bidders and JVs. Appendix D.1 gives the specification and sample rules. The fitted distributions, denoted by $\hat{F}_t^S$ and $\hat{F}_t^J$, condition on the estimated bid shift $\hat{\omega}_t$ and can be evaluated at any number of rivals.

Step 2: recovering costs. The estimated distributions of rival bids give the winning probability $W_t(y) = F_t^J(y)^m F_t^S(y)^n$ against m JV and n solo rivals. Its proportional increase from raising the log effective bid is $h_t(y) \equiv W_t'(y)/W_t(y) = m f_t^J(y)/F_t^J(y) + n f_t^S(y)/F_t^S(y)$. The first-order condition (2) then maps an interior bid into a cost relative to the engineer's estimate:

$$\widehat{\tilde{c}}_{it} \equiv \frac{b_{it}}{R_t}\left(1 - \frac{1}{\hat{h}_t(y_{it})}\right), \qquad y_{it} = \log s_{it} - \log(b_{it}/R_t), \tag{6}$$

This is the inverse-bidding step of Guerre, Perrigne and Vuong (2000), applied to the scoring rule. Given the estimated distributions of rival bids, the observed price and score reveal the

cost at which the gain in winning probability just offsets the loss in profit conditional on winning.

Step 3: bidding strength. This step estimates (α_i, σ_i) and the score-cost function q_g. We parameterize q_g as a quadratic function of the log score. Assumption 1 gives the two regression equations

$$\begin{aligned} \log \tilde{c}_{it} &= \delta_t + q_g(\log s_{it}) - \alpha_i + u_{it}, \\ \log s_{it} &= \mu_t + \sigma_i + e_{it}. \end{aligned} \tag{7}$$

If all costs were observed without censoring, least squares with firm and auction effects would estimate both (α_i, σ_i) and (δ_t, μ_t), as in Abowd, Kramarz and Margolis (1999). In practice, boundary bids reveal only cost bounds, so we cannot apply this regression directly. We therefore first estimate firm strength using the within-auction comparisons that incorporate these bounds, as described in Appendix D.3.

Comparisons of firms bidding in the same auction eliminate the common cost condition δ_t from the cost equation and the common score level μ_t from the score equation. Controlling for $q_g(\log s_{it})$ separates the cost strength α_i from cost differences associated with proposal scores. Repeated comparisons identify the firm strengths (α_i, σ_i) and the coefficients of q_g without specifying a parametric distribution for u_{it} or δ_t. Both strength components are estimated from solo bids and normalized to have mean zero across firms; q_g is normalized to zero at the median score.

Firms with few bids have imprecisely estimated effects. Passing these estimates directly to the formation stage would treat sampling noise as persistent bidding strength. We therefore use empirical Bayes shrinkage toward the cross-firm mean of zero (Morris, 1983): firms with less precise estimates are shrunk more, with the amount determined by the estimated dispersion of true strengths relative to estimation noise. Appendix D.3 gives the procedure, and Appendix D.6 reports its Monte Carlo validation. We denote the resulting strength estimates by $(\hat{\alpha}_i, \hat{\sigma}_i)$ and the posterior variance of the cost strength by V_i.[17]

[17] Let θ collect the cost strengths and the coefficients of q_g, and let $Q(\theta)$ be the comparison criterion in Appendix D.3. The quasi-posterior is proportional to $\exp\{-Q(\theta)/(2\sigma_Q^2) - \sum_i \alpha_i^2/(2\sigma_\alpha^2)\}$. Its local normal approximation gives $\alpha_i \mid \text{data} \approx N(\hat{\alpha}_i, V_i)$, with

$$V_i = 2\sigma_Q^2 \left[(H + 2\lambda D)^{-1}\right]_{ii}, \qquad \lambda = \sigma_Q^2/\sigma_\alpha^2.$$

For a potential entrant, we replace each unestimated strength component by zero, the normalized cross-firm mean, when constructing profits and formation covariates. We then take the members' componentwise maxima for potential JVs.

Thus the estimates of (α_i, σ_i) do not rely on the additional distributional assumptions used to recover δ_t in Step 4.

Step 4: auction conditions (μ_t, δ_t) and types. We next recover the common score level μ_t, the common cost condition δ_t, and residual types (e, u), holding $(\hat{\alpha}_i, \hat{\sigma}_i)$ and $\hat{q}_g$ fixed. We first estimate μ_t by averaging log scores net of the estimated score strengths $\hat{\sigma}_i$:

$$\hat{\mu}_t = \frac{1}{n_t^S} \sum_{i \text{ solo in } t} (\log s_{it} - \hat{\sigma}_i),$$

where the sum is over the n_t^S usable solo bids.[18] This calculation is possible because scores are observed without censoring.

For the common cost condition δ_t, censoring prevents the analogous average of log costs net of the estimated score-cost function and firm strengths. We use a normal model for the cost residual u_{it} and the common cost condition δ_t to incorporate the cost bounds and pool information across auctions with few bids. Let x_t collect the auction covariates.

Assumption 3 (Normal cost residuals and auction effects)**.** *The cost residual u_{it} is normally distributed with mean zero and variance $\sigma^2_{u,g}$. Accounting for uncertainty in the estimated cost strength, the cost likelihood uses*

$$\log \widehat{\tilde{c}}_{it} \mid \delta_t \sim N\big(\delta_t + \hat{q}_g(\log s_{it}) - \hat{\alpha}_i,\ \sigma^2_{u,g} + V_i\big)\,,$$
$$\delta_t \sim N(x_t'\gamma, \sigma^2_{\delta,t}).$$

An interior bid contributes the normal density of its recovered log cost, and V_i accounts for uncertainty about the cost strength.

Here H is the local curvature matrix of Q, and D is diagonal with ones on coordinates for cost strength and zeros on the coefficients of q_g. The prior variance σ^2_α and comparison-noise scale σ^2_Q are estimated as described in the appendix; σ^2_Q is distinct from the cost-residual variance estimated in Step 4.

[18] We use zero for a firm with no estimated score strength. For an auction with no usable solo bid, we use the median estimated score level in the same market and year.

We integrate over δ_t and maximize the resulting likelihood over γ, the residual variances $\sigma^2_{u,g}$ and the parameters governing $\sigma^2_{\delta,t}$. We then calculate the posterior distribution of each δ_t and denote its mean by $\hat{\delta}_t$.[19] With interior observations, this mean shrinks a precision-weighted average of adjusted log costs toward the mean predicted by auction covariates. Including the likelihood contributions of boundary bids extends this calculation to censored costs. Appendix D.4.1 specifies x_t and the variance function.

To construct a type from an interior source bid by firm i in auction t, draw δ^*_t from the normal approximation to its posterior and $\alpha^*_i \sim N(\hat{\alpha}_i, V_i)$, and form

$$
\begin{aligned}
e &= \log s_{it} - \hat{\mu}_t - \hat{\sigma}_i, \\
u &= \log \widehat{\tilde{c}}_{it} + \alpha^*_i - \delta^*_t - \hat{q}_g(\log s_{it}).
\end{aligned}
$$

Drawing the source cost condition δ^*_t and cost strength α^*_i propagates uncertainty in these components into the recovered cost residual u. These are independent draws from marginal approximations; the estimated score components $(\hat{\mu}_t, \hat{\sigma}_i)$ are held fixed. We pool the resulting solo types by market and period. Assigning a residual pair (e, u) to firm k in a target auction t' of the same market gives

$$
\begin{aligned}
\log s_{kt'} &= \hat{\mu}_{t'} + \hat{\sigma}_k + e, \\
\log \tilde{c}_{kt'} &= \hat{\delta}_{t'} + \hat{q}_g(\log s_{kt'}) - \hat{\alpha}_k + u.
\end{aligned}
$$

[19] Let $\mathcal{D}_t$ denote the cost observations in auction t, $\ell_t(v)$ their likelihood conditional on $\delta_t = v$, and $f_{\delta,t}(v)$ the fitted normal prior density. Bayes' rule gives

$$
p_t(v \mid \mathcal{D}_t) = \frac{\ell_t(v) f_{\delta,t}(v)}{\int \ell_t(r) f_{\delta,t}(r)\, dr}, \qquad \hat{\delta}_t = \int v\, p_t(v \mid \mathcal{D}_t)\, dv.
$$

If all cost observations are interior, write $r_{it} = \log \widehat{\tilde{c}}_{it} - \hat{q}_g(\log s_{it}) + \hat{\alpha}_i$ and $\varsigma^2_{it} = \hat{\sigma}^2_{u,g} + V_i$. The posterior is normal with

$$
\begin{aligned}
V_{\delta,t} &= \left(\hat{\sigma}^{-2}_{\delta,t} + \sum_i \varsigma^{-2}_{it} \right)^{-1}, \\
\hat{\delta}_t &= V_{\delta,t} \left(\frac{x_t' \hat{\gamma}}{\hat{\sigma}^2_{\delta,t}} + \sum_i \frac{r_{it}}{\varsigma^2_{it}} \right).
\end{aligned}
$$

The sums are over the observations used in the cost likelihood. With equal observation variances and a diffuse prior, this reduces to the unweighted residual average. Appendix D.4.1 gives the likelihood contributions of boundary bids.

When recovering types from historical JV bids, we replace each missing member's cost strength by zero before taking the maximum. For the score strength, we take the maximum among members with an estimate and use zero if none has one. Appendix D.4.2 gives the remaining details of the JV type construction.

Step 5: expected profits. We use the recovered types to calculate the expected bidding profit of each solo firm and potential JV at every combination of rival numbers.

For each type, we first construct its score and cost in the target auction using the estimates from Step 4. We then solve the bidding problem (1) against the fitted distributions of rival bids and average the maximized profits over types.[20] Write $\hat{\pi}^d_{b,t}(a; m, n)$ for this average for a bidder of mode $d \in \{S, J\}$ and strength a, facing m JV and n solo rivals with usable bids. The distributions of rival bids are evaluated at total bidder counts $(m, n+1)$ for a solo bidder and $(m+1, n)$ for a JV, so the bidder's own participation is included.

Formation determines the numbers of listed bidders, some of whom withdraw or submit no usable bid. We treat each listed bidder of mode d as submitting a usable bid independently with probability p^d, estimated by mode and period. Against m listed JV rivals and n listed solo rivals, the numbers of usable rivals are independent draws $K_J \sim \text{Binomial}(m, p^J)$ and $K_S \sim \text{Binomial}(n, p^S)$. The expected profit is therefore

$$\hat{\pi}^d_t(a; m, n) = p^d \, \mathbb{E}\big[\hat{\pi}^d_{b,t}(a; K_J, K_S)\big] \, .$$

The factor p^d accounts for the bidder's own probability of submitting a usable bid; otherwise its bidding profit is zero. The expectation accounts for the numbers of rivals that actually compete, with the fitted bid distributions evaluated at each realization of those numbers.

This calculation produces the profit tables for the formation stage. Given beliefs about rival numbers, equation (3) averages the table entries to obtain the expected profits of participating alone or as a JV. The formation estimator takes these tables and the estimated cost and score strengths as fixed first-stage inputs (Section 6.2).

[20] In this calculation, nonpositive maximized profits are set to zero. A lone bidder earns $R_t \max\{1 - \tilde{c}, 0\}$.

6.2 Formation stage and consistent beliefs

The formation stage fits the observed partitions—who bids alone, which firms form a JV, and who stays out—while requiring beliefs about rival numbers to agree with the distributions generated by the model's stable partitions. For auction t, the observed partition P_t^{obs} consists of solo bidders S_t, JV pairs J_t, and non-participants O_t. Let θ collect the 31 free formation parameters in equations (4) and (5), with $q = 1$ fixed.[21] The estimates of bidding strength and the profit tables $\hat{\pi}_{it}^S(m,n)$ and $\hat{\pi}_{ijt}^J(m,n)$ from the bidding stage are fixed inputs to this step.

The auction-specific states ξ and η directly shift solo and JV payoffs, respectively. They are independent standard normal variables. Firms observe both before formation, but the econometrician observes neither. We therefore condition beliefs and expected profits on both states. For each (ξ, η), $H_t^S(m, n \mid \xi, \eta)$ and $H_t^J(m, n \mid \xi, \eta)$ assign probabilities to facing m rival JVs and n rival solo bidders, for a solo bidder and a JV, respectively. These beliefs are joint distributions of rival numbers; they do not record the identities or pairings of rival firms. Equation (3), evaluated with the estimated profit tables, converts these beliefs into expected profits, denoted by $V_{it}^S(\xi, \eta; H)$ and $V_{ijt}^J(\xi, \eta; H)$.

The likelihood integrates the probability of the observed partition over the unobserved common states. Given θ and H, Proposition 1 implies that the conditional probability of the observed partition equals the probability that it is stable. Define

$$p_t(P_t^{\text{obs}} \mid \xi, \eta; \theta, H) \equiv \Pr\left(P_t^{\text{obs}} \text{ is stable} \mid \xi, \eta; \theta, H\right),$$

where the probability averages over the idiosyncratic payoff shocks and partnership availability. With ϕ denoting the standard normal density, the auction's likelihood contribution is

$$L_t(\theta, H) = \int_{\mathbb{R}^2} p_t(P_t^{\text{obs}} \mid \xi, \eta; \theta, H)\, \phi(\xi)\phi(\eta)\, d\xi\, d\eta. \tag{8}$$

Appendix E.2 derives the conditional probability from the stability conditions. The latent states are integrated out, rather than assigned a single estimated realization for each auction.

[21] Appendix E.1 records the reference project size, strength normalizations, parameter domains and numerical search ranges.

The estimation task is to maximize this likelihood subject to belief consistency. To state this restriction, write $\Gamma_\theta(H)$ for the distributions of rival numbers generated by the stable partitions under θ and H. Holding (ξ, η) fixed and denoting the resulting numbers of JVs and solo bidders by M_t and N_t, respectively, its two components are

$$\begin{aligned}
[\Gamma_\theta(H)]_t^S(m, n \mid \xi, \eta) &= \Pr(M_t = m, N_t = n + 1 \mid \xi, \eta, N_t \geq 1; \theta, H), \\
[\Gamma_\theta(H)]_t^J(m, n \mid \xi, \eta) &= \Pr(M_t = m + 1, N_t = n \mid \xi, \eta, M_t \geq 1; \theta, H).
\end{aligned} \tag{9}$$

These probabilities integrate over idiosyncratic shocks and availability while keeping both common states fixed. Subtracting one solo bidder or one JV from the respective total gives the rival numbers. Conditioning on mode presence restricts each distribution to realizations with at least one bidder of that mode. Different partitions can generate the same rival numbers: the identities of non-participants and the pairings of JV members are part of the partition but are not retained in $\Gamma_\theta(H)$. Consistency requires $H = \Gamma_\theta(H)$ conditional on the states; agreement only after averaging across states would not impose the same restriction.

We can therefore state the joint estimation problem as a likelihood maximization subject to equilibrium constraints. Let Θ be the parameter domain and $\mathcal{H}$ the set of measurable belief functions that assign nonnegative probabilities summing to one over feasible pairs of rival numbers. The mathematical program is

$$\begin{aligned}
\max_{\theta \in \Theta,\, H \in \mathcal{H}} \quad & \sum_{t=1}^{T} \log L_t(\theta, H) \\
\text{subject to} \quad & H_t^d(m, n \mid \xi, \eta) = [\Gamma_\theta(H)]_t^d(m, n \mid \xi, \eta), \\
& \text{for every } t, d, m, n \text{ and almost every } (\xi, \eta).
\end{aligned} \tag{10}$$

The constraints apply to each participation mode with positive conditional probability; no JV belief is required in an auction where JVs are prohibited. Almost every state refers to the distribution of (ξ, η). This is the mathematical program with equilibrium constraints (MPEC) formulation of Su and Judd (2012), with parameters and beliefs as unknowns.

In practice, we use nested pseudo-likelihood (NPL) to alternate between parameter estimation and belief updating, following the approach of Aguirregabiria and Mira (2007). Starting with beliefs $H^{(r)}$, a parameter step maximizes the likelihood while holding those

belief functions fixed. A belief step then uses the new parameters to generate stable partitions and moves the beliefs toward the implied distributions of rival numbers:

$$\begin{aligned} \theta^{(r+1)} &\in \arg\max_{\theta\in\Theta} \sum_{t=1}^{T} \log L_t(\theta, H^{(r)}), \\ H^{(r+1)} &= (1-\lambda)H^{(r)} + \lambda\Gamma_{\theta^{(r+1)}}(H^{(r)}), \qquad 0 < \lambda \leq 1. \end{aligned} \tag{11}$$

The parameter step integrates over the states, whereas the belief step updates the distributions of rival numbers conditional on each state. Damping, through λ, limits the change in beliefs in one round. Expected profits remain fixed within a parameter step and are recomputed after beliefs are updated.[22] Because ξ and η are continuous, the algorithm represents the belief functions on a finite grid and evaluates the consistency restrictions in (10) at those points. Appendix E.3 describes this approximation, and Appendix E.5 gives the simulation and update pseudocode.

7 Estimation Results

This section reports the bidding and formation estimates and assesses how well the model reproduces participation.[23]

7.1 Bidding stage

Bidding strength differs substantially across firms in both of its components. The standard deviation of the cost strength is 0.043 and that of the score strength 0.039: a firm one standard deviation stronger than another has costs about 4 percent lower for a proposal of the same score, or proposals with scores about 4 percent higher. Both are large in auctions

[22] Section 3.5 of Aguirregabiria and Mira (2007) applies NPL to latent market types by integrating their conditional likelihoods and updating type-specific probabilities. Here the common states are continuous. NPL is a sequential estimator; even a converged NPL fixed point need not maximize the constrained likelihood in (10).

[23] The formation coefficient tables report the final conditional likelihood optimization, holding the bidding estimates and the saved profit and belief tables fixed. The participation and price simulations, partnership classifications and posterior-state summaries reported here have not been recomputed at these refined coefficients; they use the preceding parameter vector. Appendix E.6 describes the refinement and its conditional standard errors.

Table 4. Bidding strength and observed bidding histories

	Cost α	Score σ
Log maximum project size	−0.061	−0.065
	(0.018)	(0.019)
Log number of solo bids	0.151	0.509
	(0.043)	(0.037)
Log years with solo bids	0.011	−0.015
	(0.036)	(0.034)
Log number of regions	−0.010	0.022
	(0.016)	(0.013)
Log number of work categories	−0.033	−0.025
	(0.021)	(0.017)
Mean bid year	−0.035	−0.124
	(0.014)	(0.015)
Region and work-category bid-share controls	Yes	Yes
Firms	7,123	7,123
R^2	0.019	0.205

Notes: Equal-weight firm-level OLS with the same sample and regressors in both equations. Outcomes and regressors use full-sample means and population standard deviations for standardization. Attributes cover valid solo bids with reserve at least 1 million yen over 2006–2022. Both equations include an intercept and 27 region and work-category bid-share controls. Parentheses report HC3 heteroskedasticity-robust standard errors treating firms as independent and estimated strengths as fixed; shared first-stage estimation uncertainty is not propagated. See Appendix F.1 for definitions and estimation details.

in which most winning bids lie within a few percent of the floor, and the score strength matters even at the floor, where a bidder can no longer lower its price but a higher score still raises its effective bid. The two components are nearly uncorrelated across firms, with a correlation coefficient of 0.08.

We next ask which features of firms' bidding histories are associated with the two strengths. Table 4 relates each estimated strength to the scale, frequency and breadth of solo bidding, controlling for the regional and work-category composition of bids and their average year.[24] These histories explain 20.5 percent of the variation in score strength but only 1.9 percent in cost strength. Frequent bidding is particularly closely associated with score strength: the standardized coefficient on the log number of solo bids is 0.51, compared with

[24] Appendix F.1 defines the variables and the common sample. The regressions describe associations with estimated strengths and do not identify the direction of influence between ability and participation.

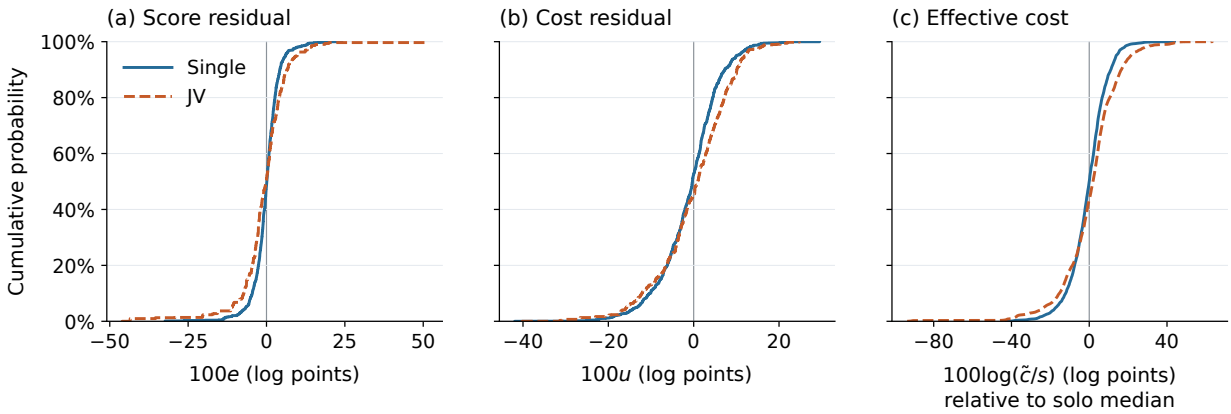


Figure 4. Recovered bidding types for solo bidders and JVs. Panels (a) and (b) show the cumulative distributions of the score residual $100e$ and cost residual $100u$. Panel (c) shows $100\log(\tilde{c}/s)$ at $\alpha = \sigma = 0$, centered in each auction by its solo-type median. Both modes use the same 396 auctions, weighted equally, and all 128 saved paired types in each relevant bank. The curves retain the full saved support.

0.15 for cost strength. The ability to produce high-scoring proposals is thus the dimension of bidding strength more closely tied to observed participation in scoring auctions.

Residual bidding types, (e_{it}, u_{it}) for solo bidders and (e_{ijt}, u_{ijt}) for JVs, also differ between the two modes. Here e is the score residual and u the cost residual. Figure 4 compares their cumulative distributions using the same 396 auctions. JV score and cost residuals are more dispersed. Equality between solo bidders and JVs is rejected for each residual marginal and for their joint distribution at the 5 percent level, with a Holm-adjusted p-value of 0.03 for each of the three tests.[25] At equal cost and score strength, the effective-cost CDFs in panel (c) cross, so this comparison does not imply a uniform cost advantage for either mode.

The resulting expected profits vary strongly with the solo state because it changes the competition that firms expect. At $(\xi, \eta) = (0, 0)$, the mean expected profit is 5.6 million yen for a potential solo entrant and 7.3 million yen for a potential JV. Holding $\eta = 0$, the mean solo profit falls from 41.5 million yen at $\xi = -2$ to 2.1 million yen at $\xi = 2$. These are the state-dependent profit tables used in the formation likelihood, conditional on the first-stage bidding estimates.

[25] Holding the first-stage estimates and residual-recovery laws fixed, we use 99 source-auction multiplier replications and Holm adjustment across the three tests. The tests use all eligible source bids, whereas the figure uses the saved type draws. They exclude first-stage estimation uncertainty; Appendix F.2 provides details.

Synergy decomposition. To locate the sources of synergy, we compare each observed JV with a solo bidder whose persistent cost and score strengths are both zero in the same auction. For strength $a = (\alpha, \sigma)$ and mode $d \in \{S, J\}$, the model gives mean log effective cost as

$$\ell_t^d(a) = \mathbb{E}_{(e,u)\sim G_g^d}\left[\delta_t + q_g^d(\mu_t + \sigma + e) - \alpha + u - (\mu_t + \sigma + e)\right], \tag{12}$$

where $q_g^S = q_g$. We evaluate this expectation at the estimates using the saved types. For JV (i, j), $a_{ij} = (\max\{\alpha_i, \alpha_j\}, \max\{\sigma_i, \sigma_j\})$. Its difference from the zero-strength solo bidder decomposes as

$$\begin{aligned} C_\alpha &= -\max\{\alpha_i, \alpha_j\}, \\ C_\sigma &= \tfrac{1}{2}\sum_{d\in\{S,J\}}\left[\ell_t^d(a_{ij}) - \ell_t^d(0,0)\right] - C_\alpha, \\ C_{q,G} &= \tfrac{1}{2}\left[\ell_t^J(0,0) - \ell_t^S(0,0) + \ell_t^J(a_{ij}) - \ell_t^S(a_{ij})\right], \\ \ell_t^J(a_{ij}) - \ell_t^S(0,0) &= C_\alpha + C_\sigma + C_{q,G}. \end{aligned}$$

The first two terms measure the cost- and score-strength contributions; the third changes both the score-cost relation and the type distribution. Table 5 averages these contributions across JVs within auctions and then equally across auctions, in log points.[26]

Relative to the zero-strength benchmark, mean log effective cost is 8.81 log points lower for the JVs. Persistent score strength contributes a reduction of 9.84 points, compared with 1.05 from cost strength; the score-cost relation and type distribution offset these gains by 2.08 points. The dominant source of synergy in this comparison is therefore the members' ability to produce high-scoring proposals. This contribution combines the strengths of the firms that select into JVs with the model's assumption that a JV uses its stronger member in each dimension.

The wider JV type distributions give synergies an uncertain component. A JV inherits persistent capabilities from its members, but its proposal and cost realization also depend on how they perform together on a particular project. The estimated dispersion is consistent

[26]The contributions average the two orders of changing strengths and switching (q, G), splitting their interaction equally. Cost strength is additive in log effective cost. The table multiplies each mean contribution by 100; negative values indicate lower JV effective cost. Appendix F.3 gives details. These are descriptive decompositions at the estimates, without sampling confidence intervals.

Table 5. JV effective-cost differences relative to zero-strength solo bidding

Contribution	Log points
Cost strength α	−1.05
Score strength σ	−9.84
Score-cost relation and types (q, G)	+2.08
Total difference	−8.81
Auctions	127
Observed JVs	312

Notes: Differences in $100\,E[\log(\tilde{c}/s)]$, JV minus solo bidding at $\alpha = \sigma = 0$; negative values denote lower JV effective cost. We use 128 saved types per bank, average JVs within each auction, and weight auctions equally. Estimates and types are fixed. Appendix F.3 gives details.

with uncertainty about the performance of a newly combined organization, although the distribution comparison does not identify its organizational source. Favorable cost and score strengths can lower effective cost, while an unfavorable type can offset that improvement. Synergies thus depend on both inherited capabilities and realized joint performance; the model allows favorable and unfavorable outcomes.

Model fit. The bidding model reproduces average prices closely overall. Under the current policy, with states at their posterior means and estimation beliefs interpolated at those states, the average winning-bid-to-reserve ratio is 89.0 percent in the model and 88.7 percent in the data. The shares of winning bids within one percent of the floor are 77 and 74 percent (Appendix Table A8). The largest projects remain harder to match: their predicted price ratio is 90.2 percent, against 94.1 percent in the data.

7.2 Formation stage

Table 6 summarizes the estimates governing solo entry and JV formation, with auction-clustered sandwich standard errors. Appendix Table A9 reports all 31 estimated parameters and the fixed relative pair-shock scale $q = 1$.

Table 6. Estimates of the formation stage

	Estimate	SE
Panel A. Bidding alone		
Capacity shortfall, λ_S	1.81	0.08
Project size, β_S	0.67	0.25
Cost strength, $\kappa_{S,\alpha}$	−0.18	0.03
Score strength, $\kappa_{S,\sigma}$	0.60	0.03
Panel B. Bidding jointly		
Capacity shortfall, λ_J	2.96	0.66
Project size, β_J	−0.40	0.14
Cost strength, $\kappa_{J,\alpha}$	−0.32	0.06
Score strength, $\kappa_{J,\sigma}$	0.40	0.07
Discount for past joint bids, γ	0.77	0.07
Availability without contact, $e^{-\tau}$	0.19	0.08
Panel C. Additional costs of observed JVs		
New JV: additional cost (σ units)	6.31	0.29
Repeat JV: additional cost (σ units)	2.32	0.26
Log likelihood −10,651.2 (128 draws); 396 auctions; 31 estimated parameters		

Notes: Selected estimates after the final conditional likelihood optimization; Appendix Table A9 reports all parameters. Payoff terms are in solo-shock units, with an estimated scale $\sigma = 197$ million yen and the relative pair-shock scale fixed at $q = 1$. Strength coefficients are per first-stage prior standard deviation. Panel C averages the total additional JV cost $K_t e^{-\gamma h_{ij}}$ over 145 new and 167 repeat observed JVs; each member bears half. These cost levels represent payoff costs rather than accounting expenditures. SEs are auction-clustered sandwich standard errors conditional on $q = 1$, the bidding-stage estimates, and the fitted profit and belief tables. Derived quantities use the delta method. Simulation-based results elsewhere use the estimates preceding this final optimization, as explained in Appendix F.5.

Bidding alone. Projects beyond a firm's previous experience discourage solo entry. The capacity coefficient is $\lambda_S = 1.81$, so a firm whose largest previous project is half the size of the current one faces a capacity cost of about $1.25\,\sigma$. Project size also reduces the solo payoff independently of this shortfall: $\beta_S = 0.67$ implies a penalty of $0.81\,\sigma$ at 10 billion yen, even for a firm with sufficient capacity.

Bidding strength remains related to solo entry after accounting for expected auction profits. Holding those profits and the other payoff terms fixed, a one-standard-deviation increase in score strength raises the solo payoff by $0.60\,\sigma$, whereas the same standardized increase in cost strength lowers it by $0.18\,\sigma$. Thus the capabilities behind high scores are associated with an additional incentive to participate, beyond their value in the auction

itself. These direct effects capture participation costs or benefits correlated with strength; production efficiency improves bidding profits, but does not imply the same additional incentive to enter. The positive score-strength term complements the stronger association of observed bidding activity with score strength in Table 4.

Bidding jointly. Large projects favor joint participation through the direct size term. The estimate $\beta_J = -0.40$ raises each member's JV payoff as project size increases, while the corresponding solo term reduces it. Past joint bids also make a JV more attractive: with $\gamma = 0.77$, one prior joint bid reduces the additional JV cost by 41 percent.

Evaluated at the observed partnerships, the mean additional JV cost is $6.31\,\sigma$ for new JVs and $2.32\,\sigma$ for repeat JVs. These are total costs split equally between the members, excluding the participation cost that each pays separately.[27] A JV forms only when its expected profit, compatibility and other payoff components make both members prefer it to bidding alone or staying out, after paying these costs.

Prior contact determines which partnerships firms can consider. Pairs that have previously participated in the same auction of the bureau and work category are always available, even if they have never bid jointly. For pairs without this contact, the estimated availability probability is 0.19, capturing the difficulty of finding a partner outside their existing contacts.

JV formation decomposition. Few potential entrants have a partner with whom both would gain from forming a JV. Table 7 classifies firms in the 326 auctions where the model permits joint participation. With equal weight on each firm–auction observation, 97.95 percent have no mutually beneficial partner even before partnership availability is imposed. Only 2.05 percent have such a candidate. Among this latter group, 17.6 percent have none available after the draws for pairs without prior contact; 9.3 percent have an available candidate but remain unmatched because their prospective partners form other JVs; and 73.1 percent form a JV.[28]

[27] At the estimated yen scale, the corresponding total additional costs are 1.24 billion and 0.46 billion yen. They are conditional on $q = 1$ and represent payoff costs rather than accounting expenditures.

[28] We classify the same 64 unconditional draws used to assess fit, holding the estimated parameters fixed. A partnership must give both members more than either bidding alone or staying out. We compare these

Table 7. JV opportunities and formation among potential entrants

Classification	Share (%)
No mutually beneficial partner	97.95
Mutually beneficial partners, none available	0.36
Available mutually beneficial partner, unmatched	0.19
Matched in a JV	1.50

Notes: Model-implied shares for 47,080 potential entrant–auction observations in the 326 auctions where the model permits JVs, averaged over 64 unconditional simulations at the estimates. A mutually beneficial partner is one with whom both firms prefer the JV to bidding alone or staying out. Availability follows the estimated contact rule. Each firm–auction observation receives equal weight. The matched category counts member firms. The four categories are exhaustive; rounding may affect their displayed sum. Appendix F.6 gives the method. These shares describe fitted-model outcomes and are not effects of removing a constraint.

The main limitation is therefore the scarcity of mutually beneficial partnerships at the estimated payoffs. Restricted access to partners without prior contact and competition for available partners prevent some of these opportunities from becoming JVs.

Model fit. The model reproduces the overall scale and incidence of participation reasonably well. Table 8 compares the data with unconditional simulations, drawing the auction-level states from their distributions. It predicts JVs in 132 auctions, compared with 127 in the data, and closely matches the total number of repeat JVs, at 168 versus 167. Observed solo-bidder counts lie within the model's central 95 percent predictive interval in 93.2 percent of auctions (Appendix F.7).

The model nevertheless tends to overpredict participation. Overall, solo participation is about 9 percent above the data, and the model predicts 185 new JVs against 145. The discrepancy is larger in the six auctions above 10 billion yen, where it predicts 31.9 JVs against 18. New partnerships are close to the data, at 15.6 against 16, but repeat partnerships are overpredicted, at 16.3 against 2. Appendix F.7 reports the detailed comparisons by project-size quartile and region–work category.

potential partnerships before and after imposing availability, then identify matches in the original stable partition. Appendix F.6 defines the categories and weights. The shares exclude parameter uncertainty and do not measure the effects of removing a constraint.

Table 8. Fit of the formation model

		Solo bids		JVs		New / repeat JVs		Auctions with JV	
	Auctions	Data	Model	Data	Model	Data	Model	Data	Model
All	396	6,159	6,694	312	353	145 / 167	185 / 168	127	132
≥5 bn yen	46	447	490	79	107	54 / 25	60 / 47	29	29
≥10 bn yen	6	4	7	18	32	16 / 2	16 / 16	6	6

Notes: Model columns are means over 64 simulated stable partitions per auction at the estimates, with the auction-level states (ξ_t, η_t) drawn from their distribution and expected profits evaluated at the drawn state. The size groups overlap. Appendix Table A10 reports fit by size quartile, region and work category.

8 Counterfactuals: Banning Joint Ventures

We compare the current eligibility rules with a ban on JVs in the 390 auctions with engineer's estimates below 10 billion yen. We exclude the six larger auctions because the model substantially overpredicts their JV participation (Section 7.2). The model predicts that allowing JVs benefits the procurer in both price and technical score in projects from 5 to below 10 billion yen. Former members' solo bids and entry by other firms partly replace the lost joint bids, but the ban raises winning prices and lowers winning scores in this group. We describe the policy comparison and outcome measures, report the participation responses, and then evaluate prices and scores together.

8.1 Counterfactual design

We remove the JV option while maintaining the estimated model's independent solo and pair shocks. Under each policy, we solve for beliefs about rival numbers and the resulting participation, holding each auction's states at their posterior means given observed participation, $(\hat{\xi}_t, \hat{\eta}_t)$.[29] A former JV member bids alone if its solo payoff is positive at the counterfactual beliefs. Participants then draw bidding types and bid against the estimated rival-bid distributions conditional on bidder counts. Both policies use the same underlying random draws; Appendix G.1 gives the numerical procedure.

[29] Estimation uses all 396 auctions; the policy comparison uses only the 390 below 10 billion yen. The simulations use the same parameter vector as the simulation-based fit in Section 7, before the final conditional optimization of the coefficient tables (Appendix E.6).

We evaluate the JV ban using the following measures of participation and procurement outcomes. Index the current policy by $k = 0$ and the ban by $k = 1$. Write the resulting partitions as $P_t^k = (S_t^k, J_t^k, O_t^k)$, with $N_t^k := |S_t^k|$ solo bidders, $M_t^k := |J_t^k|$ JVs, and nonparticipants O_t^k. Under the ban, $J_t^1 = \varnothing$ and hence $M_t^1 = 0$. Current JV members form the set $L_t^0 := \bigcup_{(i,j)\in J_t^0}\{i, j\}$. Let $\mathbb{E}_t$ integrate over the simulation draws at the fixed estimates and posterior states, coupling the two policies through the same underlying shocks. For an auction group $\mathcal{T}$, the participation measures are

$$\begin{aligned}
\text{Members bidding alone}_{\mathcal{T}} &:= 100\,\frac{\sum_{t\in\mathcal{T}} \mathbb{E}_t[|L_t^0 \cap S_t^1|]}{\sum_{t\in\mathcal{T}} \mathbb{E}_t[|L_t^0|]}, \\
\text{New solo entrants}_{\mathcal{T}} &:= \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t[|O_t^0 \cap S_t^1|], \\
\Delta\text{Independent bidders}_{\mathcal{T}} &:= \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t[N_t^1 - N_t^0 - M_t^0].
\end{aligned} \tag{13}$$

For procurement outcomes, let b_{ht}^k and s_{ht}^k denote bidder h's price in yen and technical score. In the simulation, a usable bid is a submitted bid whose price is valid, $\tau_t R_t \le b_{ht}^k \le R_t$. Listed bidders may withdraw or submit no such bid; the mode-specific submission probabilities are p^S and p^J (Section 6). Let $\mathcal{B}_t^k$ be the set of bidders submitting usable bids and define the award indicator $A_t^k := \mathbf{1}\{\mathcal{B}_t^k \neq \varnothing\}$. When $A_t^k = 1$, let w_t^k denote the winning bidder, defined by $w_t^k \in \arg\max_{h\in\mathcal{B}_t^k} s_{ht}^k/b_{ht}^k$. The winning price is $B_t^k := b_{w_t^k,t}^k$, the price submitted by bidder w_t^k; set $B_t^k := 0$ otherwise. The price and award measures are

$$\begin{aligned}
\text{Price}_{\mathcal{T}}^k &:= 100\,\frac{\sum_{t\in\mathcal{T}} \mathbb{E}_t[B_t^k]}{\sum_{t\in\mathcal{T}} R_t\mathbb{E}_t[A_t^k]}, \\
\text{Award probability}_{\mathcal{T}}^k &:= \frac{100}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \mathbb{E}_t[A_t^k].
\end{aligned} \tag{14}$$

The price measure expresses winning payments as a percentage of the engineer's estimates for awarded contracts. For both measures, we report the ban minus the current policy, $\Delta X_{\mathcal{T}} := X_{\mathcal{T}}^1 - X_{\mathcal{T}}^0$, in percentage points; a positive price change means higher procurement prices. We have two groups according to the engineer's estimates: $\mathcal{T} = \{t : R_t < 10^{10}\}$ and $\mathcal{T} = \{t : 5 \times 10^9 \le R_t < 10^{10}\}$, where R_t is measured in yen.

Table 9. Banning joint ventures in projects below 10 billion yen

	All $<$ 10 bn yen (390 auctions)	5–$<$ 10 bn yen (40 auctions)
Members bidding alone (%)	19	14
New solo entrants	0.04	0.25
Δ Independent bidders	-0.399	-0.847
Δ Winning price / estimate (pp)	-0.01	$+0.11$
Winning score change (%)	-0.31	-0.49
Δ Award probability (pp)	-0.09	-0.56

Notes: Measures are defined in equations (13), (14) and (15). Changes are the ban minus the current policy. A positive price change means that the winning price rises relative to the engineer's estimate. The score change averages each auction's percentage change in its mean winning technical score conditional on an award, weighting auctions equally. The 40-auction group is a subset of the 390-auction group. Both policies maintain independent solo and pair shocks, fix auction states at their posterior means, and use common random numbers with 64 formation draws and 16 bidding replications per auction. These predictions condition on the estimates and do not incorporate parameter uncertainty.

We also compare the technical scores of winning bids. Let $W_t^k := s^k_{w_t^k,t}$ when a contract is awarded and zero otherwise, and let $\bar{s}_t^k := \mathbb{E}_t[W_t^k]/\mathbb{E}_t[A_t^k]$ be the mean winning score conditional on an award. Because score scales differ across projects and years, we calculate the percentage change within each auction before averaging across auctions:

$$\Delta\text{Score}_{\mathcal{T}} := \frac{100}{|\mathcal{T}|}\sum_{t\in\mathcal{T}}\left(\frac{\bar{s}_t^1}{\bar{s}_t^0}-1\right). \tag{15}$$

This measure gives each auction equal weight; a positive value means higher winning scores under the ban.

8.2 Participation, prices and scores

Banning JVs removes 0.67 joint bids per auction across the 390 auctions. Each lost JV has two members, and 19 percent of these members bid alone under the ban, contributing 0.26 solo bids per auction. Firms that stay out under the current policy add another 0.04 solo bids, while some current solo bidders leave. After these responses, the net decline is 0.40 independent bidders per auction, or 2.4 percent of the current-policy number (Table 9).

In the 40 auctions from 5 to below 10 billion yen, the ban removes 1.46 joint bids per auction. Fourteen percent of their members bid alone, contributing 0.40 solo bids per auction, and firms that previously stayed out add 0.25. After accounting for current solo bidders who leave, independent bidder numbers fall by 0.85 per auction, or 7.0 percent of the current-policy number. Allowing expected competition and profits to adjust moderates this decline: holding profits at their current-policy values gives a reduction of 1.12 bidders in this subgroup, compared with 0.85 after beliefs are updated. Thus the loss of joint bids overstates the net loss of competitors, although replacement is incomplete.

Allowing JVs benefits the procurer in both price and technical score in the larger projects. In the 40 auctions from 5 to below 10 billion yen, a ban raises winning prices from 89.55 to 89.66 percent of the engineer's estimate. Winning technical scores fall by 0.49 percent, averaging the within-auction percentage changes, and mean award probabilities fall by 0.56 percentage points. Thus the replacement bids from former JV members and other firms do not fully preserve the procurement benefits of joint bidding: under the ban, the procurer pays more for proposals with lower technical scores.

Across all 390 auctions, winning scores fall even though average prices barely change. The ban reduces winning technical scores by 0.31 percent on average, while winning prices fall only from 89.05 to 89.04 percent of the engineer's estimate. Mean award probabilities also fall by 0.09 percentage points. Appendix G.2 reports the current-policy levels and additional comparisons.

The price floor helps explain why small price effects can accompany a loss in winning technical scores. If both the original and replacement winners bid at the floor, their prices are the same, while a higher technical score can still determine who wins. In the simulations, 77 percent of winning bids across the 390 auctions are within one percent of the floor under the current policy, compared with 78 percent under the ban. Evaluating JVs only through prices would therefore miss the decline in winning scores under a ban. In the larger projects, the benefits of allowing JVs appear in both outcomes: lower winning prices and higher winning technical scores.

9 Discussion

Equal sharing between JV partners is an assumption. The partners in our setting are generally firms of comparable size, so we would not expect their shares to depart greatly from an equal split. Nevertheless, we do not observe the shares they agree upon. Data on these shares would allow us to assess this assumption and study how the division of joint returns affects partner choice and participation.

The independence of the idiosyncratic payoff shocks to solo and JV entry is also a strong assumption. For example, the unexpected availability of engineers at a firm could raise both its payoff from bidding alone and the value of a JV involving that firm, creating dependence beyond observed characteristics and auction-level conditions. Allowing for such dependence would make estimation substantially more computationally demanding, because the conditional likelihood could no longer be evaluated using the current factorization. We leave this extension to future work.

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Appendix A. Institutions and Data

This appendix documents the institutions and data construction used in Sections 2–4. It provides the institutional tables, member identification and historical rules behind the formation sample, followed by the evidence on JV eligibility and the collusion screens supporting the competitive-bidding specification.

A.1 Institutional tables

Table A1 lists the thresholds of the WTO Agreement on Government Procurement for construction contracts of the central government. Table A2 reports the share of WTO-covered auctions by work category and bureau between 2012 and 2022, with the categories labeled by the codes of the bid records. The share is highest in steel-bridge construction (31 percent) and prestressed-concrete construction (15 percent) and below 10 percent in the other categories.

A.2 Identification of joint-venture members

The bid records list a JV under its own name. A project-specific JV is typically named by the abbreviated names of its members, separated by a delimiter, followed by "specified construction joint venture" or "JV". We screen every bidder record of the seven bureaus, including withdrawn and invalid bids, flag JV bidders by such name patterns, and parse each JV name into member tokens. Each token is then linked to a firm in the roster of firms that bid alone in the panel. The procedure distinguishes three kinds of evidence. First, exact matches of a token to a firm name within the same bureau, work category and period. Second, entries of a curated dictionary of abbreviations, each recording the abbreviation, the legal name of the firm, the period and scope of validity, and the evidence (company profiles, project documents or other public sources); entries that require it are confirmed against the roster of the bureau and work category. Third, direct project evidence for specific bids. Ambiguous cases are left unresolved: a token that could refer to several firms

Table A1. GPA thresholds for central-government construction contracts (tax-inclusive)

Period (fiscal years)	Threshold (million yen)
Apr 2004 – Mar 2006	730
Apr 2006 – Mar 2008	720
Apr 2008 – Mar 2010	790
Apr 2010 – Mar 2012	690
Apr 2012 – Mar 2014	580
Apr 2014 – Mar 2016	600
Apr 2016 – Mar 2018	740
Apr 2018 – Mar 2020	680
Apr 2020 – Mar 2022	680
Apr 2022 – Mar 2024	680

Sources: 2004–2016, http://www.cgr.mlit.go.jp/hattyu/pdf/keiyaku_unyou_2004.pdf; 2016–2018, https://www.mofa.go.jp/mofaj/files/000335196.pdf; 2018–2020, https://www.nikkei.com/article/DGXMZO26117570V20C18A1000000/; 2020–2022, https://www.mofa.go.jp/mofaj/annai/shocho/chotatsu/kijyungaku.html.

remains ambiguous, and a two-character abbreviation of a family name is never accepted solely because it uniquely matches the beginning of a firm's name. Parsing errors in the original records are corrected by hand with recorded evidence.

In the WTO-covered civil engineering and bridge auctions of 2012–2022 in the five bureaus with complete coverage, all named members are linked to firms for 499 of the 533 project-specific JV bids (93.6 percent). Across the panel, 10,819 bidder records carry evidence of being a JV of any type, and all named members are linked for 4,591 of them, excluding links that rely only on prefix matches. We use these 4,591 JVs to build the joint-bidding histories of pairs. Because unresolved JVs are omitted from the histories, the histories undercount past joint bids.

A.3 Histories and potential entrants

Firm and pair histories. All histories use only auctions held before the auction in question. The capacity K_i is the largest engineer's estimate among the auctions in which firm i previously submitted a valid solo bid in the same bureau and work category. For the 1,449

Table A2. Share of WTO-covered auctions by work category and bureau, 2012–2022 (percent)

Work category	Hokuriku	Kanto	Kinki	Kyushu	Shikoku	Total
CC	0.00	0.00	3.45	0.00	2.63	1.10
PC	17.39	14.29	20.59	12.73	2.91	14.67
airconditioner	1.82	7.57	5.71	1.68	3.13	4.52
boring	0.00	0.00	0.00	0.00	0.00	0.00
bridge	11.11	39.74	24.86	16.46	25.55	31.28
bridgerepair	0.00	0.00	0.00	0.00	0.00	2.38
civilengineering	0.92	3.98	5.18	1.28	1.61	2.68
construction	1.18	7.26	5.47	2.03	5.08	4.38
dredging	0.00	0.00	22.22	0.00	100.00	8.24
electricity	0.00	1.13	0.59	0.41	0.24	0.48
gardening	0.00	0.00	0.00	0.00	0.00	0.00
grout	0.00	0.00	0.00	0.00	0.00	0.00
machine	0.00	4.12	2.33	2.06	3.67	2.20
management	0.15	0.12	0.00	0.00	0.00	0.07
painting	0.00	0.00	0.00	0.00	0.00	0.00
pavement	0.00	0.00	0.35	0.00	0.00	0.05
piling	0.00	0.00	0.00	0.00	0.00	0.00
prefab	0.00	0.00	0.00	0.00	0.00	0.00
slope	0.00	0.00	1.78	0.00	0.00	0.53
substation	0.00	0.00	0.00	0.00	0.00	0.00
wood	0.00	0.00	0.00	0.00	0.00	0.00
Total	0.79	3.45	3.52	1.22	1.78	2.40

Notes: Share of the auctions of each category flagged as WTO-covered in the bid records, 2012–2022. The last row is the share over all categories; the last column covers all bureaus in the records, including bureaus not shown.

firm–auction observations for potential entrants without such a history, we impute K_i by the median among the other potential entrants of the auction whose capacity is known (each auction has at least 20 such potential entrants). The joint-bidding history n_{ijt} counts the auctions of the same bureau and work category in which i and j bid together as members of the same two-member JV. Two firms have *contact* if they previously bid in the same auction of the same bureau and work category, alone or as JV members, on an earlier date. Of the 6,680,124 pairs of potential entrants, 19.6 percent have contact; all repeat JVs and 129 of the 145 new JVs are between firms with contact.

Potential entrants. The potential entrants of an auction are firms listed, alone or as identified JV members, in an auction of the same bureau and work category on a strictly earlier date within the preceding 1,095 days, with that earlier project's engineer's estimate at least a tenth of the current project's estimate. The qualifying project must fall within this window. Historical records must have a valid date and a consistent positive engineer's estimate and refer to a competitive contract; they may include firms that withdrew or whose bids were invalid. Current participants are not added to the set if they fail the historical rule.

The window and size screen balance coverage of observed participants against the inclusion of firms whose experience is remote or confined to much smaller projects. In a comparison of rules applied to the same initial 547 auctions, a two-year window with the one-tenth size screen retains 384 auctions, compared with 396 for three years. With the three-year window, a one-quarter size screen retains 364 auctions, while one-twentieth and one-hundredth screens retain 398 and 400. Across the initial 547 auctions, the median number of potential entrants rises from 57 under the one-tenth screen to 85 and 445 under these looser screens. Thus, relaxing the screen substantially enlarges the choice sets while recovering few additional auctions. These diagnostics compare coverage, not estimates obtained under alternative rules. We also considered restricting each firm's past solo capacity to at least 20 percent of the current project's estimate, but that condition would exclude members of 14 observed JVs. We retain those options and let the capacity-shortfall terms affect their payoffs.

Formation sample. The formation sample starts from the 13 bureau–work categories and the 2006–2022 period represented in an initial sample of 450 identified JV bids in 196 auctions. For each bureau and work category, the smallest engineer's estimate in that initial sample sets the lower size threshold. We then select all competitive auctions above that threshold with a valid date and consistent engineer's estimate, giving 547 auctions. Requiring all listed firms to be identified leaves 477 auctions; requiring all project-specific JVs to have two members leaves 462. Requiring every listed firm to be in the historically constructed set

of potential entrants leaves 396. Additional checks for other JV types, duplicate appearances of a firm within an auction, conflicting engineer's estimates, and at least two potential entrants remove no further auctions in this sequence. The last restriction ensures that the observed participation outcome is defined over the same firms as the model's choice set. It excludes entire auctions rather than deleting uncovered firms and altering the observed competitive structure; the resulting sample is conditional on coverage by the historical rule.

A.4 Joint ventures by project size and bureau

Figure 1 uses the formation sample and the facility-specific reference amounts in Table 1. We identify steel bridges from the administrative work category and tunnels, dams, weirs, floodgates and utility tunnels from project names. We exclude projects whose facility is unclassified or ambiguous and those identified as maintenance or ancillary work. This assigns a reference amount to 322 of the 396 auctions; 198 lie within 50 percent of that amount. We use the engineer's estimate as the measure of project size and plot the share with a project-specific JV. The classification does not establish whether the procurer applied the usual guideline or a special-technology exception to a particular project. The pooled plot is descriptive: the mix of bureaus and facilities can also change around the reference thresholds.

Table A3 provides complementary detail by project size and bureau in the full bid panel. Kanto bridges follow the 2 billion yen threshold of the MLIT circular closely. Kyushu and Shikoku admit JVs on bridges from about 1.25 billion yen and 1 billion yen, while Kinki and Chubu rarely do at any size. For tunnels and river structures, JVs appear from about 2 to 2.5 billion yen in Kanto and Kyushu and from about 5 billion yen in Kinki. Because the bureaus' practices do not follow the circular closely except in Kanto bridges, the model admits JVs in all WTO-covered auctions except Kanto bridge auctions below 2 billion yen without a JV bidder.

Table A3. Share of auctions with a JV bidder by size and bureau (percent, number of auctions)

Panel A. Bridges; engineer's estimate in billion yen (lower bound of the bin)

Bureau	0.5–	0.75–	1.0–	1.25–	1.5–	1.75–	2.0–	2.25–	2.5–	3.0+
Kanto	0 (90)	0 (47)	0 (40)	0 (25)	13 (23)	23 (13)	100 (8)	84 (19)	100 (12)	94 (31)
Kyushu	2 (66)	12 (33)	23 (13)	86 (7)	80 (5)	50 (2)	–	–	100 (3)	80 (5)
Kinki	0 (53)	4 (24)	0 (14)	9 (11)	0 (3)	25 (4)	–	–	0 (1)	0 (4)
Shikoku	0 (21)	7 (15)	57 (7)	71 (7)	100 (2)	100 (1)	100 (1)	–	–	100 (1)
Chubu	0 (59)	0 (43)	0 (26)	0 (12)	17 (12)	0 (8)	9 (11)	100 (1)	17 (6)	50 (2)
All bureaus	0 (336)	3 (184)	6 (111)	17 (69)	24 (46)	21 (29)	48 (21)	86 (21)	74 (23)	82 (45)

Panel B. Tunnels and river structures; billion yen

Bureau	1–	2–	2.5–	3–	4–	5–	6–	8+
Kanto	0 (30)	42 (12)	56 (9)	20 (15)	50 (6)	90 (10)	86 (7)	100 (11)
Kyushu	4 (25)	18 (11)	62 (16)	50 (14)	80 (10)	80 (5)	80 (5)	–
Kinki	0 (43)	0 (13)	10 (10)	6 (18)	10 (10)	23 (13)	50 (4)	–
All bureaus	1 (136)	16 (49)	35 (46)	28 (58)	42 (45)	53 (34)	65 (20)	93 (15)

Notes: Full bid panel, 2006–2022, excluding negotiated contracts. Each cell gives the percentage of auctions with at least one project-specific JV bidder and, in parentheses, the number of auctions in the bin. Tunnels and river structures include weirs, sluices, drainage pumping stations and similar facilities.

A.5 Collusion screens

Collusion is a known concern in Japanese public procurement (Kawai and Nakabayashi, 2022; Chassang et al., 2022; Kawai et al., 2022). This subsection reports two screens applied to the bid records of the bureaus. Neither shows the patterns associated with collusion in the markets of our estimation sample, with one weak exception.

A.5.1 Marginal-bid screen

The first screen follows Kawai and Nakabayashi (2022). For losing bidders, it relates the percentage difference between the bidder's price and the winner's price to the percentage difference between their effective bids. Under competitive bidding, the relation should be smooth as a loser's effective bid approaches the winner's: which of two nearly equal bidders wins is as good as random. Under collusion, designated losers that could have won—for example, because of high technical scores—raise their prices just enough to lose, which produces a discontinuity as the difference in effective bids approaches zero. Figures A1 and A2 show the relation, estimated by local regression, by bureau and by year for the

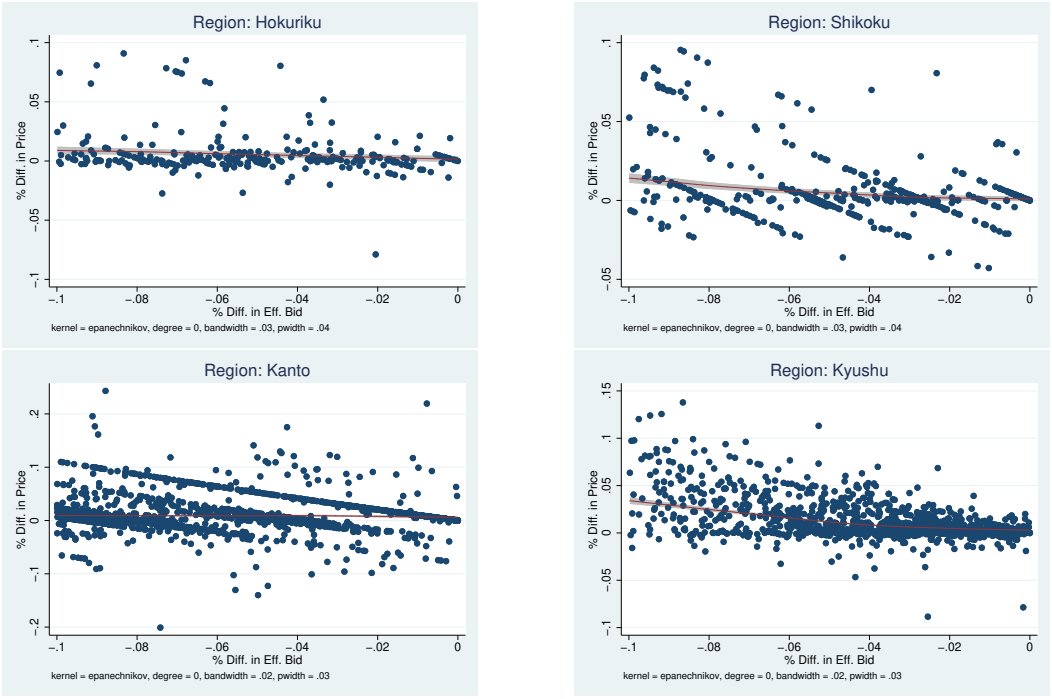


Figure A1. Marginal-bid screen by bureau (Hokuriku, Shikoku, Kanto, Kyushu).

full window of the data, 2006–2022. None of the panels shows a discontinuity at zero, in contrast with the smaller auctions studied in that literature. Our auctions are large, heavily scrutinized projects, and the firms that bid on them face strict compliance requirements.

A.5.2 Screening battery

The second approach runs a battery of five standard screens on bureau-by-work-category cells, using the full bid records (valid scoring auctions with a positive price, score and engineer's estimate):

S1 — Winner isolation. The gap between the winner's effective bid and price and the runner-up's.

S2 — Rebidding. Rank persistence across bidding rounds, and the asymmetry between the price reductions of winners and losers in re-auctioned lots.

S3 — Bid clustering and cover bids. The within-auction dispersion of normalized

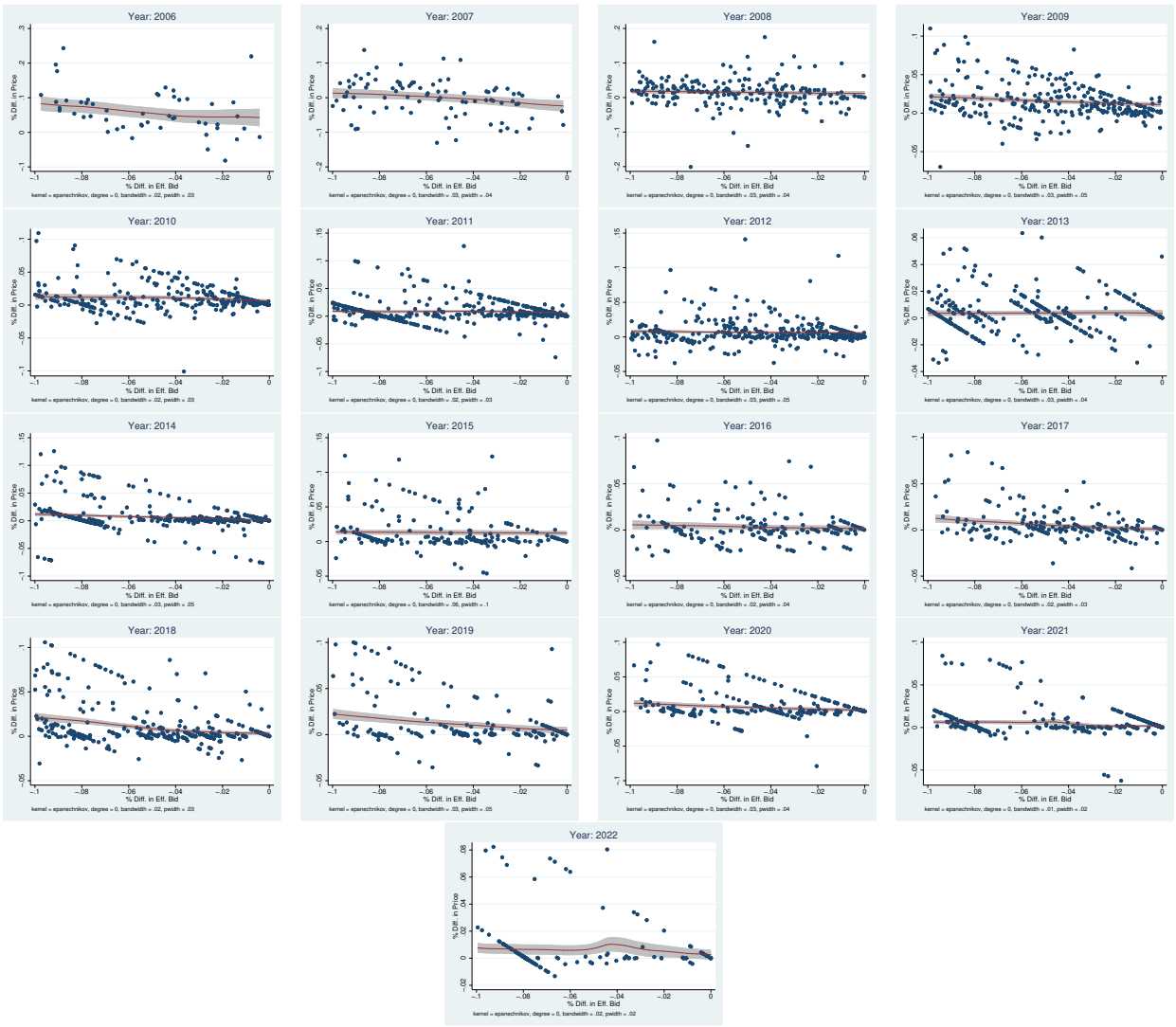


Figure A2. Marginal-bid screen by year, 2006–2022.

prices, losers clustered just above a much lower winner, and round-number bidding.

S4 — Reserve proximity. The share of winners bidding very close to the engineer's estimate, and the mass of bids just above the reference price.

S5 — Incumbency and rotation. The concentration of wins relative to participation, and the correlation between participation and wins.

Each screen averages the orientation-signed z-scores of its components, where z-scores are taken over the 56 cells with at least 500 valid scoring bids. The composite score of a cell

Table A4. Collusion screening battery: bureau-by-category cells of the estimation sample

Cell	S1	S3	S4	S5	**Composite**	**Rank/56**	**Flag**
Shikoku, bridges	−0.41	0.13	0.63	1.34	0.43	4	**YES**
Hokuriku, civil engineering	0.05	0.30	0.32	−0.05	0.26	14	no
Shikoku, civil engineering	0.26	0.61	−0.09	−0.09	0.18	19	no
Kinki, bridges	−0.14	−0.06	−0.08	0.35	0.02	26	no
Kanto, civil engineering	0.21	0.03	−0.29	−0.48	−0.00	28	no
Kanto, bridges	−0.17	0.32	0.01	−0.56	−0.10	32	no
Kinki, civil engineering	−0.16	−0.29	0.05	0.02	−0.11	33	no
Kyushu, civil engineering	−0.36	−0.45	0.07	−0.78	−0.12	36	no
Kyushu, bridges	−0.53	−0.32	0.18	−0.25	−0.23	43	no

Notes: Screen scores are orientation-signed z-scores relative to the 56 cells with at least 500 valid scoring bids; the composite is the mean of the available screens; a cell is flagged if its composite exceeds the 90th percentile of the 56 cells (0.35). S2 (rebidding) is omitted because it is undefined for the bridge cells (fewer than 20 re-auctioned lots) and enters the composite only where available. The cells of Chugoku bridges, Kanto air conditioning and Kyushu prestressed concrete, which contribute four auctions to the estimation sample, are not shown.

is the mean of its available screens, and a cell is flagged if its composite exceeds the 90th percentile of the 56 cells. Table A4 reports the cells of the markets of our sample.

Only Shikoku bridge construction is flagged, ranking 4th of 56 cells, and through the composite rule rather than any single extreme screen: no screen reaches a z-score of 2, and its composite is driven by moderate values of the rotation and reserve-proximity screens, while its winner-isolation screen is below average. The flag is a relative marker over the 56 cells, the cell is thin, and we are not aware of an adjudicated cartel case of the Japan Fair Trade Commission that involves this bureau and category in our period. Shikoku bridge construction contributes 10 of the 396 auctions of the estimation sample, and we keep it.

Appendix B. The Kanto Bridge Threshold

This appendix describes the case study in Section 4.2 in more detail and reports its robustness.

B.1 Design

We compare participation on either side of the 2 billion yen threshold, adjusting for project size and year. The data cover Kanto steel and prestressed-concrete bridge auctions in the full bid panel, 2006–2022. The main estimation window is 1.25–3.5 billion yen. For each auction-level outcome, we estimate

$$Y_t = a + \beta D_t + \theta \log(R_t/c) + \lambda_{y(t)} + u_t, \qquad D_t = \mathbf{1}\{R_t \geq c\}, \quad c = 2 \text{ billion yen}. \tag{16}$$

Here t indexes auctions, R_t is the engineer's estimate, and $y(t)$ is the auction year. The outcome Y_t is, in separate regressions, the number of solo bidders, JVs, independent bidders (solo bidders plus JVs), or participating firms (solo bidders plus JV members). The intercept a and year fixed effects $\lambda_{y(t)}$ allow participation levels to differ across years. The coefficient θ is the slope with respect to the log engineer's estimate, constrained to be the same on both sides of the threshold; β measures the discontinuous change at the threshold. The year effects are normalized to zero in the omitted year.

To examine whether the firms that form JVs switch from solo to joint bidding, we also estimate firm-level participation regressions. For each auction, potential entrants are firms that bid on a Kanto bridge within two years. We estimate separately for the 47 firms that join a JV in some Kanto bridge auction above the threshold and for the 106 other firms:

$$P_{it}^m = \alpha_i^m + \beta^m D_t + \theta^m \log(R_t/c) + \lambda_{y(t)}^m + \varepsilon_{it}^m, \qquad m \in \{\text{solo}, \text{JV}, \text{any}\}. \tag{17}$$

Here i indexes firms and P_{it}^m indicates whether firm i bids alone, bids as a JV member, or participates in either form in auction t. Each outcome and firm group has its own regression.

The firm fixed effect α_i^m absorbs persistent differences in participation across firms, while $\lambda_{y(t)}^m$ controls for year effects, normalized to zero in the omitted year. Thus β^m compares the same firms' participation across auctions on either side of the threshold, adjusting for project size and year. Standard errors in both specifications are clustered by auction.

For the firms that join JVs above the threshold, the displacement ratio is $\hat{d} = -\hat{\beta}^{\text{solo}}/\hat{\beta}^{\text{JV}}$. A ratio of one means that the decline in solo participation matches the increase in JV participation; a ratio of zero means that solo participation does not decline. Its confidence set collects the values d for which $\beta^{\text{solo}} + d\,\beta^{\text{JV}} = 0$ is not rejected at the 5 percent level, using the joint covariance of the two coefficients (a Fieller set). Of the 40 firms that join a JV above the threshold in the main window, 35 bid alone in auctions of that window below the threshold.

B.2 Robustness

Table A5 reports the estimates for alternative windows, cutoffs and periods. Moving the cutoff to 2.1 billion yen, excluding the auctions between 1.9 and 2.2 billion yen, and splitting the period all give displacement ratios between 1.34 and 1.75, with the same pattern: JV bidding rises, solo bidding falls by at least as much, and no clear change in overall participation is detected. Restricting the sample to steel bridges gives 0.85. The window from 1.5 to 3.0 billion yen gives 1.62 with less precise jumps. Windows that extend to about 5 billion yen, including the auctions of the formation sample, give smaller ratios (0.61 to 0.76) with wide confidence sets. The threshold is the one of the MLIT circular for bridges; we are not aware of other rules that change at 2 billion yen in Kanto bridge auctions, although the auctions just above and just below the threshold may differ in ways that the project-size and year controls in equations (16) and (17) do not capture. Few auctions lie near the threshold—eight between 2.0 and 2.25 billion yen—so the estimates are imprecise.

Table A5. Kanto bridge threshold: robustness

		Per auction			JV-prone firms			
Sample	Auctions	Solo	JVs	Indep. bidders	Solo	JV	Any	$\hat{d}$
Main: 1.25–3.5 bn yen	112	−3.73	+1.40	−2.33	−0.111	+0.078	−0.033	1.42
		(1.62)	(0.60)	(1.40)	(0.042)	(0.030)	(0.040)	
1.5–3.0 bn yen	75	−2.83	+0.52	−2.31	−0.052	+0.032	−0.020	1.62
		(1.98)	(0.68)	(1.68)	(0.044)	(0.034)	(0.043)	
1.0–4.0 bn yen	158	−4.50	+1.75	−2.75	−0.114	+0.089	−0.025	1.28
		(1.61)	(0.46)	(1.48)	(0.040)	(0.024)	(0.040)	
Cutoff at 2.1 bn yen	112	−3.53	+1.24	−2.29	−0.118	+0.067	−0.050	1.75
		(1.60)	(0.64)	(1.38)	(0.044)	(0.033)	(0.047)	
Excluding 1.9–2.2 bn yen	101	−5.19	+2.07	−3.12	−0.156	+0.116	−0.040	1.35
		(2.14)	(0.83)	(2.01)	(0.056)	(0.042)	(0.062)	
Steel bridges only	101	−3.24	+1.79	−1.45	−0.090	+0.105	+0.016	0.85
		(1.62)	(0.66)	(1.37)	(0.037)	(0.033)	(0.030)	
2006–2013	79	−3.40	+1.68	−1.71	−0.116	+0.081	−0.035	1.43
		(1.85)	(0.67)	(1.67)	(0.045)	(0.032)	(0.045)	
2014–2022	33	−3.66	+0.60	−3.06	−0.074	+0.055	−0.019	1.34
		(3.41)	(1.26)	(2.66)	(0.110)	(0.069)	(0.101)	
Formation sample, 1.58–5.1 bn yen	58	−1.77	+1.28	−0.48	−0.041	+0.067	+0.026	0.61
		(1.53)	(0.70)	(1.22)	(0.037)	(0.034)	(0.028)	

Notes: Jumps at the threshold (standard errors clustered by auction), estimated as in Table 3. Firm-level columns are changes in the probabilities of bidding alone, in a JV, and at all, for firms that join a JV in some Kanto bridge auction above 2 billion yen. $\hat{d}$ is the displacement ratio, minus the solo jump over the JV jump.

Appendix C. Formation Model

With equal sharing and strict rankings, the greedy procedure gives the unique stable partition conditional on beliefs. This appendix proves Proposition 1, compares our treatment of competition with that of Uetake and Watanabe (2020), and derives the pair-shock normalization. It does not establish uniqueness of beliefs.

C.1 Proof of Proposition 1

Fix an auction and drop the index t. Let $\mathcal{N}$ be the set of potential entrants and $\mathcal{E}$ the set of available pairs. Write $\bar{u}_i = \max\{0, u_i^S\}$ for the best option potential entrant i can take alone and $w_e = u_{ij}^J$ for the common payoff of the members of pair $e = (i,j) \in \mathcal{E}$. Because the payoffs are continuously distributed, with probability one: $u_i^S \neq 0$ for all i; the values w_e are distinct across pairs; and $w_e \neq \bar{u}_i$ for all e and i. We assume these properties throughout. Let

$$\mathcal{E}^* = \{e = (i,j) \in \mathcal{E} : \ w_e > \max\{\bar{u}_i, \bar{u}_j\}\}$$

be the set of available pairs that both members prefer to their individual options.

Lemma 1. *In every stable partition, every JV belongs to $\mathcal{E}^*$.*

Proof. If i and j bid together in a stable partition, condition (i) of Definition 1 gives $w_{ij} = U_i \geq \bar{u}_i$ and $w_{ij} = U_j \geq \bar{u}_j$, and the inequalities are strict with probability one. □

Uniqueness. We argue by induction on the number of potential entrants. If $\mathcal{E}^*$ is empty, Lemma 1 implies that no stable partition has a JV, and condition (i) requires every potential entrant to take its best individual option, which is unique because $u_i^S \neq 0$. This is also the outcome of the greedy procedure. Suppose now that $\mathcal{E}^*$ is not empty, and let $e_1 = (i,j)$ be the pair in $\mathcal{E}^*$ with the highest value. Take any stable partition and suppose that i and j do not bid together in it. Potential entrant i then either takes an individual option, so that $U_i \leq \bar{u}_i < w_{e_1}$, or bids with some $k \neq j$, in which case $(i,k) \in \mathcal{E}^*$ by Lemma 1 and

$U_i = w_{ik} < w_{e_1}$ by the definition of e_1. In both cases $U_i < w_{e_1}$, and likewise $U_j < w_{e_1}$. Since (i, j) is available, it blocks the partition, which contradicts condition (ii). Hence i and j bid together in every stable partition, which is also the first step of the greedy procedure.

Remove i and j. The restriction of a stable partition to $\mathcal{N} \setminus \{i, j\}$ is stable in the problem with potential entrants $\mathcal{N} \setminus \{i, j\}$ and the available pairs among them: condition (i) involves each potential entrant alone, and a pair that blocked the restriction would block the original partition. The set of mutually acceptable pairs of the smaller problem is $\mathcal{E}^*$ restricted to $\mathcal{N} \setminus \{i, j\}$, because whether a pair is acceptable depends only on its own value and on its members' individual options; so the remaining steps of the greedy procedure are exactly the greedy procedure of the smaller problem. By the induction hypothesis, the smaller problem has a unique stable partition, equal to the outcome of its greedy procedure. Therefore every stable partition of the original problem consists of the JV (i, j) and the greedy outcome of the smaller problem, that is, of the outcome of the greedy procedure.

Existence. It remains to show that the outcome G of the greedy procedure is stable. Condition (i) holds: every JV of G belongs to $\mathcal{E}^*$, and every other potential entrant takes its best individual option. For condition (ii), suppose that an available pair (k, l), not a JV of G, satisfies $w_{kl} > U_k$ and $w_{kl} > U_l$. Since $U_k \geq \bar{u}_k$ and $U_l \geq \bar{u}_l$ under G, the pair belongs to $\mathcal{E}^*$. The procedure stops only when no pair of $\mathcal{E}^*$ has two unmatched members, so at least one of k and l is matched in G. Consider the first step at which one of them is matched, say k with m, so that $U_k = w_{km}$. At that step neither k nor l was matched, so (k, l) was among the pairs from which the procedure chose the one with the highest value, and therefore $w_{km} > w_{kl}$. This contradicts $w_{kl} > U_k$. Hence G is stable. □

Remarks. The argument is the one for stable roommates problems with globally ranked pairs (Abraham et al., 2008), extended to individual outside options. Existence also follows from the general result that aligned preferences guarantee a stable outcome (Pycia, 2012); uniqueness comes from the strict common ranking of the pairs. It does not rely on the normality of the shocks or on the form of the payoffs, only on the members of a JV receiving

the same payoff. Unequal sharing rules break it: if member i received a share $\theta \neq 1/2$ of the JV's value, firms would no longer rank JVs in the same order, and stable partitions could fail to be unique.

C.2 Comparison with Uetake and Watanabe

Like our model, Uetake and Watanabe (2020, Sections 3.1–3.2) jointly model participation and partner choice, allow firms to enter alone or stay out, and require mutual consent for a merger. Their stable outcome is defined conditional on *estimation functions*, which map each firm's choice to a conjectured number of operating firms. They use myopic estimation functions: other firms' behavior is held fixed when a firm considers a deviation, while its own choice can affect the number of operating firms. In estimation, they require these conjectures to be consistent with the observed number of operating firms. Their approach therefore also distinguishes stability conditional on expectations from the requirement that those expectations be consistent with outcomes.

Our treatment of competition differs in what firms hold fixed and what consistency requires. A firm evaluates a solo or JV option using the state-conditional distributions of rival numbers in H, and holds these distributions fixed when considering a deviation. The payoff u^J_{ijt} used to evaluate a possible JV therefore does not change with the bidder composition that would result from that deviation, including the fate of any former partners. The condition $H = \Gamma(H)$ requires these beliefs to match the distribution of rival numbers generated across realizations of the shocks and availability. It does not require beliefs to equal the realized bidder counts in each auction or to incorporate the change in competition following each individual or pair deviation. Competition affects formation through expected bidding profits and the consistency of beliefs, while the conditional formation problem retains fixed option payoffs.

The matching structure also differs. Uetake and Watanabe match potential entrants with incumbents, allowing the two parties to value the same merger differently. Their model can have multiple stable outcomes, and they use the extremal stable allocations to bound payoffs

for estimation. Here firms can pair with others from the same set of potential entrants, and both members receive the same net payoff from any given JV. Together with strict rankings, this common-payoff restriction yields the unique stable partition in Proposition 1, conditional on H. It removes the need to select among stable partitions at given beliefs, but it does not establish uniqueness of the equilibrium beliefs themselves.

C.3 Normalization of the pair shocks

Remark 1 explains why we adjust the pair shocks for the number of potential partners. To see the effect without an adjustment, consider a firm with m available partners and hold all observed payoffs fixed. If each partner supplies an independent standard normal draw, the largest draw satisfies

$$\Pr\left(\max_{1\le j\le m} \zeta_j \le z\right) = \Phi(z)^m,$$

where Φ is the standard normal distribution function. As m increases, this probability falls for every finite z: the largest draw shifts upward even though the distribution of each individual draw is unchanged. For illustration, the probability of at least one draw above two is about 21 percent with ten partners and 90 percent with one hundred. Thus a larger candidate set can make joint bidding more attractive solely by increasing the chance of an exceptionally favorable shock.

The concern is related to the market-size normalizations in Menzel (2015), who adjusts outside options and the scale of idiosyncratic utilities along a growing sequence of matching markets. These adjustments prevent the extremes of a growing set of alternatives from dominating the outside option and keep observed and unobserved payoff components relevant for choices. Our adjustment uses the same extreme-value reasoning to align the location and spread of the best pair shock across finite auctions at a fixed reference number of partners.

For $m \ge 2$ independent standard normal draws, the usual quantile and inverse-hazard normalizations are

$$b_m = \Phi^{-1}(1 - 1/m), \qquad a_m = \frac{1 - \Phi(b_m)}{\phi(b_m)} = \frac{1/m}{\phi(b_m)},$$

where ϕ is the standard normal density. They give the Gumbel limit

$$\Pr\left(\frac{\max_{1\le j\le m}\zeta_j - b_m}{a_m} \le z\right) \longrightarrow \exp\{-e^{-z}\} \qquad \text{as } m \to \infty;$$

see Gasull, Jolis and Utzet (2015) for these normalizing constants and the normal extreme-value approximation. We use b_m and a_m as approximate location and scale parameters for the maximum.

To apply this approximation, let $d_t = 2(H_t + p_0 U_t)/A_t$ be the average expected number of available partners per potential entrant in auction t, where A_t is the number of potential entrants, H_t and U_t are the numbers of pairs of potential entrants with and without contact, and p_0 is a fixed reference value of the availability probability from a preliminary estimate. Approximating the number of independent draws by d_t, we use the normal extreme-value location $b_t = \Phi^{-1}(1 - 1/d_t)$ and scale $a_t = (1/d_t)/\phi(b_t)$. Let d^* be the median of d_t across auctions, and let b^* and a^* be the same functions evaluated at d^*. We set

$$r_t = \frac{a^*}{a_t}, \qquad l_t = r_t b_t - b^*.$$

The approximate location and scale of the largest normalized draw are then $r_t b_t - l_t = b^*$ and $r_t a_t = a^*$. The constants are fixed before estimation; across the 396 auctions, r_t ranges from 0.83 to 1.16 and l_t from -0.73 to 0.84.

At the reference, $r_t = 1$ and $l_t = 0$, so q is the standard deviation of the pair payoff shock relative to the solo shock. In another auction, $q\nu_{ijt}$ has mean $-ql_t$ and standard deviation qr_t. The main specification fixes $q = 1$, leaving mean $-l_t$ and standard deviation r_t. Applying the same r_t and l_t to every pair preserves the equality of the two members' payoffs. The adjustment aligns the location and scale of the normal extreme-value approximation, evaluated at average expected rather than firm-specific realized partner counts; it does not make the finite-sample distributions of the maxima or the outcomes of the matching game identical. Firms still differ in their observed payoffs and available partners, and a preferred partner must also agree to the match.

Appendix D. Bidding-Stage Estimation

The formation stage uses bidding strengths and expected auction profits recovered from the bid records. This appendix gives the estimation details behind these inputs, including cost bounds from floor and reserve bids, and then assesses the cost-strength estimator in a Monte Carlo experiment.

D.1 Rival bid distributions

Under the bidding environment described in Section 5.1, the winning probability $W_t(y) = F_t^J(y)^m F_t^S(y)^n$ treats the log effective bids of rivals as independent draws, with one distribution for each mode. If bidders observed the identities of their rivals, the bid of each rival would follow its own distribution given its strength, the winning probability would be the product of these distributions, and h_t would be the sum of their reverse hazards. The distributions of rival bids must account for the common score and cost conditions (μ_t, δ_t) that bidders know. A distribution that averages over these conditions misstates the winning probability, and adding the cost effect δ_t to the recovered costs afterwards cannot repair the inversion, because h_t enters it nonlinearly.

Assumption 2 approximates an equilibrium object rather than specifying a primitive: it represents the distributions of rival bids by a parametric shape, under which effective bids are log-normal. The shift ω_t summarizes the effects of the common score and cost conditions (μ_t, δ_t) on mean log effective bids. Allowing for these conditions, which bidders know, follows the treatment of unobserved auction heterogeneity by Krasnokutskaya (2011). We estimate ω_t from bids rather than calculate it from a known function of μ_t and δ_t. Its independence from the numbers of bidders lets the effects of these numbers be estimated from comparisons across auctions. It rules out association between these numbers and the auction conditions (μ_t, δ_t) represented by ω_t, including association induced by participation responding to expected profits.[30]

[30] Parametric bid distributions with unobserved auction heterogeneity are used, for example, by Athey, Levin and Seira (2011), who specify Weibull bid distributions with Gamma-distributed auction heterogeneity;

We specify $\mu^d(X_t, m, n) = x_t'\beta^d + p^d(\log R_t, m, n)$. The vector x_t contains market effects, a quadratic trend in the fiscal year and the log floor ratio $\log \tau_t$, and p^d is a quadratic polynomial in $(\log R_t, m, n)$ with all interactions. In the bidding stage, a market is a bureau-by-work-category cell of the bid panel. The log scale $\log \sigma_t^d$ is the sum of a market effect and a quadratic trend in the fiscal year, because the dispersion of bids within auctions fell over the period, and $\log \sigma_{\omega,t}$ is quadratic in the year. The coefficients and scales differ between solo bidders and JVs, whose types and strengths come from different distributions, while ω_t is common to both.

Market effects in the mean are estimated for cells with at least 30 solo bids (50 JV bids) and pooled within the bureau (within the work category for JVs) otherwise; scale effects are estimated for cells with at least 300 solo bids (150 JV bids), and pooled otherwise. The trends are specific to the mode. When the floor is not recorded, the log floor ratio is replaced by its bureau-by-year median and an indicator for the missing floor is included. Because bidders do not observe their rivals' identities, μ^d contains no characteristics of particular rivals.

The estimation sample consists of bids with a positive score and a first recorded price between one half and the full engineer's estimate, in scoring auctions of fiscal years 2006 to 2022 with at least two such bids. We drop auctions with such a bid by a bidder of unidentified mode, as well as 11 auctions with a recorded score below 90, below the 100 points that every qualified bidder receives. The sample contains 267,060 bids (261,400 solo and 5,660 JV) in 48,826 auctions. M_t and N_t count the JV and solo bids in this sample. Bids at the floor and at the reserve enter at their observed values, because they are draws from the bid distributions. Listed bidders that withdraw or submit no valid price enter the profit calculation through the usable-bid probabilities in Appendix D.5.

The model of Assumption 2 is a linear model with an auction random effect and error variances specific to the mode, the market and the year. We estimate all its parameters jointly

Jofre-Bonet and Pesendorfer (2003) also estimate a parametric (Weibull) bid distribution before inverting first-order conditions. Log-normal distributions of private values or costs are used by Laffont, Ossard and Vuong (1995), Hong and Shum (2003) and Bhattacharya, Roberts and Sweeting (2014).

on the pooled solo and JV bids. Its log likelihood has a closed form, auction by auction, and we maximize it with the coefficients of the means concentrated out by generalized least squares and an analytic gradient in the variance parameters. Given the estimates, the posterior distribution of ω_t given a set $\mathcal{B}$ of bids of auction t is normal with mean and variance

$$\hat{\omega}_{t,\mathcal{B}} = \frac{\sum_{l\in\mathcal{B}}(y_{lt} - \hat{\mu}_{lt})/\hat{\sigma}_l^2}{1/\hat{\sigma}_{\omega,t}^2 + \sum_{l\in\mathcal{B}} 1/\hat{\sigma}_l^2}, \qquad v_{t,\mathcal{B}} = \frac{1}{1/\hat{\sigma}_{\omega,t}^2 + \sum_{l\in\mathcal{B}} 1/\hat{\sigma}_l^2},$$

where $\hat{\mu}_{lt} = \hat{\mu}^{d(l)}(X_t, M_t, N_t)$ and $\hat{\sigma}_l = \hat{\sigma}_t^{d(l)}$. The profit tables of the formation auctions use all bids of the auction. For the pairwise estimator, $\mathcal{B}$ excludes both bids of the pair. Excluding only the bidder's own bid would keep its residual out of its own winning probability but put each bidder's bid into the other's recovered cost, so that the two trimmed values would no longer be exchangeable. In auctions with two bids, the leave-pair-out posterior is the prior. To reflect this uncertainty in the inversion, the costs of a pair are recovered from the predictive distribution of rival bids, normal with mean $\hat{\mu}^d + \hat{\omega}_{t,\mathcal{B}}$ and variance $\hat{\sigma}_t^{d\,2} + v_{t,\mathcal{B}}$. The types use the posterior that excludes the bidder's own bid, with the same predictive variance.

At the estimates, the within-auction scale is 0.057 for solo bids and 0.059 for JV bids at the median, and the standard deviation of the shift is 0.046 in 2014. The expected-profit calculations evaluate the estimated distributions at every number of rivals, holding the auction's estimated shift fixed.

D.2 Cost bounds from boundary bids

A bid at the floor or the reserve reveals a bound on cost rather than its value. Bids cluster just above the recorded floor rather than at it, and some lie below it: of the 232,909 bids with a recorded floor, 2,010 are within one yen of the floor price, 72,283 are below $1.01\,\tau_t R_t$ and 4,404 are below the floor. We therefore treat every bid below $(1+\epsilon)\tau_t R_t$, with $\epsilon = 0.01$,

as a floor bid. Such a bid reveals

$$\tilde{c}_{it} \leq U_{it}, \qquad U_{it} = (1+\epsilon)\tau_t \left(1 - \frac{1}{\hat{h}_t(y_{it}^{\epsilon})}\right), \tag{18}$$

where $y_{it}^{\epsilon} = \log s_{it} - \log((1+\epsilon)\tau_t)$ is the log effective bid at the top of the band, and $h_t = W_t'/W_t$. The bound is evaluated at the top of the band, so the calculation does not require a bid classified as a floor bid to equal the recorded floor price. The derivation below establishes the bound for bids recorded only as inside or outside this band.

Omitting the positive scale R_t, the derivative of bidding profit is

$$\frac{d}{dy}\{(se^{-y} - \tilde{c})W_t(y)\} = W_t(y)\{(se^{-y} - \tilde{c})h_t(y) - se^{-y}\}.$$

Under Assumption 2, the reverse hazard of each normal distribution of rival bids decreases with y, so $h_t(y)$ decreases with y whenever there is at least one rival. Dividing the profit derivative by the positive quantity $W_t(y)se^{-y}$, its sign is that of

$$\Psi_t(y, \tilde{c}) = \left(1 - \frac{\tilde{c}e^{y}}{s}\right) h_t(y) - 1, \qquad \frac{\partial \Psi_t}{\partial y} = -\frac{\tilde{c}e^{y}}{s} h_t(y) + \left(1 - \frac{\tilde{c}e^{y}}{s}\right) h_t'(y) < 0$$

where the price exceeds cost. Thus profit can change from increasing to decreasing at most once as the log effective bid rises, and lowering cost raises Ψ_t at every y. For any level $\bar{y}$, the costs for which the optimal log effective bid is at least $\bar{y}$ (a price at most $R_t s e^{-\bar{y}}$) are therefore those with $\Psi_t(\bar{y}, \tilde{c}) \geq 0$, that is, $\tilde{c} \leq se^{-\bar{y}}(1 - 1/h_t(\bar{y}))$. With $\bar{y} = y^{\epsilon}$ this gives (18); the argument does not use the floor itself, so it also holds if bids below the floor are allowed but penalized, since a penalty below the floor only makes such bids less attractive. At the reserve, the log effective bid is at its lower bound $y^R = \log s$, and the derivative must be nonpositive, giving

$$\tilde{c} \geq L_{it}, \qquad L_{it} = 1 - \frac{1}{\hat{h}_t(y_{it}^R)}. \tag{19}$$

A bid above the reserve is invalid and does not enter the estimation of the distributions of rival bids, but it reveals that the bidder's cost is at least this lower bound. Scores for these

bids are recorded only in Kyushu; elsewhere we use the median log score of the auction.

The result uses positive costs and at least one rival; a lone bidder has $h_t = 0$ and no inversion. With ties or atoms, derivatives may not describe the best response. Let $W_{it}(y)$ then denote the actual winning probability, including the tie-breaking rule. The admissible costs are those satisfying

$$(s_{it}e^{-y_{it}} - \tilde{c})W_{it}(y_{it}) \geq (s_{it}e^{-y} - \tilde{c})W_{it}(y) \quad \text{for every feasible } y, \tag{20}$$

which are linear in cost and define its admissible interval.

The bound U_{it} is not positive when $\hat{h}_t(y^{\epsilon}_{it}) \leq 1$, which happens when the top of the floor band lies far in the upper tail of the distribution of rival bids: no positive cost then leads to a floor bid. For an interior bid, this is not a contradiction: the bidder simply has no lower censoring point. A floor bid with $U_{it} \leq 0$, or an interior bid with $\hat{h}_t(y_{it}) \leq 1$, which implies a nonpositive cost, cannot be rationalized given $\hat{F}_t$; we count these as violations and drop the pairs that contain them. With the posterior that excludes the bidder's own bid, 1,190 of the 156,800 interior solo bids imply a nonpositive cost and 485 of the 70,820 floor bids a nonpositive bound, while 8,872 interior bids have no lower censoring point.

D.3 Cost and score strengths

D.3.1 Cost strengths

We estimate the cost strengths and the score-cost function in equation (7) by comparing solo bidders in the same auction. The function q_g is quadratic in the log score and normalized to zero at the median score. Its linear coefficient is specific to the market and to the periods 2006–2007 and 2008–2022, between which the scale of scores changed; its quadratic coefficient is specific to the market. Markets with fewer than 3,000 solo bids are pooled within the work category, or into one group. The cost strengths are normalized to mean zero across firms, the mean of their prior. Identification requires that firms be connected through common auctions, that scores vary within auctions and that comparisons remain after trimming. JV

bids are not used to estimate the strengths. Because types are independent of the formation-stage shocks, observed bids are not selected on their residuals, and bids in the formation auctions can also be used.

Without censoring, the cost equation would be a regression with firm and auction effects (Abowd, Kramarz and Margolis, 1999). Floor bidders, however, are the low-cost draws and bidders at or above the reserve are the high-cost draws, so dropping them or treating their bounds as costs distorts the comparison. We instead compare two solo bidders after moving their recovered log costs into the range in which both would be observed. This is the symmetric trimming of Honoré (1992), extended to censoring on both sides with observation-specific censoring points by Alan et al. (2014), with the bidders of an auction in the role of the time periods of an individual.

For a solo bidder i in auction t, let $\underline{z}_{it} = \max\{\log U_{it}, \log(0.8\,\tau_t)\}$ and $\overline{z}_{it} = \log L_{it}$ be its lower and upper censoring points, which are known for every bidder, and let z_{it} be its recovered log cost $\max\{\log \tilde{c}_{it}, \log(0.8\,\tau_t)\}$ if it bids in the interior, $\underline{z}_{it}$ if it bids in the floor band and $\overline{z}_{it}$ if it bids at or above the reserve. The point $\log(0.8\,\tau_t)$ is an additional, artificial censoring point that does not depend on the bids. It limits the influence of bids far in the upper tail of the estimated distribution of rival bids, whose recovered costs can be close to zero: 2.4 percent of the 156,800 interior solo bids have recovered costs below $0.8\,\tau_t$. For two solo bidders i and j in the auction, let $\Delta = \Delta_{ijt}(\theta) = q_g(\log s_{it}) - q_g(\log s_{jt}) - (\alpha_i - \alpha_j)$ be the difference between their cost indices, where θ collects the strengths and the coefficients of q_g. In the units of bidder j, bidder i's latent log cost is $z^*_{it} - \Delta$ and is observed between $\underline{z}_{it} - \Delta$ and $\overline{z}_{it} - \Delta$. Moving both bidders' values into the common observable range gives the trimmed difference

$$m_{ijt}(\Delta) = \min\{\max\{z_{it} - \Delta, \underline{z}_{jt}\}, \overline{z}_{jt}\} - \min\{\max\{z_{jt}, \underline{z}_{it} - \Delta\}, \overline{z}_{it} - \Delta\}$$

for $\underline{z}_{it} - \overline{z}_{jt} < \Delta < \overline{z}_{it} - \underline{z}_{jt}$, when the ranges overlap, and $m_{ijt}(\Delta) = 0$ otherwise. At the true parameters, the two terms are the values of $v_{it} = \delta_t + u_{it}$ and v_{jt}, both censored to the same interval. Under the assumptions on types in Section 5.1 and Assumption 1(a), they

are exchangeable given the bidders' scores, strengths and censoring points, so

$$E\big[m_{ijt}(\Delta_{ijt}(\theta_0))\,(w_{it} - w_{jt})\big] = 0, \tag{21}$$

where w_{it} collects the regressors of bidder i's cost index: its firm indicator and the terms of $q_g(\log s_{it})$. The common cost condition δ_t enters both terms equally and does not need to be estimated. For two interior bidders whose values lie in the common range, the trimmed difference is the difference of their residuals. A pair of floor bidders carries no information, and a floor bidder is compared with another bidder only above its bound. In the implementation, the estimated bid shift $\hat{\omega}_t$ enters both recovered costs nonlinearly, so exchangeability holds only approximately even when both bids are excluded from its estimation and its posterior variance is included. The Monte Carlo experiment in Appendix D.6 includes these steps.

The moment condition is the first-order condition of the criterion

$$Q(\theta) = \sum_t n_t^{-1} \sum_{i<j} R_{ijt}(\Delta_{ijt}(\theta)), \qquad R'_{ijt} = -2m_{ijt}.$$

Writing z_i, z_j, $\underline{z}_i$, $\overline{z}_i$, $\underline{z}_j$, $\overline{z}_j$ for the values of the pair and $(x)_+ = \max\{x, 0\}$,

$$R_{ijt}(\Delta) = (\Delta - z_i + z_j)^2 - (z_i - \overline{z}_j - \Delta)_+^2 - (\Delta - z_i + \underline{z}_j)_+^2 - (\Delta - \overline{z}_i + z_j)_+^2 - (\underline{z}_i - z_j - \Delta)_+^2$$

for Δ inside the overlap interval, and R_{ijt} is constant outside it. Without upper censoring ($\overline{z} = +\infty$), R_{ijt} is the convex trimmed least squares criterion of Honoré (1992) with observation-specific censoring points: it equals $(z_i - z_j - \Delta)^2$ when both values lie in the interior of the common range. With upper censoring, it is the criterion of Alan et al. (2014) for two-sided censoring. It is piecewise quadratic, with negative curvature near the ends of the overlap interval, and flat outside it. Because each $\Delta_{ijt}(\theta)$ is linear in θ, the criterion is a sum of such functions of linear indices.

To the criterion we add the penalty $\lambda \sum_i \alpha_i^2$. Consider the quasi-posterior proportional

to $\exp\{-Q(\theta)/(2\sigma_Q^2)\}$ times a normal prior for strength with variance σ_α^2, whose variance we estimate by empirical Bayes (Morris, 1983). Its mode is the penalized estimator with $\lambda = \sigma_Q^2/\sigma_\alpha^2$, and, approximating the criterion by a quadratic function, the posterior variance of α_i is $V_i = 2\sigma_Q^2[(H + 2\lambda D)^{-1}]_{ii}$, where H is the Hessian of the criterion over the pairs in its positive-curvature region and D is diagonal with ones on the strength coordinates and zeros on the coefficients of q_g. With interior bids only, the weighted criterion is the within-auction sum of squared residuals, and with $\sigma_Q^2 = \sigma_u^2$ these are the posterior mean and variance of strength when strength and residuals are normal. Trimming makes the compared values less dispersed than the residuals, so that $\sigma_Q^2 = \sigma_u^2$ overstates the noise in the criterion and shrinks the estimates too much. We calibrate σ_Q^2 by the information equality instead: the outer product S of the scores summed within auctions should equal $2\sigma_Q^2 H$, and we set $\sigma_Q^2 = \sum_i S_{ii}/(2\sum_i H_{ii})$ over the firms with comparisons. The empirical Bayes update of the prior variance is $\hat{\sigma}_\alpha^2 = F^{-1}\sum_i(\hat{\alpha}_i^2 + V_i)$ over the F firms with comparisons. Because most firms have few comparisons, iterating this update converges slowly. After each fit at λ, we therefore compute the fixed point of the update under the quadratic approximation, in which the fit at λ' solves $(H+2\lambda' D)\theta(\lambda') = (H+2\lambda D)\hat{\theta}$ with D selecting the strengths, refit there, and stop when the exact update changes λ by less than 2 percent. At the estimates, the noise scale is $\sigma_Q = 0.075$, against the bid-weighted residual standard deviation 0.093 of the residual model in Appendix D.4.1, the estimated standard deviation of strength is $\hat{\sigma}_\alpha = 0.043$ and $\lambda = 3.04$; two fits were needed. The penalty also keeps the estimates of firms with few bids out of the flat regions of the criterion. For the 316 firms without positive-curvature comparisons at the estimate, the criterion is flat in their own strength near the estimate, which is the point of the flat set closest to zero, and their posterior variance is the prior variance; for the 183 firms outside the largest connected set of the comparison graph, the prior also sets the level of their component. We minimize the penalized criterion by Newton steps on its positive-curvature part followed by quasi-Newton iterations with the analytic gradient. The criterion uses 1,086,696 pairs of 7,728 firms in 40,150 auctions. We estimate the sampling variance of the score-function coefficients with the sandwich formula, using the

penalized Hessian over the pairs in the positive-curvature region and the outer product of the scores summed within auctions.

D.3.2 Score strengths

Scores are not censored, so their persistent firm component can be estimated by within-auction least squares. Let $\mathcal{S}$ be the usable solo bids with an identified bidder in auctions with at least two such bids: 261,065 bids of 7,499 firms in 48,447 auctions. The criterion $\sum_t n_t^{-1} \sum_{i<j} \{(\log s_{it} - \log s_{jt}) - (\sigma_i - \sigma_j)\}^2 + \lambda_\sigma \sum_i \sigma_i^2$ is the within-auction sum of squared residuals of $\log s_{it} = \mu_t + \sigma_i + e_{it}$ plus the penalty. With $\lambda_\sigma = \sigma_e^2/\sigma_\sigma^2$, its minimizer is the posterior mean of σ when $\sigma_i \sim N(0, \sigma_\sigma^2)$ and $e_{it} \sim N(0, \sigma_e^2)$, with posterior variance $2\sigma_e^2[(H_\sigma + 2\lambda_\sigma I)^{-1}]_{ii}$, where H_σ is the Hessian of the sum of squares. Because the model is linear and the scores are not censored, we estimate σ_e^2 and σ_σ^2 by the expectation–maximization updates of restricted maximum likelihood, $\sigma_e^2 \leftarrow \{\mathrm{RSS} + \sigma_e^2 \,\mathrm{tr}[H_\sigma(H_\sigma + 2\lambda_\sigma I)^{-1}]\}/(N - T)$ and $\sigma_\sigma^2 \leftarrow F^{-1} \sum_i \{\hat{\sigma}_i^2 + 2\sigma_e^2[(H_\sigma + 2\lambda_\sigma I)^{-1}]_{ii}\}$, computed with one eigendecomposition of H_σ. The estimates are $\hat{\sigma}_\sigma = 0.039$ and $\hat{\sigma}_e = 0.043$, so persistent differences in scores are about as large as their variation within firms. The score strengths are normalized to mean zero across firms, and the score level μ_t is the mean over the auction's solo bidders of $\log s_{it} - \hat{\sigma}_i$. We use all usable solo bids for this average, setting the strength to zero for firms without an estimate. For a target auction with no usable solo bid, we use the median estimated score level among auctions of the same market and year. A JV's score is that of its stronger member under a_{ij}. Regressing a JV's log score relative to $\hat{\mu}_t$ on the score strengths of its members gives 0.86 (0.33) on the higher score strength and -0.28 (0.25) on the lower score strength for the 438 bids of project-specific JVs whose members are both estimated, with standard errors clustered by auction. Cost and score strengths are nearly uncorrelated across firms: the correlation of $\hat{\alpha}_i$ and $\hat{\sigma}_i$ among the potential entrants of the formation sample is 0.08.

D.4 Cost residual distributions and types

D.4.1 Cost likelihood with boundary bids

We extend the cost likelihood in Section 6.1 to bids that reveal only cost bounds. We retain the normal residual distribution in Assumption 3 and the normal random effect $\delta_t \sim N(x_t'\gamma, \sigma_{\delta,t}^2)$. The covariates x_t include market effects, $\log R_t$, $\log \tau_t$ and a quadratic trend, and $\log \sigma_{\delta,t}$ is quadratic in the year. Given $\hat{\theta}$, an interior bid contributes the density of its recovered log cost, a floor bid the probability that cost lies below its bound and a bid at or above the reserve the probability that cost lies above its bound. The residual variances and the distribution of the common cost conditions δ_t maximize

$$\mathcal{L}_c = \prod_t \int \left[\prod_{i \in \mathcal{I}_t} \frac{1}{\varsigma_{it}} \phi\left(\frac{z_{it} - m_{it}}{\varsigma_{it}} \right) \prod_{i \in \mathcal{F}_t} \Phi\left(\frac{\underline{z}_{it} - m_{it}}{\varsigma_{it}} \right) \times \prod_{i \in \mathcal{U}_t} \left\{ 1 - \Phi\left(\frac{\overline{z}_{it} - m_{it}}{\varsigma_{it}} \right) \right\} \right] \frac{1}{\sigma_{\delta,t}} \phi\left(\frac{\delta_t - x_t'\gamma}{\sigma_{\delta,t}} \right) d\delta_t, \tag{22}$$

with $m_{it} = \delta_t + \hat{q}_g(\log s_{it}) - \hat{\alpha}_i$ and $\varsigma_{it}^2 = \sigma_{u,g}^2 + V_i$, where $\mathcal{I}_t$, $\mathcal{F}_t$ and $\mathcal{U}_t$ are the interior bids, the bids at the floor or below the artificial point, and the bids at or above the reserve of auction t, and V_i is the posterior variance of firm i's strength. Integrating over δ_t as a random effect avoids the downward bias of the residual variance that free auction effects would cause with few bids per auction, and gives a posterior distribution of δ_t for every auction, including those whose bids are all at the floor. We use the bids of the firms whose strength is identified through comparisons, evaluate the integral by Gauss–Hermite quadrature and maximize with an analytic gradient. The residual standard deviation is 0.081 in the largest market, Kyushu civil engineering. Assumption 3 is used for this residual model and for the costs beyond the censoring bounds when constructing types; it plays no role in the estimates of strength.

D.4.2 Recovered types

A solo type is drawn from a source bid by firm i in auction t of the same market and period as follows: δ_t from the normal approximation to its posterior; the bidder's cost strength from $N(\hat{\alpha}_i, V_i)$; and the residual as $u = z_{it} + \alpha_i - \delta_t - \hat{q}_g(\log s_{it})$ for an interior bid, or from $N(0, \hat{\sigma}^2_{u,g})$ truncated above at $\underline{z}_{it} + \alpha_i - \delta_t - \hat{q}_g(\log s_{it})$ for a floor bid (below at the corresponding upper point for a bid at the reserve). Drawing δ_t and α_i propagates uncertainty in the source cost condition δ_t and cost strength α_i into the recovered residuals. The type records $e = \log s_{it} - \hat{\mu}_t - \hat{\sigma}_i$ and u, and for firm k in a target auction t' of the same market

$$\begin{aligned} &\log s_{kt'} = \hat{\mu}_{t'} + \sigma_k + e, \\ &\log \tilde{c}_{kt'} = \delta_{t'} + \hat{q}_g(\log s_{kt'}) - \alpha_k + u, \qquad c_{kt'} = R_{t'} \tilde{c}_{kt'}, \end{aligned} \tag{23}$$

with $\delta_{t'}$ at its posterior mean. At the drawn source values of δ_t and α_i, a type satisfies its recovered cost equality or censoring bound for the distribution of rival bids used in the inversion. Using target posterior means gives a plug-in prediction and does not guarantee reproduction of the source bid. Each bank has 128 types. Solo banks are formed by market and period (2006–2007 and 2008–2022); when a market has fewer than 200 source bids in a period, the bank pools the markets of the same work category. JV banks are formed by JV market group, pooling across periods. JV types come from the JV residual model for a pair of firms i and j: $\log \tilde{c} = \delta_t + \kappa^J_h + q^J(\log s - m^J) - \max\{\alpha_i, \alpha_j\} + u^J$, with a level κ^J_h for each of four JV market groups (Kyushu civil engineering, other civil engineering, bridges, other), a common slope, a normal residual with its own variance, the stronger members' strengths, and δ_t integrated over its posterior from the solo bids, or its fitted prior when no posterior is available. In constructing a JV type, we draw the source cost condition δ_t from the corresponding normal approximation or prior and keep the members' estimated strengths fixed; a JV type records its score relative to $\hat{\mu}_t$ net of the higher score strength of its members. For the cost strength, we replace each missing member estimate by zero before taking the maximum. For the score strength, we take the maximum over members with an estimate, using zero if none has one. The term $\kappa^J_h + q^J(\log s - m^J)$ specifies the function

$q_g^J(\log s)$ in Section 5.1. We check the stronger-member rule by letting the JV's log cost load on its members' strengths with free coefficients, $-(a_H \max\{\alpha_i, \alpha_j\} + a_L \min\{\alpha_i, \alpha_j\})$, for JVs whose members' strengths are both estimated. For project-specific JVs (418 bids), $\hat{a}_H = 0.05$ and $\hat{a}_L = -0.09$, with standard errors 0.23 and 0.06. Recovered JV costs thus vary little with the members' cost strengths estimated from their solo bids, while JV scores follow the stronger member's score strength (Appendix D.3). The profit tables keep the stronger-member rule of Section 5.1 for both components.

Assumptions 2 and 3 restrict different objects, the equilibrium distribution of bids and the distribution of cost residuals, and need not be jointly consistent: normal cost residuals do not in general generate normal log effective bids. We use these specifications as approximations at different stages: the bid distribution recovers interior costs and censoring bounds, and the residual model estimates common cost conditions δ_t and completes censored costs. The resulting type banks are not constrained to regenerate the fitted bid distribution.

We interpret δ_t as a common project characteristic known to firms before formation. It is conditioned on along with the other auction characteristics, and is not an additional unobserved formation state to integrate over. Transferred types need not reproduce the bid distribution of the target auction exactly; Appendix F.4 compares the implied prices with the data.

D.5 Expected bidding profits

The resulting inputs are firm strengths $a_i = (\hat{\alpha}_i, \hat{\sigma}_i)$, the strengths a_{ij} of pairs, and the arrays $\pi_t^S(a; m, n)$ and $\pi_t^J(a; m, n)$ on the product of a grid for cost strength (from -0.15 to 0.15 in steps of 0.025) and a grid for score strength (from -0.18 to 0.12 in steps of 0.03), interpolated bilinearly. For each unestimated strength component of a potential entrant, we use zero before calculating its profits and formation covariates and the componentwise maxima for potential JVs. For the formation auctions, τ_t is the recorded floor in 373 of the 396 auctions and the median of the fiscal year in the others; ω_t is the posterior mean from all bids for the 380 auctions in the bid sample and its prior mean otherwise. For δ_t, we use its

posterior mean from the solo bids when available and its fitted prior mean otherwise. The formation stage counts every listed bidder, including firms that withdraw or submit no valid price, about 30 percent of the listed solo bidders in the formation auctions, whereas the rival numbers used to estimate bid distributions count usable bids. We treat each listed bidder as submitting a usable bid independently with probability p^d, by mode and period; a bidder that does not earns nothing, and rivals that do not do not compete. The probabilities are the ratios of usable to listed bidders in the formation auctions included in the bid estimation sample: 0.81, 0.69 and 0.67 for solo bidders, and 0.90, 0.75 and 0.72 for JVs, in 2006–2010, 2011–2016 and 2017–2022, respectively. The profit against m listed JVs and n listed solo bidders is the binomial average

$$\pi_t^d(a; m, n) = p^d \sum_{m' \le m} \sum_{n' \le n} \binom{m}{m'} \binom{n}{n'} (p^J)^{m'} (1-p^J)^{m-m'} (p^S)^{n'} (1-p^S)^{n-n'} \, \pi_{b,t}^d(a; m', n'),$$

where $\pi_{b,t}^d(a; m', n')$ is the bidding-stage profit against m' JVs and n' solo bidders with usable bids, the maximized value of (1) averaged over the bank and evaluated with $\hat{F}^d(\cdot \mid I_t, M', N')$ at the counts that include the bidder. Nonpositive maximized profits are set to zero. A lone bidder bids the reserve and earns $R_t \max\{1-\tilde{c}, 0\}$. Strength enters the formation covariates through the two components $\hat{\alpha}_i/\hat{\sigma}_\alpha$ and $\hat{\sigma}_i/\hat{\sigma}_\sigma$, with separate coefficients. For a potential JV, we use the maximum of the two members' strengths in each component and the same first-stage scales. The formation likelihood and belief updates condition on these first-stage inputs.

D.6 Monte Carlo validation

To check the cost-strength estimator, including the estimation of the distributions of rival bids and the bid shift ω_t, we simulate 12,000 auctions with the numbers of solo bidders of the data, drawn from 4,000 firms whose activity varies, with $\sigma_\alpha = 0.035$, $\sigma_u = 0.04$, a common cost condition δ_t with standard deviation 0.05, a score coefficient of 0.5 and floor ratios of 0.85, 0.88 and 0.91. Bidders best-respond to normal distributions of rival bids whose mean

and scale, for each number of bidders and floor ratio, equal the mean and standard deviation of the log effective bids they generate. About 28 percent of the simulated bids are at the floor and 11 percent at the reserve, close to the shares among the solo bids compared in the data (27 and 14 percent, most of the latter above the reserve). We then estimate the distributions of rival bids, recover costs and bounds and estimate strength as in the data, including the empirical Bayes step. Panel A of Table A6 reports averages over three replications of the estimated score coefficient, the estimated standard deviation of strength and, by number of bids per firm, the slope of the regression of true on estimated strength, which is one for a well-calibrated posterior mean, and the correlation between them. Panel B raises the residual standard deviation to 0.09 and lowers that of strength to 0.03, so that, as in the data, the residual is the larger source of variation; there, 20 percent of the bids are at the reserve.

The estimator is close to calibrated in every group, and it recovers the score coefficient and the dispersion of strength. Treating bids at the reserve as interior compresses the estimates, and dropping floor bids or treating their bounds as costs compresses them further and moves the score coefficient toward zero. Recovering the costs with the posterior that excludes only the bidder's own bid overstates the dispersion of strength, and so, to a lesser extent, does ignoring the posterior variance of the shift. Panel B shows why the noise scale is calibrated: with the residual variance in the penalty, even at the true σ_α, the estimates are shrunk too much, and the more so the fewer bids a firm has.

Table A6. Monte Carlo experiment: the estimator of bidding strength

	Score		Slope, true on estimated			Correlation		
	coefficient	$\hat{\sigma}_\alpha$	1–4 bids	5–19	≥ 20	1–4 bids	5–19	≥ 20
A. $\sigma_\alpha = 0.035$, $\sigma_u = 0.04$								
Estimator	0.49	0.035	0.95	0.96	0.97	0.67	0.87	0.95
Known rival distributions	0.48	0.034	1.01	1.01	1.00	0.70	0.89	0.96
Plug-in scale	0.48	0.037	0.90	0.91	0.92	0.66	0.86	0.95
Own bid left out	0.45	0.041	0.81	0.83	0.84	0.65	0.85	0.95
All bids of the auction	0.49	0.035	0.95	0.96	0.99	0.68	0.87	0.95
Reserve bids as interior	0.36	0.029	1.18	1.19	1.19	0.66	0.86	0.95
Floor bids dropped	0.16	0.025	1.34	1.34	1.37	0.53	0.77	0.92
Floor bounds as costs	0.05	0.021	1.63	1.61	1.63	0.58	0.81	0.93
B. $\sigma_\alpha = 0.030$, $\sigma_u = 0.09$								
Estimator	0.47	0.029	0.99	0.98	1.02	0.42	0.62	0.84
Penalty $\sigma_u^2/\sigma_\alpha^2$	0.46	–	1.62	1.40	1.22	0.42	0.63	0.84

Notes: Averages over 3 (panel A) and 3 (panel B) replications of 12,000 auctions and 4,000 firms; true score coefficient 0.5. Floor bids are 27 and 28 percent of the bids, and bids at the reserve 11 and 20 percent, in panels A and B. The slope is that of the regression of true on estimated strength among the firms with the given number of bids; a well-calibrated posterior mean has slope one. Estimator: costs of each pair from the shift estimated without both bids and the predictive scale; censoring at the floor band, at the reserve and at $\log(0.8\,\tau_t)$; penalty $s^2/\hat{\sigma}_\alpha^2$ with the noise scale s^2 of the criterion and $\hat{\sigma}_\alpha$ from the empirical Bayes step. Every row of panel A repeats the empirical Bayes step. Known rival distributions: the distributions used by the bidders, without the artificial censoring point. Plug-in scale: the scale without the posterior variance of the shift. Own bid left out: each bidder's cost from the shift estimated without its own bid. Reserve bids as interior: their bound used as the cost, without upper censoring. Floor bids dropped: floor bids removed and no lower censoring. Floor bounds as costs: bounds used as costs, without lower censoring. Penalty $\sigma_u^2/\sigma_\alpha^2$: the residual variance in place of s^2, at the true σ_α.

Appendix E. Formation Estimation

The likelihood assigns a probability to each observed partition using the model's stability conditions. After recording the parameter domains, this appendix derives that probability, describes its numerical integration and belief updates, and reports the remaining numerical discrepancies. The final subsection distinguishes the conditional refinement used for the coefficient tables and standard errors from the estimates used for the simulations.

E.1 Parameter domains and numerical search ranges

We set the reference project size to $\bar{R} = 3$ billion yen and divide the cost and score strengths by their first-stage prior standard deviations, $\hat{\sigma}_\alpha = 0.042884$ and $\hat{\sigma}_\sigma = 0.038750$. Each direct strength coefficient measures the payoff effect of a one-prior-standard-deviation increase in its component, in units of the solo payoff shock. The reference $\bar{R}$ is the fixed knot of the project-size regressor.

The two project-size coefficients β_S and β_J and the four direct strength coefficients $\kappa_{S,\alpha}$, $\kappa_{S,\sigma}$, $\kappa_{J,\alpha}$ and $\kappa_{J,\sigma}$ range over the real line. The cost levels, capacity coefficients, repeat-partner discount and contact parameter τ are nonnegative. Market and size effects enter the cost levels exponentially; we normalize the reference market and size effects to zero.

For numerical optimization, we fix $q = 1$ and search over $c_S \in [0, 6]$ and each non-reference log market or size coefficient in $[-8, 8]$. The JV common-state scale has domain $c_J \in [0, \infty)$: no finite upper bound is imposed. The remaining nonnegative cost and capacity parameters have no finite upper bound. We distinguish these numerical ranges from the unrestricted size and strength effects; the ranges do not establish identification of a parameter that lies close to a boundary.

E.2 The conditional probability of a partition

Fix auction t, both common states (ξ, η), parameters θ, and beliefs H. The expected profits in (3) then determine the means of the option payoffs. Write

$$u_{it}^S = a_{it}(\xi, \eta; \theta, H) + \varepsilon_{it}, \qquad u_{ijt}^J = \bar{m}_{ijt}(\xi, \eta; \theta, H) + s_t \zeta_{ijt}, \qquad s_t = q r_t.$$

Here $\bar{m}_{ijt}$ includes the location adjustment $-q l_t$, so both ε_{it} and ζ_{ijt} are standard normal. Both payoff means can depend on both common states through expected profits. We suppress their arguments below. Let $\mathcal{E}_t$ be the set of unordered pairs of potential entrants, and let p_{ij} be a pair's availability probability: one for a pair with prior contact, $e^{-\tau}$ without contact, and zero if joint bidding is prohibited.

To evaluate $P_t^{\text{obs}} = (S_t, J_t, O_t)$, assign payoff zero to its non-participants, a solo payoff U_i to each solo bidder, and a common payoff U_{ij} to the two members of each observed JV. A non-participant must have a nonpositive solo payoff. Each observed bidder must have a positive assigned payoff, and each observed JV must be available. Conditional on these requirements, stability additionally requires each JV member to prefer the JV to bidding alone and each other available pair to fail to block. For a pair with assigned payoffs U_i, U_j, the probability of no block is

$$B_{ij}(v) = 1 - p_{ij} + p_{ij} \Phi\left(\frac{v - \bar{m}_{ijt}}{s_t} \right), \qquad v = \max\{U_i, U_j\}.$$

The maximum enters because a pair blocks only if its common payoff exceeds the assigned payoffs of both members.

These conditions give a convenient factorization of the conditional likelihood. Define the probability of solo nonparticipation, positive assigned payoffs, and availability of the observed JVs as

$$C_t(\xi, \eta; \theta, H) = \prod_{i \in O_t} \Phi(-a_{it}) \prod_{i \in S_t} \Phi(a_{it}) \prod_{(i,j) \in J_t} p_{ij} \Phi\left(\frac{\bar{m}_{ijt}}{s_t} \right).$$

Let $\mathbb{E}_+$ denote expectation over independent draws of the observed bidders' payoffs from their respective normal distributions conditional on being positive. There is one draw for each solo bidder and one shared draw for each JV. Then

$$p_t(P_t^{\text{obs}} \mid \xi, \eta; \theta, H) = C_t(\xi, \eta; \theta, H)\, \mathbb{E}_+\Bigg[\prod_{(i,j)\in J_t} \Phi(U_{ij} - a_{it})\Phi(U_{ij} - a_{jt}) \times \prod_{(i,j)\in \mathcal{E}_t \setminus J_t} B_{ij}(\max\{U_i, U_j\}) \Bigg]. \tag{24}$$

For a JV member, $U_i = U_{ij}$; for a non-participant, $U_i = 0$. Independence of the underlying shocks and availability draws gives the product after conditioning on these assigned payoffs. In particular, a pair of two non-participants contributes $B_{ij}(0)$ without any additional payoff integration. In this exact expression its mean $\bar{m}_{ijt}(\xi, \eta; \theta, H)$, including expected bidding profit, is evaluated at the same two common states as every other option. The numerical reference-state approximation for these pairs is described below.

E.3 Finite approximation of the common states

The belief restriction and the likelihood integral use different finite sets of states. We represent beliefs at the $K = 25$ pairs (ξ_k, η_k) formed from the Cartesian product of $\{-2, -1, 0, 1, 2\}$ with itself. At each point, let h_{tk} collect the distributions of rival numbers for solo bidders and JVs, and let $\mathbf{h} = (h_{tk})_{t,k}$. The associated expected profits are

$$v_{itk}^S = \sum_{m,n} h_{tk}^S(m,n)\hat{\pi}_{it}^S(m,n), \qquad v_{ijtk}^J = \sum_{m,n} h_{tk}^J(m,n)\hat{\pi}_{ijt}^J(m,n).$$

Write $\Gamma_{\theta,tk}(h_{tk})$ for the distributions generated with these profits while fixing both states at (ξ_k, η_k). The first-stage profits for a given pair of rival numbers do not depend on the common formation states, so the same cached bidding calculations are reused at all 25 points.

The likelihood instead uses $D = 128$ state pairs for each auction. For each marginal distribution, transform the midpoints $(d - 1/2)/D$, $d = 1, \ldots, D$, by Φ^{-1}. Independently

permute the two lists within each auction to obtain (ξ_{td}, η_{td}), with equal weight $1/D$. These are 128 pairs, not the Cartesian product of two 128-point lists. The permutations are held fixed across parameter and belief updates. At each such pair, bilinearly interpolate the expected profits from the four surrounding belief-grid points. For profit evaluation, a coordinate outside the grid is held at its nearest endpoint. These convex weights preserve nonnegative profits and correspond to mixtures of the saved beliefs. The actual normal state draws remain unchanged in the direct payoff terms; only the coordinates used to evaluate the profit grid are bounded. These operations approximate the state dependence of expected profits separately from the integration over the normal states.

Write the conditional partition probability using interpolated profits as $\widetilde{p}_t(P_t^{\text{obs}} \mid \xi, \eta; \theta, \mathbf{h})$. Before simulating the remaining payoff integral, the finite counterpart of (10) is

$$\begin{aligned} \max_{\theta \in \Theta,\, \mathbf{h} \in \mathcal{H}_K} \quad & \sum_{t=1}^{T} \log \left[\frac{1}{D} \sum_{d=1}^{D} \widetilde{p}_t(P_t^{\text{obs}} \mid \xi_{td}, \eta_{td}; \theta, \mathbf{h}) \right] \\ \text{subject to} \quad & h_{tk} = \Gamma_{\theta,tk}(h_{tk}), \qquad t = 1, \ldots, T, \quad k = 1, \ldots, K. \end{aligned} \tag{25}$$

Here $\mathcal{H}_K$ is the product of probability simplexes over feasible pairs of rival numbers. If $A_t = |\mathcal{N}_t|$, solo counts satisfy $2m + n \leq A_t - 1$ and JV counts satisfy $2m + n \leq A_t - 2$, in addition to eligibility restrictions. Conditions for a mode with zero probability are omitted. The restrictions are evaluated at the 25 belief points, not at the 128 integration draws. Increasing the integration draws alone therefore does not refine the belief grid or reduce the discrepancies in its consistency conditions.

E.4 Simulation of the likelihood

We integrate the states and the observed bidders' positive payoffs jointly. With N_t solo bidders and M_t JVs, take D points of a scrambled Sobol sequence in $N_t + M_t$ dimensions, and associate its dth point with (ξ_{td}, η_{td}). For a payoff with normal mean μ and standard

deviation s, map a coordinate $u \in (0,1)$ to

$$U = \mu + s\Phi^{-1}(1 - (1-u)\Phi(\mu/s)) .$$

This draws from the payoff distribution conditional on $U > 0$. Use $(\mu, s) = (a_{it}, 1)$ for a solo bidder and $(\bar{m}_{ijt}, qr_t)$ for a JV, with the means evaluated at the same state pair. There is one positive-payoff vector for each state pair, rather than a separate 128-draw inner integration at each pair.

Let $W_{td}(\theta, \mathbf{h})$ be the product in (24) evaluated at this payoff vector, multiplied by its conditioning probability C_t. The simulated objective is

$$\widehat{Q}(\theta, \mathbf{h}) = \sum_{t=1}^{T} \log\left[\frac{1}{D}\sum_{d=1}^{D} W_{td}(\theta, \mathbf{h})\right] .$$

Conditioning the proposal on positive payoffs avoids draws incompatible with observed participation. The factors $\Phi(a_{it})$ and $\Phi(\bar{m}_{ijt}/s_t)$ in C_t restore the probabilities of those conditioning events. This is the importance-sampling component of the calculation; it does not change the normal distribution of the common states. The average is taken before the logarithm because each auction has a single unobserved pair of common states. We evaluate products in logarithms and compute the mixture with a log-sum-exp calculation. With the draw banks and profits fixed, the likelihood is continuous and piecewise differentiable in the parameters; ties in the maximum of two assigned payoffs can create kinks.

For a pair of two observed non-participants, we retain a reference-state approximation to expected bidding profit: $V^J_{ijt}(\xi, \eta)$ is replaced by $V^J_{ijt}(0,0)$ in its no-blocking term. Both states have prior mean zero. Thus this approximation uses profit at the prior-mean state, rather than setting profit to zero; it is not generally equal to profit averaged over states. The direct common payoff term $c_J\eta$ remains. We evaluate the sum of these pairs' log no-blocking probabilities at 21 Chebyshev points on $[-2.7, 2.7]$ and use barycentric polynomial interpolation at the likelihood's η draws. The posterior-state calculation uses 512 joint draws; its draws beyond this interval use the same polynomial for extrapolation. All other

options use profits interpolated at both states. This reference-state shortcut is confined to these likelihood terms; unconditional formation simulations and policy calculations use the appropriate two-state profits for all options.

E.5 NPL implementation and pseudocode

A parameter step holds all node-specific beliefs fixed and maximizes $\widehat{Q}$. The next belief step fixes the resulting parameters and simulates formation separately at every (t, k). Both ξ_k and η_k remain fixed; only firm shocks, pair shocks and partnership availability are drawn. These draws are unconditional on the observed partition and are not restricted to positive assigned payoffs. Proposition 1 supplies the stable partition for each draw.

For $B = 64$ such draws, let (M_{tkb}, N_{tkb}) be the simulated bidder counts. The empirical update for solo beliefs is

$$\widehat{\Gamma}^S_{\theta,tk}(h_{tk})(m, n) = \frac{\sum_{b=1}^B \mathbf{1}\{M_{tkb} = m, N_{tkb} = n + 1\}}{\sum_{b=1}^B \mathbf{1}\{N_{tkb} \geq 1\}},$$

and the JV update uses $\mathbf{1}\{M_{tkb} = m + 1, N_{tkb} = n\}$ in the numerator and $\mathbf{1}\{M_{tkb} \geq 1\}$ in the denominator. Each draw in which the mode occurs receives equal weight, without additional weighting by its bidder count. If a legally possible mode never appears, its previous conditional belief is retained to continue the iteration, but its consistency residual is unavailable, not zero.

The implementation uses the following finite sequence, with fixed random-number streams throughout.

1. **Initialize.** Construct the 25 belief points and 128 integration pairs. Initialize from saved two-state beliefs, and apply two updates with both states fixed and damping weight 0.5 before the first parameter step.

2. **Construct profits.** For every (t, k), average the fixed first-stage profit tables using $h^{(r)}_{tk}$. Hold the resulting profit arrays fixed in the parameter step.

3. **Estimate parameters.** Starting from the previous parameter vector, maximize $\widehat{Q}(\theta, \mathbf{h}^{(r)})$. At each trial vector, interpolate profits to the integration pairs, transform the positive-payoff draws and average the joint likelihood weights before taking logarithms.

4. **Update beliefs.** Holding the estimated parameters fixed, generate 64 stable partitions at every belief point and calculate the conditional distributions of rival numbers. Set

$$h_{tk}^{\text{new}} = (1-\lambda)h_{tk}^{\text{old}} + \lambda\widehat{\Gamma}_{\theta,tk}(h_{tk}^{\text{old}}).$$

   Recompute expected profits after each update. Perform two such updates with $\lambda = 0.5$ per parameter round, and repeat Steps 2–4 for five rounds.

5. **Finish the iteration.** At the last parameter vector, perform 16 additional belief updates with $\lambda = 0.1$. Then refine the parameter estimates once more with those final profits fixed. Return this parameter–belief pair.

6. **Record discrepancies.** At the returned parameters, simulate the conditional belief map again without updating the saved beliefs. Report differences in the numbers of rival JVs, rival solo bidders and total rivals, total variation of the distributions of rival numbers, and the parameter-step optimization residual. Flag modes without simulated support. A small change after damping alone is not a consistency check.

The likelihood and belief simulations use 128 and 64 draws, respectively. In this sequence, parameter optimization uses a function tolerance of 10^{-8}, with scaled projected-gradient tolerances of 10^{-3} during the alternating rounds and 3×10^{-4} in the last parameter step. The fixed coordinate $q = 1$ is excluded from the gradient stopping criterion. Mixing discards belief cells with probability at most 10^{-6} and renormalizes. The finite iteration budget bounds computation; completion of the parameter optimization does not establish that every belief restriction is satisfied. We report its remaining discrepancies separately from the optimizer residual. Neither calculation establishes a global solution of (25), uniqueness of beliefs, or convergence of NPL.

The parameter vector returned by this sequence has a scaled projected-gradient norm of 2.12×10^{-4}, with no free parameter at a search bound. At that vector, the belief check has simulated support for 9,838 of 9,900 solo auction–node combinations and 7,192 of 8,150 combinations in which JVs are legally possible. The remaining 62 solo and 958 JV combinations have unmeasured residuals. At each node, we average absolute discrepancies across auctions using the simulated probability that the relevant mode occurs as the weight. The largest such mean discrepancy in total rivals across the 25 nodes is 0.361 for solo beliefs and 0.579 for JV beliefs over the full sample, and 1.07 and 1.84 among the six largest projects. The largest individual discrepancies are 4.35 and 9.00 rivals, and the largest total-variation distances are 0.98 and 1.00, respectively. Thus the finite updates leave material discrepancies in some auction–state combinations, especially for large projects.

E.6 Conditional refinement and standard errors

To obtain the coefficients and standard errors reported in Tables 6 and A9, we subsequently reoptimized the formation likelihood while holding the bidding estimates and the saved profit and belief tables fixed. A one-dimensional search along the weak participation-cost direction provided a starting point for joint L-BFGS-B optimization of the same 31 free coefficients, with the parameter domains in Appendix E.1 and the same 128 integration draws. The function tolerance was 10^{-13} and the projected-gradient target 10^{-7}. The optimizer stopped on the function-tolerance criterion after 122 iterations, with a scaled projected-gradient norm of 2.07×10^{-6} and no free parameter at a search bound.

This conditional optimization raises the simulated log likelihood from $-10{,}656.14$ to $-10{,}651.18$. The base participation cost falls from 64.27 to 2.20, while the solo size coefficient falls from 0.97 to 0.67. We did not update beliefs or recompute their residuals at the refined coefficients. The participation and price simulations, partnership classifications, posterior-state summaries and policy results retain the preceding parameter vector and have not been recomputed after this refinement. Neither parameter vector is claimed to solve the equilibrium restrictions exactly.

Standard errors cluster the formation score by auction, conditional on $q = 1$, the bidding-stage estimates, and the saved profit and belief tables. Let $\widehat{g}_t = \nabla_\theta \log \widehat{L}_t$ be the score of auction t for the 31 free coefficients at the refined estimate, $\bar{g} = T^{-1} \sum_t \widehat{g}_t$, and $\widehat{A} = -\sum_t \nabla_\theta \widehat{g}_t$, with $T = 396$. We use the centered sandwich covariance

$$\widehat{V} = \widehat{A}^{-1} \left\{ \frac{T}{T-1} \sum_{t=1}^{T} (\widehat{g}_t - \bar{g})(\widehat{g}_t - \bar{g})' \right\} \widehat{A}^{-1}.$$

The auction scores are analytic; central differences of the same analytic gradient, with the integration draws held fixed, give the symmetrized Hessian. The information matrix is positive definite on the free-parameter subspace, and halving the difference step changes every reported coefficient and derived-quantity SE by less than 0.1 percent. Derived quantities, including contact probabilities and mean JV costs, use the delta method. These are local conditional standard errors: they do not propagate bidding-stage estimation error or changes in beliefs with the formation parameters, and they have not been propagated to the reported policy predictions.

Appendix F. Additional Estimation Results

This appendix documents the comparisons and diagnostics reported in Section 7, following its order. The bidding results cover the correlates of strength, differences in residual types and effective costs, and price fit. The formation results give the full coefficient table, the classification of partnership opportunities, and the detailed participation fit.

F.1 Correlates of bidding strength

Table 4 describes which features of firms' observed solo bidding are associated with the two estimated strengths. The sample contains 7,123 firms with both strength estimates and at least one valid, nonduplicated solo bid with an engineer's estimate of at least one million yen during 2006–2022. The size restriction excludes small recorded amounts that include unit-price contracts; it applies only to these descriptive histories. Each firm receives equal weight. For the common covariate vector X_i, we estimate by OLS

$$\frac{\hat{\alpha}_i - \overline{\hat{\alpha}}}{\mathrm{sd}(\hat{\alpha})} = c_\alpha + X_i' b_\alpha + r_{\alpha i},$$
$$\frac{\hat{\sigma}_i - \overline{\hat{\sigma}}}{\mathrm{sd}(\hat{\sigma})} = c_\sigma + X_i' b_\sigma + r_{\sigma i}.$$

The bars and standard deviations refer to this firm sample. The standard deviations of estimated cost and score strength are 0.0290 and 0.0327, respectively, rather than the prior standard deviations used to scale the formation coefficients. The vector X_i contains the logs of maximum project size, the number of solo bids, the number of distinct calendar years with a solo bid, the numbers of bidding regions and work categories, together with the mean year of the firm's bids. It also includes six regional and 21 work-category bid shares, omitting one category from each group. All covariates are standardized over the same firm sample. Because both equations use the same observations and regressors, conventional seemingly unrelated regressions yield the same coefficient estimates as OLS. The reported HC3 standard errors allow heteroskedastic regression residuals across firms, treating firms

as independent and the estimated strengths as fixed. They do not propagate first-stage estimation uncertainty or its dependence across firms.

The conditional association with project size depends on the size measure and firm sample. Restricting the number of observed bids changes both the precision of estimated strengths and the composition of firms, so the comparison does not isolate either source of the difference. All histories cover the observation period as a whole; the regressions describe associations between estimated ability and participation rather than a direction of causation.

F.2 Tests of type distributions

The distribution tests compare solo and JV score residuals e, cost residuals u, and their joint distribution for the same mix of auctions as Figure 4. We give each of the 396 formation auctions equal weight for both modes, assigning it the source pools described in Appendix D.4.2. A pool's weight is therefore the share of target auctions that use it. The 128 types per bank used in the figure and profit tables approximate these distributions. For inference, we instead use all eligible source bids, averaging their recovered distributions within each pool and integrating over the posterior and censored-cost distributions described in that appendix. Censored-cost integration uses 16-point Gauss–Hermite quadrature.

For each comparison, let $\hat{\Delta}(x)$ be the JV CDF minus the solo CDF, with x a score residual, a cost residual, or a residual pair. We measure the difference by the finite-grid Cramér–von Mises-type statistic

$$T = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \hat{\Delta}(x)^2.$$

Each marginal grid contains 99 points at the first through 99th percentiles of the equally weighted mixture of the two recovered marginal distributions; cost quantiles are interpolated from a 513-point CDF evaluation. The joint grid contains 21×21 points, selecting 21 evenly spaced indices from each marginal grid. These grids remain fixed across replications.

Sampling variation comes from the auctions supplying the source bids. In each of 99

replications, we draw independent exponential weights with mean one for these auctions and normalize their overall mean to one. Every bid of an auction receives the same weight in both modes and in every pool where it appears. We renormalize the bid weights within each pool, keeping its target weight fixed. With $\hat{\Delta}_b^*(x)$ the resulting CDF difference, the bootstrap statistic is centered under equality:

$$T_b^* = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \left[\hat{\Delta}_b^*(x) - \hat{\Delta}(x)\right]^2.$$

The p-value is $(1 + \#\{b : T_b^* \geq T\})/100$, with minimum 0.01. We apply Holm's adjustment across the three tests.

Inference is conditional on the adopted first-stage estimates, source selection, pooling rules, and posterior recovery distributions, all held fixed. It excludes first-stage estimation uncertainty and assumes independence across source auctions conditional on the modeled persistent effects; it does not allow additional serial dependence from repeated firms. The tests concern recovered residuals under the maintained inversion and censoring models. Rejection of joint equality can follow from unequal marginals and does not by itself establish different copulas or a causal effect of forming a JV.

F.3 Decomposing effective-cost differences

Table A7 reports the decomposition for three solo benchmarks, including the zero-strength comparison in Table 5. We evaluate every observed JV at the componentwise maxima of its members' estimated strengths. The first benchmark is a solo bidder with $\alpha = \sigma = 0$; the second is an equal mixture of the two members evaluated individually at their own strengths. Both comparisons use the 312 JVs in 127 auctions. The third benchmark uses the observed solo bidders in the same auctions, restricting the comparison to 122 auctions with 302 JVs and 1,686 solo bidder listings. Within each comparison, we first average over the 128 types in each relevant bank, then over the JV pairs, member firms, or solo bidders within an auction, and finally give each auction equal weight. Evaluating the members individually preserves

Table A7. Sources of JV effective-cost differences

Contribution (log points)	Solo benchmark		
	Zero strength	JV members individually	Observed solo bidders
Cost strength α	−1.05	−0.86	−0.94
Score strength σ	−9.84	−1.29	−0.95
Score-cost relation and types (q, G)	+2.08	+2.47	+2.56
Total difference	−8.81	+0.32	+0.66
Auctions	127	127	122
Observed JVs	312	312	302

Notes: Differences in $100\,E[\log(\tilde{c}/s)]$, JV minus solo benchmark; negative values denote lower effective cost. The zero-strength benchmark sets $\alpha = \sigma = 0$. The member benchmark averages the two members' individually evaluated solo outcomes. The observed-solo benchmark uses the 1,686 solo bidder listings in the 122 auctions containing both modes. Each column holds its auction sample fixed, weights auctions equally, and averages the relevant bidders within each auction. The JV uses the componentwise maxima of its members' strengths. Contributions average the two orders of changing strengths and switching (q, G); the third row changes both the score-cost relation and residual law. Saved types and estimates are held fixed. Appendix F.3 gives details; no sampling confidence intervals are reported.

the nonlinear effect of score strength; it differs from evaluating a bidder at their average strength.

Using the mean log effective cost $\ell_t^d(a)$ defined in (12), compare an observed JV (i, j), with strength a_{ij}, to a solo bidder with strength $a = (\alpha, \sigma)$ in the same auction. We change strengths and the bidding specification in both possible orders and average the contributions. Changing d switches both the score-cost relation, q_g or q_g^J, and the joint residual distribution G_g^d. Since cost strength enters additively as $-\alpha$, the three contributions are

$$\begin{aligned} C_\alpha &= \alpha - \max\{\alpha_i, \alpha_j\}, \\ C_\sigma &= \frac{1}{2} \sum_{d \in \{S,J\}} \big[\ell_t^d(a_{ij}) - \ell_t^d(a)\big] - C_\alpha, \\ C_{q,G} &= \frac{1}{2}\big[\ell_t^J(a) - \ell_t^S(a) + \ell_t^J(a_{ij}) - \ell_t^S(a_{ij})\big]. \end{aligned} \tag{26}$$

Their sum is $\ell_t^J(a_{ij}) - \ell_t^S(a)$. Averaging the two orders splits equally any interaction between

score strength and the score-cost relation and types.

For the zero-strength benchmark, evaluate each contribution at $a = (0, 0)$. For the member benchmark, average its values at $a = a_i$ and $a = a_j$ with equal weight; for the observed-solo benchmark, average over $a = a_k$ for $k \in S_t$. We then average across observed JVs within each auction and equally across the relevant auctions. Applying the same weights to all three contributions preserves their sum as the mean effective-cost difference. Multiplying by 100 expresses them in log points, with a negative difference indicating lower JV effective cost. This decomposition conditions on the estimated bidding model and type banks and does not include sampling uncertainty. The member benchmark averages separate solo outcomes rather than taking the better of two competing bids, and the comparisons do not identify individual pairs' causal gains from forming a JV.

The solo benchmark matters because firms that bid alone in these projects also have strong persistent capabilities. In the member comparison, taking the stronger component lowers mean log effective cost by 2.15 points, but the JV's conditional score-cost distribution adds 2.47 points, leaving a net increase of 0.32. Relative to the solo bidders actually participating in the same auctions, the cost- and score-strength contributions are similar, and the net increase is 0.66 points. The large score contribution against the zero-strength reference therefore does not establish an equally large gain over the members' solo alternatives or other participating firms.

F.4 Bidding model fit

Table A8 compares the model's prices with the data under the current policy, fixing the auction-level states at their posterior means and interpolating the estimation beliefs at those states. The mean ratio of the winning bid to the engineer's estimate is 89.0 percent in the model and 88.7 percent in the data; the shares within one percent of the floor are 77 and 74 percent. In the six auctions above 10 billion yen, the corresponding price ratios are 90.2 and 94.1 percent, and the floor shares are 50 and 33 percent. The correlation across auctions between the simulated mean and observed winning-bid ratio is 0.74. Thus average prices

Table A8. Bidding stage: prices and the floor

		Winning bid / engineer's estimate (%)		Winners within 1% of the floor (%)	
	Auctions	Data	Model	Data	Model
All 396 auctions	396	88.7	89.0	74	77
≥5 bn yen	46	90.3	89.7	70	81
≥10 bn yen	6	94.1	90.2	33	50
Size quartile 1	99	88.5	89.1	72	70
Size quartile 2	99	88.6	88.9	71	72
Size quartile 3	99	88.4	88.6	79	83
Size quartile 4	99	89.5	89.3	76	83

Notes: The estimated formation model under the current policy, with states at their posterior means and estimation beliefs interpolated at those states. There are 32 formation draws and eight bidding replications per auction. Data use the first recorded bid of the marked winner when it lies between half of the reserve and the reserve. Floors are observed for 373 auctions and imputed for 23. Means condition on a usable winning bid.

are close overall, while the model still predicts lower prices and more binding floors in the largest projects.

F.5 All formation parameters

Table A9 reports all 31 estimated coefficients and the fixed relative pair-shock scale $q = 1$, including the separate size effects and the two strength effects for each participation mode. The participation-cost and JV-cost shifts are logarithmic: they multiply the corresponding base costs. The reference market–size cell has no auctions, so its base participation cost must be read together with the regional and size shifts. The cost coefficients and their yen equivalents condition on the relative scale restriction; fixing q does not make them directly measured resource costs.

The coefficient tables use the final conditional refinement, whereas the participation and price simulations, partnership classifications, posterior-state summaries and policy results use the preceding parameter vector. Appendix E.6 describes the refinement and the conditional standard errors; the reported simulations have not been recomputed at the refined coefficients.

Table A9. All parameters of the formation stage

	Estimate	SE
Participation cost, base cell $\bar{f}$	2.200	0.817
log shift: Kanto bridges	−1.058	0.140
log shift: Kinki civil engineering	−0.198	0.088
log shift: Kyushu civil engineering	−0.438	0.171
log shift: other bridges	−1.285	0.353
log shift: other civil engineering	−0.621	0.379
log shift: size quartile 2	−0.338	0.360
log shift: size quartile 3	−0.512	0.366
log shift: size quartile 4	−0.757	0.402
Capacity shortfall, λ_S	1.808	0.077
Project size, β_S	0.669	0.246
Cost strength, $\kappa_{S,\alpha}$	−0.176	0.028
Score strength, $\kappa_{S,\sigma}$	0.603	0.031
Solo state s.d., c_S	0.636	0.028
Expected profit, $1/\sigma$ (per million yen)	0.00508	0.00122
Additional JV cost K, quartile 1	7.198	0.920
Additional JV cost K, quartile 2	5.941	0.440
Additional JV cost K, quartile 3	6.095	0.343
Additional JV cost K, quartile 4	5.901	0.478
log shift: Kanto bridges	0.015	0.061
log shift: Kinki civil engineering	0.208	0.054
log shift: Kyushu civil engineering	0.115	0.067
log shift: other bridges	−0.085	0.112
log shift: other civil engineering	0.110	0.097
Discount for past joint bids, γ	0.771	0.073
Contact parameter τ	1.643	0.400
Capacity shortfall, λ_J	2.959	0.657
Project size, β_J	−0.401	0.136
Cost strength, $\kappa_{J,\alpha}$	−0.323	0.062
Score strength, $\kappa_{J,\sigma}$	0.397	0.068
Pair-shock scale, $\log q$ (fixed)	0.000	—
JV state s.d., c_J	0.424	0.052

Notes: The 31 estimated coefficients and the fixed value $\log q = 0$. Participation-cost log shifts multiply $\bar{f}$ by $\exp(\psi)$; JV-cost log shifts multiply K by $\exp(\delta)$. The reference market is Kanto civil engineering, and the reference size is quartile 1. Strength coefficients multiply each component divided by its first-stage prior standard deviation. Parameter domains are in Appendix E.1. SEs are auction-clustered sandwich standard errors conditional on $q = 1$, the bidding-stage estimates, and the fitted profit and belief tables. The fixed parameter has no estimated SE.

F.6 Partnership opportunities and realized matches

Table 7 distinguishes firms without a mutually beneficial partnership from those whose attractive partnerships are unavailable or whose prospective partners match elsewhere. We classify the same 64 unconditional simulation draws used to assess fit, drawing the auction states and idiosyncratic shocks and evaluating every solo and JV option with the fitted state-dependent profits. The unit is a potential entrant in a particular auction and draw. The classification covers 47,080 firm–auction observations in the 326 auctions where the model permits JVs. The remaining 70 auctions and 3,056 observations are excluded because joint participation is prohibited by the model's eligibility rules.

In eligible auction t and draw r, let $\mathcal{E}^0_{tr}$ contain the firms i for which some partner j permitted by the model satisfies

$$u^{J,(r)}_{ijt} > \max\{0, u^{S,(r)}_{it}, u^{S,(r)}_{jt}\}.$$

This condition requires both members to prefer the JV to bidding alone or staying out, before imposing partnership availability. Let $\mathcal{E}^a_{tr}$ contain the firms with at least one such partner available through prior contact or the availability draw, and let $\mathcal{E}^J_{tr}$ contain the firms matched in the simulated stable partition. Since $\mathcal{E}^J_{tr} \subseteq \mathcal{E}^a_{tr} \subseteq \mathcal{E}^0_{tr}$, the four categories are

$$\mathcal{N}_t \setminus \mathcal{E}^0_{tr}, \qquad \mathcal{E}^0_{tr} \setminus \mathcal{E}^a_{tr}, \qquad \mathcal{E}^a_{tr} \setminus \mathcal{E}^J_{tr}, \qquad \mathcal{E}^J_{tr}.$$

A firm in the third category has an available, mutually beneficial partner, but each such partner receives at least as much from another match. The classification leaves the stable partition unchanged.

The shares give equal weight to each eligible firm–auction observation and each draw. To compare availability and partner competition among firms with an attractive partnership, we instead divide the counts in the last three categories by the total count in $\mathcal{E}^0_{tr}$ across auctions and draws. With this common denominator, the shares are 17.62, 9.29 and 73.08

Table A10. Formation model fit by project size, region and work category

		Solo bids		JVs		New / repeat JVs		Auctions with JV	
	Auctions	Data	Model	Data	Model	Data	Model	Data	Model
Size quartile 1	99	1,463	1,561	21	25	9 / 12	13 / 12	14	15
Size quartile 2	99	1,643	1,795	67	72	31 / 36	34 / 38	29	28
Size quartile 3	99	1,676	1,893	88	71	30 / 58	33 / 38	38	32
Size quartile 4	99	1,377	1,445	136	184	75 / 61	104 / 80	46	57
Kanto, civil engineering	69	971	1,052	66	120	48 / 18	70 / 50	30	36
Kanto, bridges	58	779	914	134	106	34 / 100	34 / 72	40	32
Kinki, civil engineering	99	1,807	1,904	24	30	18 / 6	16 / 14	12	19
Kyushu, civil engineering	50	889	938	37	45	22 / 15	34 / 11	20	19
Other bridges	104	1,527	1,617	30	35	16 / 14	22 / 13	19	19
Other civil engineering	16	186	268	21	17	7 / 14	9 / 8	6	7

Notes: Model columns are means over 64 simulated stable partitions per auction at the estimates, with the auction-level states (ξ_t, η_t) drawn from their distribution and expected profits evaluated at the drawn state. The size quartiles and the region–work category groups each partition the full sample of 396 auctions. Table 8 reports the full sample and the two large-project groups.

percent, respectively.

Payoffs of unavailable pairs are latent draws from the fitted model. The classification therefore describes partnership opportunities under its payoff and availability specifications, including the fixed relative pair-shock scale $q = 1$. It does not estimate how many additional JVs would form if a barrier were removed, because changing availability can also change competing matches. The reported shares condition on the estimated parameters and do not include sampling uncertainty.

F.7 Formation model fit

The aggregate fit masks differences across project sizes and region–work categories. Table A10 reports these comparisons using the same unconditional simulations as Table 8. For example, the model predicts 120 JVs against 66 in Kanto civil engineering, but 106 against 134 in Kanto bridges. Across all auctions, it predicts 132.5 auctions with a JV against 127 in the data, and 2.66 JVs per auction with a JV against 2.46.

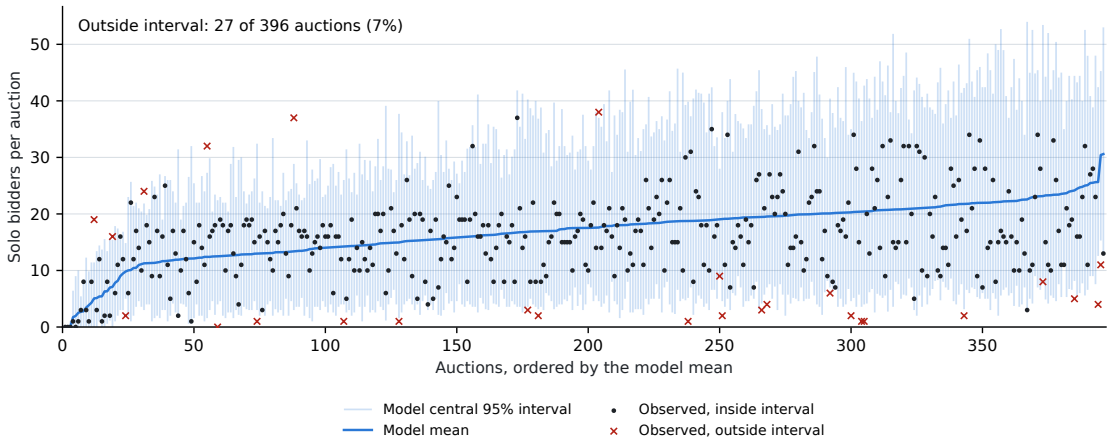


Figure A3. Solo bidders per auction: data and model. Observed solo bidders and the mean and central 95 percent interval of the unconditional predictive distribution from 64 simulations, with auctions ordered by the model mean. The auction-level states are drawn from their distributions. The interval describes variation in simulated outcomes at the estimates, not parameter uncertainty.

Figure A3 compares each auction's observed solo bidders with the model's predictive distribution. Of the 396 auctions, 27 (6.8 percent) lie outside the central 95 percent predictive interval. The standard deviation of observed minus predicted solo bidders is 7.6, and the root mean predictive variance is 8.4. These fit measures hold the estimated parameters fixed.

The policy calculations in Section 8.1 condition on posterior mean auction states inferred from observed participation, whereas the fit comparisons here draw states from their distributions. Restricting policy outcomes to auctions below 10 billion yen does not change the estimation sample or these unconditional fit comparisons.

Appendix G. Counterfactual Computation and Additional Results

This appendix describes the policy calculations in Section 8 and compares results with fixed expected profits and alternative initial beliefs. The simulations cover the 390 auctions below 10 billion yen and use the parameter vector preceding the conditional refinement, as explained in Appendix E.6.

G.1 Counterfactual equilibria

Each auction's beliefs and entry depend on its own expected profits and states, so we solve its fixed point separately from those of the other auctions. Starting from convex bilinear mixtures of the estimation beliefs at the posterior mean states $(\hat{\xi}_t, \hat{\eta}_t)$, with either coordinate outside the grid clamped to its boundary, an iteration computes expected profits, simulates 64 stable partitions, and forms the conditional distributions of rival numbers. For a solo bidder these are the numbers of JVs and of other solo bidders in draws with at least one solo bidder; for a JV they are the numbers of other JVs and solo bidders in draws with at least one JV. We mix the implied distributions with the current beliefs, initially with weight 0.5, reducing the weight when changes fail to shrink.

For the 390 auctions below 10 billion yen, the stopping rule requires changes in mean independent bidders below 0.02 overall and 0.05 among auctions from 5 to below 10 billion yen. It also checks the numbers of JV rivals, solo rivals and total rivals separately for both focal modes. Within each size group, the mean absolute difference between believed and simulated rival numbers, weighted by the simulated frequency of the focal mode, must be below 0.15. A mode absent from all simulation draws has no measured conditional residual; it is not assigned a zero error. We retain its previous conditional belief. The iteration cap is 60; it bounds computation rather than defines convergence.

Under a JV ban, the support of each distribution of rival numbers is restricted to zero JVs. If no firm bids alone in any draw, we evaluate the solo option again at zero rivals

before accepting a zero-entry outcome. This prevents an inherited conditional belief from sustaining nonparticipation without checking the profitable deviation to entry. Both policies use common random numbers. The expectations in equation (14) average the winning payment, assigning zero to a draw without a usable bid, and the indicator that at least one usable bid is submitted. The simulation applies the mode-specific usable-bid probabilities from Section 6; positive formation-stage bidder counts alone do not guarantee an award. The price denominator therefore includes the award probability, and changes in it can change the weights attached to different auctions. For the score comparison in equation (15), we retain the technical score of the same winning bidder in each draw. We calculate each auction's mean winning score conditional on an award under each policy, then average the within-auction percentage changes with equal auction weights.

The formation estimates and posterior states are retained from the 396-auction estimation sample; policy outcomes are aggregated over the 390 auctions below 10 billion yen. Each auction has its own belief fixed point, so restricting the reporting population does not alter the retained auctions' solutions. We also compute the ban with beliefs held at their current-policy values and solve both policies from beliefs built from similar auctions, to distinguish entry responses from sensitivity to the starting beliefs.

Rival bid distributions in the counterfactual. Bidders best-respond to the estimated distributions of rival bids conditional on the numbers of JVs and solo bidders, $\hat{F}_t(\cdot \mid m, n)$. The policy changes these counts and the identities and strengths of participating bidders. We do not reestimate the conditional bid distributions or solve an asymmetric bidding equilibrium under each policy. This approximation matters for the interpretation of the small average price effects and remains a limitation of the counterfactual exercise.

G.2 Additional policy results

Policy outcomes. Table A11 reports current-policy participation, JV wins and award probabilities, and winning prices under both policies, for the two reporting groups.

Table A11. Current-policy outcomes and winning prices below 10 billion yen

	All < 10 bn yen (390 auctions)	5–< 10 bn yen (40 auctions)
Solo bidders	15.75	10.69
JVs	0.67	1.46
JV wins (%)	7	18
Winning price / estimate, current policy (%)	89.05	89.55
Winning price / estimate, ban (%)	89.04	89.66
Award probability (%)	99.75	99.47

Notes: Measures are defined in equations (13) and (14). The first three rows and the award probability report current-policy levels; winning prices are shown for both policies. The winning price is expressed as a percentage of the engineer's estimate. JV wins is the share of awarded contracts won by a JV. The 40-auction group is a subset of the 390-auction group. Both policies maintain independent solo and pair shocks, fix auction states at their posterior means, and use common random numbers with 64 formation draws and 16 bidding replications per auction. These predictions condition on the estimates and do not incorporate parameter uncertainty.

Fixed expected profits. Holding expected profits fixed at their current-policy values gives a decline of 0.415 independent bidders per auction across the 390 auctions, compared with 0.399 after beliefs are updated. In auctions from 5 to below 10 billion yen, the corresponding declines are 1.118 and 0.847. Thus equilibrium adjustment has a larger effect on the participation comparison in the upper size group.

Alternative initial beliefs. Starting the policy calculations from beliefs constructed from comparable auctions gives changes in winning prices, as a percentage of the engineer's estimate, of -0.006 percentage points overall and $+0.141$ points in the upper size group, compared with -0.013 and $+0.112$ in the main calculation. The signs are unchanged, and the largest difference is 0.029 points. These comparisons check initialization at the stated simulation precision, not uniqueness of equilibrium.